

# A Kohlrausch or Weibull Exponent Below One Admits a Classical Mixture

## Pollard's Criterion as a Licence Condition, Two Predictions Refutable Against Existing Literature, and a Self-Audit of the Framework That Produced Them

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I consolidate into a single document a programme previously distributed across three papers, reordered by logical dependency rather than by chronology so that the surviving core is separable from the speculation built on it. Every equation was reimplemented in a dependency-free numerical core and forced to evaluate; every published number was recomputed from the equations or the accompanying data files.

**Exact identities [EXACT].** The  $\Lambda$ -hierarchy postulate is not irreducible: since  $\tau_P E_P = \hbar$  and  $\ell_P E_P = \hbar c$  hold identically it reduces to  $\tau(\Lambda) = \hbar/E$  and  $\ell(\Lambda) = c\tau(\Lambda)$ , so at  $E = mc^2$  the hierarchy length *is* the reduced Compton wavelength. The accessible information satisfies  $I_Q(\gamma) = (\gamma/\tau)^\beta$  exactly, and a distinguished point at  $D/L = 1/2$  carries four coincident properties: exactly one bit, the entanglement sudden-death point for  $C_0 = 1$ , the maximum of  $D\bar{C}$  at  $L^2/4$ , and a Bures arc of  $\pi/3$ . All invariant in  $\beta$ ,  $\tau$  and  $L$ .

**Verification [OBS].** The conservation law  $D + C = 1$  holds to machine precision across 650 circuits on two IBM backends and  $\sim 5.5 \times 10^6$  shots;  $\beta < 1$  is the invariant no platform violates.

**No-go theorem [THM].** For a power-law bath and *any* pulse-sequence filter  $g$ ,  $\Gamma(t) = A t^{1+\alpha} \int u^{-2-\alpha} g(u) du$ : the exponent is independent of  $g$ . Since  $1 - \alpha/2 < 1 \leq 1 + \alpha$ , the relation  $\beta = 1 - \alpha/2$  is unreachable by any filter-function route, and Axiom 3 is relabelled from theorem to postulate.

**Errata [ERRATUM].** The critical qubit number is  $n^* = 1.848$ , not 2.6. The LIGO ringdown significance, with the internal error scaled by  $\sqrt{\chi^2/\nu} = 1.669$  as the catalogue's own goodness of fit requires, is  $12.5\sigma$ ; we record that our first correction of it,  $20.9\sigma$ , was itself wrong for neglecting that scaling, and that the originally published  $13.8\sigma$  was closer to the defensible value than our correction. A reported 18 kHz emergence mode lies below the Rayleigh limit of its own data. The published comoving density of remnants corresponds to  $\Omega = 0.063$ , not to the 0.12 it is stated to reproduce.

**Transferable criteria [THM].** Three results export, each with the observation that would retire it. Pollard's criterion, used as a licence condition:  $\beta \leq 1$  certifies that the curve admits a positive rate density and  $\beta > 1$  proves none exists, so in reliability engineering an infant-mortality exponent certifies batch heterogeneity while a wear-out exponent forbids it under any partition. The corrected diffusion bridge  $\beta = (1 + \alpha_{\text{diff}})/2$ , a correction inside the framework but a *prediction* outside it, whose domain forces  $\beta \in (1/2, 1]$  and which, composed with the criterion, yields the falsifiable statement that superdiffusive transport forbids reading the relaxation of the same medium as a distribution of rates. And the additive coordinate  $I_Q = -\ln(1 - D/L)$ , which quantifies the information a saturation clamp destroys.

We state plainly the inference this work does *not* license: a fitted  $\beta < 1$  is not evidence of quantum behaviour. What the theorem establishes is *representability* of the curve, not identification of the mechanism — the map from mechanism to curve is many-to-one — so the correct statement is that the data are compatible with a classical mixture, which is precisely why they cannot be adduced as evidence against one.

**The endpoint [CONJ].** The framework culminates in a dark-matter candidate: Planck-scale black-hole remnants at  $M_{\min} = m_P/2\sqrt{\pi} = 6.140 \mu\text{g}$ , an exact identity rather than the approximation  $0.28 m_P$  under which it was published. Evaluating the scenario shows the usual dismissal to be right for the wrong reason. The remnant carries 148.6 J of kinetic energy at galactic velocity and one transit is expected per  $0.86 \text{ km}^2\cdot\text{yr}$ , so rarity is not the obstacle; but the mean free path in rock is  $10^{23} \text{ ly}$  and the probability of one interaction while crossing the Earth is  $10^{-32}$ , so the energy is never deposited. The candidate is cross-section-limited, not exposure-limited, and the two demand opposite experiments. Four falsifiers are given, two requiring no new instrument.

All results are reproducible from the accompanying implementation (354 tests, 37 published-number checks).

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## Contents

|                                                                       |           |                                                                 |           |
|-----------------------------------------------------------------------|-----------|-----------------------------------------------------------------|-----------|
| <b>I The framework</b>                                                | <b>4</b>  | <b>VII. Hardware Verification of Conservation Laws</b>          | <b>18</b> |
| I. Introduction                                                       | 4         | A. Experimental Setup                                           | 18        |
| A. The object                                                         | 5         | B. Results: $D + C = 1$ Verification                            | 18        |
| B. What this paper claims, in decreasing order of confidence          | 5         | C. $\gamma$ -Sweep Verification                                 | 18        |
| C. The inference this paper does not license                          | 5         | D. Evidence Level                                               | 18        |
| D. Reproducibility                                                    | 5         | <b>VIII. Discovery of Three Decoherence Regimes</b>             | <b>18</b> |
| II. Complete Mathematical Framework                                   | 5         | A. Multi-State Experimental Design                              | 18        |
| A. Axiomatic Foundation                                               | 5         | B. Three Regimes Identified                                     | 19        |
| B. The Deviation Operator                                             | 7         | C. Regime 1: Exponential ( $\beta \approx 1$ )                  | 19        |
| C. Transition Rate and Dynamical Structure                            | 7         | D. Regime 2: Stretched Exponential ( $\beta < 1$ )              | 19        |
| D. The Noise Functional and Lindblad Connection                       | 8         | E. Regime 3: Revival Dynamics                                   | 19        |
| E. Coherence Metrics and Operational Measures                         | 9         | F. Physical Interpretation                                      | 19        |
| F. Parameter Space Geometry                                           | 9         | <b>IX. Theoretical Framework: Lindblad Classification</b>       | <b>19</b> |
| G. Information-Theoretic Structure                                    | 10        | A. Noise Spectrum and $\alpha$ Exponent                         | 19        |
| H. Summary of Mathematical Results                                    | 10        | B. Regime Classification Theorem                                | 20        |
| III. Information Geometry and the Entropic Phase Space                | 10        | C. Topology $\rightarrow$ Effective Noise                       | 20        |
| A. The Space of Quantum States as a Riemannian Manifold               | 11        | D. Revival Condition                                            | 20        |
| B. The Information-Decoherence Functional                             | 11        | E. Experimental Validation and Circularity Correction           | 20        |
| C. The Entropic Phase Space                                           | 12        | F. BMRB Circularity Analysis                                    | 20        |
| D. The Quantum-Classical Transition as a Geometric Flow               | 13        | G. $\eta$ Independence Measurement                              | 20        |
| E. The $\Lambda$ -Hierarchy as Scale Fibration                        | 13        | <b>X. Scaling Laws</b>                                          | <b>20</b> |
| F. Information Conservation in the Phase Space                        | 14        | A. Initial Validation Results: Parameter Scaling                | 20        |
| IV. Exact Identities                                                  | 14        | B. Discovery: $\beta(n)$ Scaling Law                            | 21        |
| A. The $\Lambda$ -hierarchy is the Compton scale                      | 14        | C. Timescale Scaling: $\tau(n)$                                 | 21        |
| B. Accessible information is the reduced argument                     | 15        | D. Hardware Power-Law (Sec. VI)                                 | 21        |
| C. The binary threshold                                               | 15        | E. Quantum-Classical Boundary                                   | 21        |
| <b>II Verification</b>                                                | <b>16</b> | F. $\beta_{\text{gate}}$ vs $\beta_{\text{idle}}$ Discrepancy   | 21        |
| V. Simulation Validation <a href="#">[FIT]</a>                        | 16        | <b>III Structure and limits</b>                                 | <b>21</b> |
| A. Experimental Setup                                                 | 16        | <b>XI. Physical-Chemistry <math>\Lambda</math> Hierarchy</b>    | <b>21</b> |
| B. Fit Results                                                        | 16        | A. Fundamental Postulate                                        | 21        |
| C. Model Selection                                                    | 17        | B. Nuclear Validation (NNDC)                                    | 21        |
| D. Robustness Ranking                                                 | 17        | C. Atomic Validation (NIST)                                     | 21        |
| VI. Hardware Validation <a href="#">[FIT]</a> / <a href="#">[OBS]</a> | 17        | D. Molecular Validation                                         | 22        |
| A. Foundational Results: Power-Law Universality                       | 17        | E. Multi-Scale Hierarchy                                        | 22        |
| B. Conservation Law: $D + C = 1$                                      | 17        | F. Standard Particle Content Table                              | 22        |
| C. Phase 29 Experiment: 400-Circuit v2.0                              | 18        | G. Majorana Fermion Predictions                                 | 22        |
| D. Cross-Platform Validation                                          | 18        | <b>XII. Information Conservation and Non-Markovian Backflow</b> | <b>22</b> |
|                                                                       |           | A. Information Conservation                                     | 22        |
|                                                                       |           | 1. Level 1: Algebraic identity                                  | 22        |
|                                                                       |           | 2. Level 2: Von Neumann information conservation from unitarity | 22        |
|                                                                       |           | 3. Level 3: Operational (accessible) information                | 24        |
|                                                                       |           | B. Quantum Accessible Information                               | 24        |
|                                                                       |           | C. Non-Markovian Information Backflow                           | 24        |
|                                                                       |           | 1. The apparent contradiction                                   | 24        |
|                                                                       |           | 2. Noise accumulation and time mapping                          | 24        |
|                                                                       |           | D. Experimental Signatures                                      | 26        |

|                                                                    |    |                                                                  |    |
|--------------------------------------------------------------------|----|------------------------------------------------------------------|----|
| XIII. A No-Go Theorem for the Filter-Function Route                | 26 | XIX. Transient Emergence Phenomena                               | 36 |
| A. Statement                                                       | 26 | A. Spectral Signatures                                           | 36 |
| B. Numerical verification                                          | 26 | B. DFS Encapsulation                                             | 37 |
| C. The two published routes                                        | 26 | C. Golden Ratio in Hardware                                      | 37 |
| D. What survives, and what this settles                            | 27 | D. Emergence Threshold                                           | 37 |
| E. Why $\beta < 1$ needs no spectral derivation                    | 27 | E. Container State Rankings                                      | 37 |
| F. Scope of the theorem                                            | 27 | XX. Anti-Quantum Formalization                                   | 37 |
| XIV. Corrections to the Derivations                                | 28 | A. Deformation Structure                                         | 37 |
| A. The fluctuator coupling density                                 | 28 | B. Spectral Predictions                                          | 37 |
| B. A missing hypothesis in the basic-properties theorem            | 28 | C. Connection to Emergence                                       | 38 |
| C. An expansion written in the wrong order                         | 28 | <b>V Transferable criteria</b>                                   | 38 |
| D. An invalidated backflow criterion                               | 29 | XXI. Transferable Criteria                                       | 38 |
| E. Relabelling the restatement of Axiom 3                          | 29 | A. The mixture gate                                              | 38 |
| F. A closed question                                               | 29 | B. A falsifiable prediction linking transport to relaxation      | 39 |
| XV. Errata to Published Numbers                                    | 29 | C. Channel counting and the additive coordinate                  | 40 |
| A. The critical qubit number                                       | 29 | D. What does not transfer                                        | 40 |
| B. The gravitational-wave significance                             | 30 | E. Summary of transfers                                          | 40 |
| C. The primary quantity                                            | 30 | <b>VI The gravitational frontier</b>                             | 40 |
| D. The choice decides whether the framework predicts negentropy    | 30 | XXII. Gravitational Connections                                  | 40 |
| XVI. Spectral Resolution as an Audit Criterion                     | 31 | A. LIGO Ringdown Analysis                                        | 40 |
| A. The criterion                                                   | 31 | B. Mercury Perihelion Precession — 1PN Verification              | 41 |
| B. Application to the reported emergence mode                      | 31 | C. Formal Lorentz Invariance                                     | 41 |
| C. Two incompatible analyses                                       | 32 | D. Gravitational Coalescence as Entropic Duality                 | 42 |
| D. What survives                                                   | 32 | E. Gravity as Spatial Decoherence Operator                       | 42 |
| E. A note on interpretation                                        | 32 | F. Anti-Entropic Information Condensation                        | 43 |
| XVII. Two Limits That Discriminate Between Conventions             | 32 | G. Emergent Gravity Conjecture                                   | 44 |
| A. The biological limit, and a correction that rescues it          | 32 | H. Black-Hole $C \rightarrow Q$ Transition                       | 44 |
| B. The cosmological limit selects a convention                     | 33 | I. Dark-Matter Scenario — Superseded                             | 45 |
| C. A numerical coincidence, and why it is not reported as a result | 33 | XXIII. The Dark-Matter Candidate                                 | 45 |
| <b>IV Extensions</b>                                               | 34 | A. The chain, with each link labelled                            | 45 |
| XVIII. Biological Extension: Programmed Decoherence                | 34 | B. An exact identity hidden behind an approximation              | 45 |
| A. The Biological Scale in the $\Lambda$ -Hierarchy                | 34 | C. Erratum: the published number density                         | 46 |
| B. The Decoherence-Selection Principle                             | 34 | D. The phenomenology, and why the usual verdict is the wrong one | 46 |
| C. Anomalous Diffusion as Biological Decoherence                   | 35 | E. Falsifiers                                                    | 46 |
| D. Information Transformation: Quantum $\rightarrow$ Biological    | 35 | F. Verdict                                                       | 47 |
| E. The Entropic-Spatial Duality in Biology                         | 36 | <b>VII Closing</b>                                               | 47 |
| F. Predictions for Biological Systems                              | 36 | XXIV. Open Problems                                              | 47 |
|                                                                    |    | A. The derivation of Axiom 3                                     | 47 |
|                                                                    |    | B. The two meanings of $\alpha$                                  | 47 |
|                                                                    |    | C. The information-condensation fixed point                      | 47 |

|                                                    |    |                                                          |    |
|----------------------------------------------------|----|----------------------------------------------------------|----|
| D. Bidirectionality against the unique attractor   | 48 | e. Bell chain states                                     | 57 |
| E. The reported spectral mode                      | 48 | 4. Error mitigation                                      | 57 |
| F. A bridge between state space and physical space | 48 | a. Readout error correction                              | 57 |
| G. A methodological remark                         | 48 | b. Dynamical decoupling                                  | 57 |
|                                                    |    | c. Post-selection                                        | 57 |
|                                                    |    | 5. Fitting procedure                                     | 57 |
| XXV. Discussion                                    | 48 | E. Raw Data Tables                                       | 58 |
| A. Implications for Quantum Computing              | 48 | 1. Foundational data summary                             | 58 |
| B. Cross-Backend Discoveries                       | 49 | 2. Phase 21 raw data                                     | 58 |
| 1. The $\beta$ anomaly                             | 49 | 3. Phase 26C GHZ decay data                              | 58 |
| 2. THM-26.1 cross-backend tension                  | 49 | 4. Phase 26E summary                                     | 58 |
| 3. Implications for universality                   | 49 | 5. Phase 24 $\Lambda$ sample data                        | 58 |
| C. Observable Emergence from the SDF               | 49 | 6. Phase 29 data                                         | 58 |
| 1. The observable decay formula                    | 49 |                                                          |    |
| 2. Significance                                    | 49 | F. Negative Results and Lessons Learned                  | 58 |
| D. Information Conservation — Three Levels         | 50 | 1. Phase 25: Failed universality test                    | 59 |
| E. Biological Decoherence Interpretation           | 50 | 2. Dicke state anomalies                                 | 59 |
| F. Connection to Paper 1 Predictions               | 50 | 3. $\sqrt{n/2}$ hypothesis refutation                    | 59 |
| G. Falsifiability and Future Tests                 | 50 | 4. Circularity in $\alpha \leftrightarrow \beta$ mapping | 59 |
| H. Limitations and Caveats                         | 51 | 5. BMRB circularity confirmed (Phase 29)                 | 59 |
| 1. Experimental limitations                        | 51 | 6. T1/T2 proxy inadequacy (Phase 29)                     | 59 |
| 2. Theoretical limitations                         | 51 |                                                          |    |
| 3. Scope limitations                               | 51 | G. LIGO Ringdown Catalog                                 | 60 |
|                                                    |    | 1. Analysis methodology                                  | 60 |
| XXVI. Conclusion                                   | 52 | 2. BBH ringdown fit results                              | 60 |
| A. What survives                                   | 52 | 3. Catalog breakdown                                     | 60 |
| B. What does not                                   | 52 |                                                          |    |
| C. The inference this work does not license        | 52 | H. Information Conservation: Formal Derivation           | 60 |
| D. The decisive test, which we do not perform      | 52 | 1. Framework setup                                       | 60 |
| E. On the endpoint                                 | 52 | 2. Conservation theorem: proof                           | 60 |
| F. On the method                                   | 52 | 3. Von Neumann entropy connection                        | 61 |
|                                                    |    | 4. Implications                                          | 61 |
| A. Register of Documented Inconsistencies          | 53 | a. Information redistribution, not destruction           | 61 |
| B. Reproducibility                                 | 53 | b. Consistency with unitarity                            | 61 |
| 1. The implementation                              | 53 | c. Black hole information paradox                        | 62 |
| 2. Selected verifications                          | 53 | d. Operational interpretation                            | 62 |
| 3. Reproducing the audit                           | 53 |                                                          |    |
| 4. Scope and limitations of the audit              | 53 | References                                               | 62 |
| C. Mathematical Derivations                        | 54 |                                                          |    |
| 1. Lindblad derivation of $\beta = 1 - \alpha/2$   | 54 |                                                          |    |
| 2. Information conservation: $D + C = 1$           | 55 |                                                          |    |
| 3. Accessible quantum information                  | 55 |                                                          |    |
| 4. Scaling law derivation                          | 56 |                                                          |    |
| 5. Lorentz invariance of the decoherence function  | 56 |                                                          |    |
| 6. Inverse function                                | 56 |                                                          |    |
| D. Experimental Details                            | 56 |                                                          |    |
| 1. IBM Quantum backends                            | 56 |                                                          |    |
| 2. Circuit configurations                          | 56 |                                                          |    |
| 3. State preparation protocols                     | 56 |                                                          |    |
| a. GHZ states                                      | 56 |                                                          |    |
| b. W states                                        | 57 |                                                          |    |
| c. Cluster states                                  | 57 |                                                          |    |
| d. Dicke states                                    | 57 |                                                          |    |

## Part I

# The framework

### I. Introduction

This paper consolidates into one document a programme previously distributed across three: the original formulation of the stretched-exponential decoherence framework, its hardware verification and extension, and the executable audit that corrected both. The consolidation is not editorial. Presenting the results in the order in which they were *obtained* obscured which of them survive; presenting them in the order in which they *depend on each other* makes the surviving core small, explicit, and separable from the speculation built on top of it.

### A. The object

The framework studies a single functional. For a quantum system coupled to an environment, the decoherence accumulated after interaction parameter  $\gamma$  is taken to be

$$D(\gamma) = L \left[ 1 - \exp \left( - \left( \frac{\gamma}{\tau} \right)^\beta \right) \right], \quad (\star)$$

with  $L$  the asymptotic ceiling,  $\tau$  a characteristic scale and  $\beta$  the stretching exponent. The form is Kohlrausch's, and its use as a relaxation law predates quantum information by more than a century. Nothing in this paper claims the functional as new. What the paper establishes concerns its exact structure, the limits of what can be inferred from a fitted  $\beta$ , and where the published derivations do and do not hold.

### B. What this paper claims, in decreasing order of confidence

The document is organised so that this ordering is also the reading order.

- **[EXACT] Identities** (Part I). The  $\Lambda$ -hierarchy reduces to  $\tau = \hbar/E$  and  $\ell = \lambda_C$ ; the accessible information collapses to  $I_Q = (\gamma/\tau)^\beta$ ; and a distinguished point at  $D/L = 1/2$  carries four coincident exact properties. Verified at machine precision.
- **[OBS] Verification** (Part II). The conservation law  $D + C = 1$  holds across 650 circuits on two IBM backends and  $\sim 5.5 \times 10^6$  shots;  $\beta < 1$  is the robust empirical invariant across every platform tested.
- **[THM] Limits** (Part III). A no-go theorem shows the temporal exponent of dephasing is independent of the pulse sequence, which removes the derivation previously claimed for Axiom 3; a spectral-resolution criterion retires a reported emergence mode; and two published numbers are corrected, including one whose first correction, by us, was itself wrong.
- **[THM] Transferable criteria** (Part V). What the work exports to fields that fit stretched exponentials without reference to this framework, each stated with the observation that would retire it.
- **[CONJ] The gravitational frontier** (Part VI), ending in the dark-matter candidate. Speculative throughout, and labelled as such link by link.

### C. The inference this paper does not license

One result inverts a common reading and is placed here rather than in the conclusion, because it governs how the rest should be interpreted.

By Pollard's criterion,  $\exp(-x^\beta)$  is completely monotone if and only if  $\beta \leq 1$ , and by Bernstein's theorem this holds if and only if the curve is the Laplace transform of a *positive* measure. A fitted  $\beta \leq 1$  therefore certifies that the data are reproducible by a classical mixture of

exponential decays. Every platform in this corpus reports  $\beta < 1$ .

A stretched exponential is consequently *not* evidence of quantum behaviour. In the regime where this framework fits best, it certifies compatibility with classicality. The framework's value lies in what it forbids, and Sec. XXI A makes that forbidding quantitative.

### D. Reproducibility

Every numerical claim in this document is recomputed from the equations by an accompanying dependency-free implementation: 354 regression tests and 37 checks that recompute published numbers from the data files they came from. Where a published number could not be reproduced it is recorded as an erratum rather than adjusted, and where a correction of ours proved wrong it is recorded as such. Appendix B gives the procedure.

## II. Complete Mathematical Framework

This section develops the mathematical structure of the Stretched-exponential Decoherence Framework (SDF) from first principles. Rather than postulating the functional form of  $D(\gamma)$ , I show that it emerges as the *unique* solution to a physically motivated differential equation. This section contains all definitions, theorems, and proofs that underpin the remainder of the paper.

### A. Axiomatic Foundation

I begin with three physical axioms from which the entire framework follows.

**Definition 1** (Decoherence Functional). *For a quantum system  $\mathcal{S}$  with Hilbert space  $\mathcal{H}_S$ , coupled to an environment  $\mathcal{E}$  with Hilbert space  $\mathcal{H}_E$ , the decoherence functional is a map  $D : \mathbb{R}_+ \rightarrow [0, L]$  parametrized by the noise exposure  $\gamma \geq 0$ . The coherence complement is  $C(\gamma) \equiv 1 - D(\gamma)$ .*

**Axiom 1** (Bounded Decoherence) [DEF]. *The decoherence functional  $D$  satisfies:*

- (i)  $D(0) = 0$  (initial purity),
- (ii)  $\lim_{\gamma \rightarrow \infty} D(\gamma) = L \leq 1$  (asymptotic limit),
- (iii)  $D \in C^\infty(\mathbb{R}_+)$  and  $D$  is strictly monotone increasing.

*The parameter  $L \in (0, 1]$  represents the maximum decoherence achievable by the system–environment coupling. For a system that fully decoheres,  $L = 1$ .*

**Axiom 2** (Generalized Decay) [DEF]. *The functional  $D$  satisfies the first-order ordinary differential equation:*

$$\boxed{\frac{dD}{d\gamma} = \frac{\beta}{\tau} \left( \frac{\gamma}{\tau} \right)^{\beta-1} (L - D)} \quad (1)$$

*where  $\beta \in (0, 2]$  is the stretching exponent encoding environmental noise structure, and  $\tau > 0$  is the characteristic decoherence scale.*



The ODE (1) has a transparent physical reading: the rate of decoherence at noise exposure  $\gamma$  is proportional to the *remaining coherence*  $(L - D)$ , modulated by a power-law kernel  $(\gamma/\tau)^{\beta-1}$  that encodes memory effects. For  $\beta = 1$  this reduces to the memoryless (Markovian) decay  $dD/d\gamma = (L - D)/\tau$ .

**Axiom 3** (Environmental Encoding — Noise-Structure Duality) **[DEF]**. *The stretching exponent  $\beta$  encodes the noise power spectrum  $S(\omega)$  of the environment through:*

$$\beta = 1 - \frac{\alpha}{2}, \quad S(\omega) \propto \frac{1}{\omega^\alpha}, \quad \alpha \in [0, 2] \quad (2)$$

*This maps the noise exponent  $\alpha$  bijectively onto  $\beta \in (0, 1]$ : white noise ( $\alpha = 0$ ) gives  $\beta = 1$  (Markovian),  $1/f$  noise ( $\alpha = 1$ ) gives  $\beta = 1/2$  (deeply stretched), and the limit  $\alpha \rightarrow 2$  corresponds to  $\beta \rightarrow 0$  (frozen dynamics).*

I emphasize that Axiom 3 establishes a *duality* between the frequency-domain description (noise spectrum) and the time/noise-domain description (stretched exponent), unifying the physics of the environment with the mathematical form of the decoherence functional.

**Theorem 1** (Existence and Uniqueness — from Axioms to the SDF Equation). **[THM]**

*Axioms 1 and 2 admit a unique solution:*

$$D(\gamma) = L \left[ 1 - \exp\left(-\left(\frac{\gamma}{\tau}\right)^\beta\right) \right] \quad (3)$$

*Proof.* Separate variables in the ODE (1):

$$\frac{dD}{L - D} = \frac{\beta}{\tau} \left(\frac{\gamma}{\tau}\right)^{\beta-1} d\gamma. \quad (4)$$

Integrate both sides. The left-hand side gives  $-\ln(L - D) + C_1$ , while the right-hand side yields  $(\gamma/\tau)^\beta + C_2$ . Combining constants:  $-\ln(L - D) = (\gamma/\tau)^\beta + C$ . Apply the initial condition  $D(0) = 0$  (Axiom 1i):  $-\ln L = C$ , so that  $-\ln(L - D) = (\gamma/\tau)^\beta$ . Exponentiating:  $\frac{L-D}{L} = \exp\left(-\left(\frac{\gamma}{\tau}\right)^\beta\right)$ , hence  $D(\gamma) = L[1 - \exp(-(\gamma/\tau)^\beta)]$ .

*Uniqueness.* Write the ODE as  $dD/d\gamma = f(\gamma, D)$  with  $f(\gamma, D) = (\beta/\tau)(\gamma/\tau)^{\beta-1}(L - D)$ . For  $\beta \geq 1$ ,  $f$  is Lipschitz in  $D$  on any compact subset of  $\mathbb{R}_+ \times [0, L]$ , and the Picard–Lindelöf theorem guarantees uniqueness. For  $0 < \beta < 1$ ,  $f(\gamma, D)$  has an integrable singularity at  $\gamma = 0^+$ ; uniqueness follows from the Osgood criterion since  $|f(\gamma, D_1) - f(\gamma, D_2)| \leq (\beta/\tau)(\gamma/\tau)^{\beta-1}|D_1 - D_2|$  and the coefficient is integrable near  $\gamma = 0$ .  $\square$

*Remark.* Equation (3) is *not* an ansatz. It is the unique solution to the physical ODE (1) subject to boundary conditions dictated by Axiom 1. This deductive structure parallels Newton’s derivation of Keplerian orbits from the inverse-square law: the functional form is a theorem, not a hypothesis.

**Theorem 2** (Basic Properties). **[THM]**

*The functional  $D(\gamma)$  in Eq. (3) satisfies:*

- (i)  $D(0) = 0$  (quantum boundary),
- (ii)  $\lim_{\gamma \rightarrow \infty} D(\gamma) = L$  (classical limit),
- (iii)  $\partial D/\partial \gamma > 0$  for all  $\gamma > 0$  (strict monotonicity),
- (iv)  $D \in C^\infty(\mathbb{R}_+)$  (smoothness),
- (v)  $D$  is strictly concave for  $\beta \leq 1$ .

*Proof.* Properties (i)–(iv) follow by direct computation from Eq. (3): (i)  $D(0) = L(1 - e^0) = 0$ . (ii)  $\lim_{\gamma \rightarrow \infty} D = L(1 - 0) = L$ . (iii) From (1),  $dD/d\gamma > 0$  since each factor  $\beta$ ,  $\tau^{-1}$ ,  $(\gamma/\tau)^{\beta-1}$ , and  $(L - D)$  is positive for  $\gamma > 0$ . (iv)  $D$  is a composition of  $C^\infty$  functions on  $\mathbb{R}_+$ .

For (v), compute the second derivative:

$$\frac{d^2 D}{d\gamma^2} = \frac{L\beta}{\tau^2} \left(\frac{\gamma}{\tau}\right)^{\beta-2} e^{-(\gamma/\tau)^\beta} \left[ (\beta - 1) - \beta \left(\frac{\gamma}{\tau}\right)^\beta \right]. \quad (5)$$

For  $\beta \leq 1$ , both terms in brackets are non-positive (the first is  $\leq 0$  and the second is  $< 0$  for  $\gamma > 0$ ), giving  $d^2 D/d\gamma^2 < 0$ : strict concavity.  $\square$

**Theorem 3** (Inverse Function). **[THM]**

*The map  $D : [0, \infty) \rightarrow [0, L]$  is a bijection. Its inverse is:*

$$\gamma = \tau \left[ -\ln\left(1 - \frac{D}{L}\right) \right]^{1/\beta} \quad (6)$$

*Proof.* Strict monotonicity (Theorem 2(iii)) guarantees that  $D$  is injective. Since  $D$  is continuous with  $D(0) = 0$  and  $\lim_{\gamma \rightarrow \infty} D = L$ , the intermediate value theorem gives surjectivity onto  $[0, L)$ . Solving Eq. (3) for  $\gamma$ :

$$\begin{aligned} 1 - D/L &= e^{-(\gamma/\tau)^\beta} \\ -\ln(1 - D/L) &= (\gamma/\tau)^\beta \\ \gamma &= \tau [-\ln(1 - D/L)]^{1/\beta}. \end{aligned} \quad \square$$

**Theorem 4** (Scale Invariance). **[THM]**

*For any  $\lambda > 0$ :*

$$D(\lambda\gamma; \lambda\tau, \beta, L) = D(\gamma; \tau, \beta, L) \quad (7)$$

*That is,  $D$  depends on  $\gamma$  and  $\tau$  only through the dimensionless ratio  $\gamma/\tau$ .*

*Proof.* Direct substitution:  $D(\lambda\gamma; \lambda\tau, \beta, L) = L[1 - \exp(-(\lambda\gamma/\lambda\tau)^\beta)] = L[1 - \exp(-(\gamma/\tau)^\beta)] = D(\gamma; \tau, \beta, L)$ .  $\square$

*Physical consequence.* Scale invariance implies that the SDF is insensitive to the choice of units for the noise parameter  $\gamma$ ; only the ratio  $\gamma/\tau$  is physically meaningful. This parallels the scale invariance of dimensional analysis in classical mechanics and constrains the form of any extension of the framework.

## B. The Deviation Operator

The SDF makes a sharp, testable prediction: ideal (simulated) decoherence is exponential ( $\beta = 1$ ), while hardware decoherence is stretched ( $\beta < 1$ ) due to the non-trivial noise spectrum of real environments. I now formalize this observation.

**Definition 2** (Ideal Decoherence). *In the Markovian limit (white noise,  $\alpha = 0$ , hence  $\beta = 1$ ):*

$$D_{\text{ideal}}(\gamma) = L \left[ 1 - \exp\left(-\frac{\gamma}{\tau_{\text{id}}}\right) \right] \quad (8)$$

**Definition 3** (Hardware Decoherence). *For a real environment with noise spectrum  $S(\omega) \propto 1/\omega^\alpha$ :*

$$D_{\text{hw}}(\gamma) = L \left[ 1 - \exp\left(-\left(\frac{\gamma}{\tau_{\text{hw}}}\right)^{\beta_{\text{hw}}}\right) \right], \quad \beta_{\text{hw}} = 1 - \alpha/2 \quad (9)$$

**Definition 4** (Deviation Field). *The deviation field captures the pointwise difference:*

$$\Delta(\gamma; \alpha) \equiv D_{\text{hw}}(\gamma) - D_{\text{ideal}}(\gamma) \quad (10)$$

When  $\tau_{\text{hw}} = \tau_{\text{id}} \equiv \tau$  (matched scales), this isolates the pure effect of stretching.

**Theorem 5** (Deviation Properties). *[THM]*

*For matched scales  $\tau_{\text{hw}} = \tau_{\text{id}} = \tau$  and  $\alpha > 0$ :*

- (i)  $\Delta(0; \alpha) = 0$  (preserved initial conditions),
- (ii)  $\lim_{\gamma \rightarrow \infty} \Delta(\gamma; \alpha) = 0$  (asymptotic equivalence),
- (iii)  $\exists! \gamma_{\text{max}}(\alpha) > 0$  such that  $|\Delta|$  is maximized (“deviation peak”),
- (iv) The deviation integral satisfies  $I_\Delta \equiv \int_0^\infty |\Delta(\gamma; \alpha)| d\gamma = O(\alpha)$  for small  $\alpha$ .

*Proof.* (i) Both  $D_{\text{hw}}(0) = 0$  and  $D_{\text{ideal}}(0) = 0$ , so  $\Delta(0) = 0$ .

(ii) Both limits equal  $L$ , so  $\Delta(\gamma \rightarrow \infty) \rightarrow L - L = 0$ .

(iii)  $\Delta$  is continuous, vanishes at  $\gamma = 0$  and  $\gamma \rightarrow \infty$ , and is not identically zero for  $\alpha > 0$  (since  $e^{-(\gamma/\tau)^\beta} \neq e^{-\gamma/\tau}$  when  $\beta \neq 1$ ). By the extreme value theorem (Weierstrass), the continuous function  $|\Delta|$  on  $[0, M]$  attains its maximum; since  $\Delta(0) = 0$  and  $\Delta(M) \rightarrow 0$  while  $\Delta$  is not identically zero, this maximum occurs at an interior point. Uniqueness of  $\gamma_{\text{max}}$  follows from the analysis of  $\partial\Delta/\partial\gamma = 0$ , which yields a single transcendental equation with unique solution for  $\beta$  close to 1.

(iv) Taylor-expand  $e^{-(\gamma/\tau)^\beta}$  around  $\beta = 1$ . Writing  $\beta = 1 - \alpha/2$ :

$$e^{-(\gamma/\tau)^\beta} = e^{-\gamma/\tau} \left[ 1 + \frac{\alpha}{2} \frac{\gamma}{\tau} \ln \frac{\gamma}{\tau} + O(\alpha^2) \right].$$

Hence  $\Delta(\gamma; \alpha) = -L \cdot (\alpha/2) \cdot e^{-\gamma/\tau} \cdot (\gamma/\tau) \ln(\gamma/\tau) + O(\alpha^2)$ , and  $I_\Delta = O(\alpha)$  follows by dominated convergence.  $\square$

**Theorem 6** (Deviation Bound). *[THM]*

$$\sup_{\gamma \geq 0} |\Delta(\gamma; \alpha)| \leq L \cdot \frac{\alpha}{2} \cdot e^{-1} [1 + O(\alpha)] \quad (11)$$

*Proof.* At the supremum,  $\partial\Delta/\partial\gamma = 0$ . From the first-order expansion (proof of Theorem 5(iv)), the extremum of  $g(x) = x \ln x \cdot e^{-x}$  (where  $x = \gamma/\tau$ ) occurs near  $x = 1$ , where  $|g(1)| = 0$  but the deviation expression involves  $|x \ln x| \cdot e^{-x}$ . The function  $h(x) = |x \ln x| \cdot e^{-x}$  on  $(0, \infty)$  attains its maximum at  $x^* = e^{-W(1)} \approx 0.278$  (Lambert  $W$  branch), giving  $h(x^*) \leq e^{-1}$ . Hence  $\sup |\Delta| \leq L \cdot (\alpha/2) \cdot e^{-1} + O(\alpha^2)$ .  $\square$

**Corollary 1.** *The hardware deviation is bounded by:*

- $|\Delta|_{\text{max}} \leq 0.184 L$  for pure  $1/f$  noise ( $\alpha = 1$ ,  $\beta = 0.5$ ),
- $|\Delta|_{\text{max}} \leq 0.092 L$  for  $\alpha = 0.5$  ( $\beta = 0.75$ ),
- $|\Delta|_{\text{max}} \leq 0.037 L$  for  $\alpha = 0.2$  ( $\beta = 0.9$ ).

*These bounds are tight: numerical evaluation confirms that the first-order estimate is accurate to within 5% for  $\alpha \leq 0.5$ .*

### Physical Interpretation: The Deviation Operator

The deviation field  $\Delta(\gamma; \alpha)$  quantifies exactly how much information is redistributed *differently* due to environmental noise structure. In ideal simulation, decoherence follows the geodesic path (pure exponential). On real hardware, the noise spectrum *bends* this path, creating a stretched trajectory. The deviation field  $\Delta$  measures this bending at each noise exposure  $\gamma$ .

The bounded nature of  $\Delta$  (Theorem 6) ensures that hardware decoherence never diverges catastrophically from the ideal model — it is a perturbative deformation, not a qualitative change. This is why the SDF framework can describe both regimes with a single equation.

## C. Transition Rate and Dynamical Structure

**Definition 5** (Transition Rate). *The instantaneous decoherence rate is:*

$$R(\gamma) \equiv \frac{dD}{d\gamma} = \frac{L\beta}{\tau} \left(\frac{\gamma}{\tau}\right)^{\beta-1} \exp\left(-\left(\frac{\gamma}{\tau}\right)^\beta\right) \quad (12)$$

**Theorem 7** (Rate Properties — The Dynamical Trichotomy). *[THM]*

*The transition rate  $R(\gamma)$  has the following properties:*

- (i)  $R(\gamma) > 0$  for all  $\gamma > 0$  (irreversibility),
- (ii) For  $\beta < 1$ :  $\lim_{\gamma \rightarrow 0^+} R(\gamma) = +\infty$  (singular onset — instant response),

- (iii) For  $\beta = 1$ :  $R(0) = L/\tau$  (finite onset — Markovian),
- (iv) For  $\beta > 1$ :  $R(0^+) = 0$  (quiescent onset — sub-Markovian),
- (v) For  $\beta > 1$ ,  $R$  has a unique maximum at

$$\gamma_{\text{peak}} = \tau \left( \frac{\beta - 1}{\beta} \right)^{1/\beta} \quad (13)$$

with peak rate

$$R_{\text{max}} = \frac{L\beta}{\tau} \left( \frac{\beta - 1}{\beta} \right)^{(\beta-1)/\beta} \exp\left(-\frac{\beta - 1}{\beta}\right). \quad (14)$$

*Proof.* (i) Each factor in Eq. (12) is positive for  $\gamma > 0$ .

(ii) For  $\beta < 1$ , the factor  $(\gamma/\tau)^{\beta-1} \rightarrow \infty$  as  $\gamma \rightarrow 0^+$ , while  $e^{-(\gamma/\tau)^\beta} \rightarrow 1$ , so  $R \rightarrow \infty$ .

(iii) For  $\beta = 1$ :  $R(\gamma) = (L/\tau)e^{-\gamma/\tau}$ , hence  $R(0) = L/\tau$ .

(iv) For  $\beta > 1$ :  $(\gamma/\tau)^{\beta-1} \rightarrow 0$  as  $\gamma \rightarrow 0^+$ , so  $R(0^+) = 0$ .

(v) Setting  $dR/d\gamma = 0$  and writing  $u = (\gamma/\tau)^\beta$ :

$$\frac{dR}{d\gamma} \propto (\beta-1)(\gamma/\tau)^{\beta-2} - \beta(\gamma/\tau)^{2\beta-2}/\tau = 0 \implies u = \frac{\beta-1}{\beta},$$

giving  $\gamma_{\text{peak}} = \tau[(\beta-1)/\beta]^{1/\beta}$ . Substitution into  $R$  yields (14). This is a unique maximum since  $R(0^+) = 0$ ,  $R(\infty) = 0$ , and  $R > 0$  in between.  $\square$

*Physical significance.* The trichotomy  $\{\beta < 1, \beta = 1, \beta > 1\}$  is the *mathematical origin* of the three empirically observed decoherence regimes:

- **Stretched regime** ( $\beta < 1$ ): Initial rapid decoherence followed by power-law slowing. Characteristic of systems with  $1/f^\alpha$  noise ( $\alpha > 0$ ). Observed on IBM Quantum hardware.
- **Exponential regime** ( $\beta = 1$ ): Constant rate decay. The Markovian limit. Observed in ideal noise-free simulations.
- **Compressed/revival regime** ( $\beta > 1$ ): Delayed onset with accelerating decoherence. Signature of structured environments with finite correlation time. Observed in revival phenomena (Sec. VIII).

**Theorem 8** (Asymptotic Behavior). [THM]

The decoherence functional admits the following asymptotic expansions:

(i) Early-time ( $\gamma \ll \tau$ ):

$$D(\gamma) = L \left( \frac{\gamma}{\tau} \right)^\beta - \frac{L}{2} \left( \frac{\gamma}{\tau} \right)^{2\beta} + O((\gamma/\tau)^{3\beta}) \quad (15)$$

(ii) Late-time ( $\gamma \gg \tau$ ):

$$D(\gamma) = L - L \exp\left(-\left(\frac{\gamma}{\tau}\right)^\beta\right) \quad (16)$$

with the correction term vanishing super-exponentially.

(iii) Half-decoherence point:  $D(\gamma_{1/2}) = L/2$  at

$$\gamma_{1/2} = \tau (\ln 2)^{1/\beta} \quad (17)$$

*Proof.* (i) Taylor expand  $e^{-x} = 1 - x + x^2/2 - \dots$  with  $x = (\gamma/\tau)^\beta$ .

(ii) Direct from Eq. (3).

(iii) Set  $1 - D/L = 1/2$ , hence  $(\gamma/\tau)^\beta = \ln 2$ , so  $\gamma_{1/2} = \tau(\ln 2)^{1/\beta}$ .  $\square$

#### D. The Noise Functional and Lindblad Connection

I now derive the SDF equation from the standard open quantum systems formalism [1, 2], establishing a rigorous connection between Axiom 3 and the Lindblad master equation [3].

**Theorem 9** (Lindblad Derivation of Stretched-Exponential Decoherence). [THM]

Consider a qubit undergoing pure dephasing under the time-local Lindblad master equation:

$$\frac{d\rho}{dt} = -i[H, \rho] + \sum_k \gamma_k(t) \left( L_k \rho L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho\} \right) \quad (18)$$

with time-dependent dephasing rates  $\gamma_k(t)$  determined by the noise spectral density  $S(\omega)$ . For pure dephasing ( $L_k = \sigma_z$ ), the off-diagonal coherence evolves as:

$$\rho_{01}(t) = \rho_{01}(0) e^{-\Phi(t)} \quad (19)$$

where the dephasing function  $\Phi(t)$  is given by the noise integral:

$$\Phi(t) = \int_0^\infty \frac{S(\omega)}{\pi} \frac{\sin^2(\omega t/2)}{(\omega/2)^2} d\omega \quad (20)$$

For a power-law noise spectrum  $S(\omega) = A/\omega^\alpha$  with infrared cutoff  $\omega_{\text{ir}}$  and ultraviolet cutoff  $\omega_{\text{uv}}$ , in the regime  $1/\omega_{\text{uv}} \ll t \ll 1/\omega_{\text{ir}}$ :

$$\Phi(t) \sim c_\alpha t^{2-\alpha} \quad (21)$$

where  $c_\alpha$  depends on  $A$  and cutoff parameters:

$$c_\alpha = \frac{A}{\pi} \Gamma(1-\alpha) \sin\left(\frac{\pi\alpha}{2}\right) \quad \text{for } 0 < \alpha < 1,$$

and for the special case  $\alpha = 1$  (pure  $1/f$  noise):  $\Phi(t) \sim (A/\pi) t^2 \ln(\omega_{\text{uv}} t)$ .

Hence the decoherence function takes the stretched-exponential form:

$$W(t) \equiv e^{-\Phi(t)} \sim \exp\left(-\left(\frac{t}{\tau_{\text{eff}}}\right)^{2-\alpha}\right) \quad (22)$$

with  $\tau_{\text{eff}} = c_\alpha^{-1/(2-\alpha)}$ . Identifying  $\beta = 2-\alpha$  in the physical dephasing exponent, or equivalently  $\beta = 1 - \alpha_{\text{SDF}}/2$  in the SDF convention where  $\alpha_{\text{SDF}} \in [0, 2)$ , recovers the SDF equation (3).



*Proof.* Starting from Eq. (20) with  $S(\omega) = A/\omega^\alpha$ , split the integral at  $\omega_0 = 1/t$ :

$$\Phi(t) = \frac{A}{\pi} \int_{\omega_{\text{ir}}}^{1/t} \frac{\sin^2(\omega t/2)}{\omega^\alpha (\omega/2)^2} d\omega + \frac{A}{\pi} \int_{1/t}^{\omega_{\text{uv}}} \frac{\sin^2(\omega t/2)}{\omega^\alpha (\omega/2)^2} d\omega. \quad (23)$$

For  $\omega t \ll 1$ :  $\sin^2(\omega t/2)/(\omega/2)^2 \approx t^2$ , contributing  $\sim At^2 \int \omega^{-\alpha} d\omega$ . For  $\omega t \gg 1$ :  $\sin^2(\omega t/2) \rightarrow 1/2$  on average, contributing  $\sim 2A \int \omega^{-2-\alpha} d\omega$ . The dominant term for  $0 < \alpha < 2$  scales as  $\Phi(t) \propto t^{2-\alpha}$ , with the proportionality constant given by the gamma-function expression. Inverting gives  $W(t) = \exp(-(t/\tau_{\text{eff}})^{2-\alpha})$ , which is precisely the SDF form with  $\beta_{\text{phys}} = 2 - \alpha$ .  $\square$

*Convention note.* The mapping between the physical noise exponent  $\alpha$  and the SDF stretching exponent  $\beta$  depends on the specific dephasing protocol. For free-induction decay of transmon qubits with  $1/f^\alpha$  flux noise (the primary experimental context of this paper), the appropriate filter function yields  $\beta = 1 - \alpha_{\text{SDF}}/2$  as stated in Axiom 3. The consistency of this convention is validated experimentally in Sec. VI and Sec. IX.<sup>1</sup>

### E. Coherence Metrics and Operational Measures

The abstract decoherence functional  $D(\gamma)$  connects to experiment through operationally defined metrics.

**Definition 6** (Leakage Metric). *For an  $n$ -qubit system, the leakage-based decoherence is:*

$$D_{\text{leak}}(n) = 1 - P(\text{valid subspace}) \quad (24)$$

where  $P(\text{valid})$  is the probability of measuring outcomes in the expected computational subspace.

**Definition 7** (Stabilizer Metric). *For a stabilizer state with generators  $\{K_1, \dots, K_n\}$ :*

$$D_{\text{stab}}(n) = 1 - \frac{1}{n} \sum_{i=1}^n \langle K_i \rangle \quad (25)$$

**Definition 8** (Fidelity Metric). *Given target state  $\rho_{\text{target}}$  and measured state  $\rho_{\text{meas}}$ :*

$$D_{\text{fid}} = 1 - F(\rho_{\text{meas}}, \rho_{\text{target}}) \quad (26)$$

where  $F(\rho, \sigma) = (\text{Tr} \sqrt{\sqrt{\rho} \sigma \sqrt{\rho}})^2$  is the Uhlmann fidelity.

**Proposition 1** (Metric Ordering). *For stabilizer states measured in the computational basis:*

$$D_{\text{fid}} \leq D_{\text{leak}} \leq D_{\text{stab}} \quad (27)$$

*Proof.* The fidelity  $F \geq P(\text{valid})$  since fidelity accounts for coherent overlap within the valid subspace as well as population. Hence  $D_{\text{fid}} = 1 - F \leq 1 - P(\text{valid}) = D_{\text{leak}}$ . The stabilizer metric averages over all generators, each of which can detect errors not reflected in the global valid-subspace probability, hence  $D_{\text{leak}} \leq D_{\text{stab}}$ .  $\square$

All three metrics satisfy the SDF equation (3) with the same  $(\beta, L)$  but potentially different  $\tau$  values reflecting the sensitivity of each metric to the underlying decoherence process. This metric independence of the functional form (but not the scale) is a non-trivial prediction validated in Sec. VI.

### F. Parameter Space Geometry

The SDF parameters form a structured geometric space whose properties constrain experimental inference and physical interpretation.

**Definition 9** (SDF Parameter Space). *The SDF parameter space is the open manifold:*

$$\mathcal{P} = \{(\tau, \beta, L) : \tau > 0, \beta \in (0, 2], L \in (0, 1]\} \quad (28)$$

with natural coordinates  $(\ln \tau, \beta, L)$ .

**Definition 10** (Fisher Information Metric). *Given  $N$  independent measurements  $\{d_i\}$  at noise exposures  $\{\gamma_i\}$ , the likelihood is  $p(\{d_i\}|\boldsymbol{\theta})$  with  $\boldsymbol{\theta} = (\tau, \beta, L)$ . The Fisher information matrix is:*

$$[\mathbf{F}]_{ij} = \mathbb{E} \left[ \frac{\partial \ln p}{\partial \theta_i} \frac{\partial \ln p}{\partial \theta_j} \right] \quad (29)$$

This induces a Riemannian metric on  $\mathcal{P}$ :

$$ds_{\text{CR}}^2 = \sum_{i,j} F_{ij} d\theta_i d\theta_j \quad (30)$$

known as the Cramér–Rao metric.

**Theorem 10** (Fisher Matrix Structure). *[THM]*

*For Gaussian measurement noise with variance  $\sigma^2$  and  $N$  measurements uniformly distributed in  $\gamma \in [0, \gamma_{\text{max}}]$ , the diagonal Fisher information elements scale as:*

$$F_{\tau\tau} \propto \frac{N\beta^2 L^2}{\sigma^2 \tau^2} \quad (31)$$

$$F_{\beta\beta} \propto \frac{NL^2}{\sigma^2} \left\langle \left( \frac{\gamma}{\tau} \right)^{2\beta} \left( \ln \frac{\gamma}{\tau} \right)^2 \right\rangle \quad (32)$$

$$F_{LL} \propto \frac{N}{\sigma^2} \quad (33)$$

*Proof.* From the SDF model  $D(\gamma; \boldsymbol{\theta})$  with Gaussian likelihood:

$$\ln p = -\frac{1}{2\sigma^2} \sum_i [d_i - D(\gamma_i; \boldsymbol{\theta})]^2 + \text{const.}$$

The Fisher matrix elements are  $F_{ij} = \sigma^{-2} \sum_k (\partial D / \partial \theta_i) (\partial D / \partial \theta_j) |_{\gamma_k}$ .

<sup>1</sup> Throughout this work, I adopt the SDF convention  $\beta = 1 - \alpha/2$  (Axiom 3), which relates to the Lindblad-derived exponent  $\beta_{\text{Lindblad}} = 2 - \alpha$  via the identity  $\beta_{\text{SDF}} = \beta_{\text{Lindblad}}/2$  for pure dephasing channels.

Computing partial derivatives from Eq. (3):

$$\begin{aligned}\frac{\partial D}{\partial \tau} &= -\frac{L\beta}{\tau} \left(\frac{\gamma}{\tau}\right)^\beta e^{-(\gamma/\tau)^\beta}, \\ \frac{\partial D}{\partial \beta} &= L \left(\frac{\gamma}{\tau}\right)^\beta \ln\left(\frac{\gamma}{\tau}\right) e^{-(\gamma/\tau)^\beta}, \\ \frac{\partial D}{\partial L} &= 1 - e^{-(\gamma/\tau)^\beta}.\end{aligned}$$

Substituting into  $F_{ij}$  and replacing sums by integrals for large  $N$  yields the stated scalings.  $\square$

*Physical interpretation.* The Fisher metric on  $\mathcal{P}$  measures the *statistical distinguishability* of nearby decoherence processes. Regions of large  $F_{ij}$  correspond to parameter ranges where experiments are maximally informative; regions of small  $F_{ij}$  indicate parameter degeneracies. Key observations:

- $F_{\beta\beta}$  peaks when  $\gamma_{\max} \sim \tau$ : experiments must probe the transition region to constrain  $\beta$  effectively.
- $F_{\tau\tau} \propto \beta^2$ : the decoherence scale  $\tau$  is hardest to determine for small  $\beta$  (deeply stretched), where the transition is gradual.
- The off-diagonal element  $F_{\tau\beta}$  induces a  $\tau$ - $\beta$  correlation, visible as “banana-shaped” confidence contours in parameter estimation (Sec. V).

**Proposition 2** (Geodesics in Parameter Space). *In the limit  $\gamma_{\max} \gg \tau$  and for fixed  $L$ , the Fisher metric on the  $(\ln \tau, \beta)$  submanifold is approximately conformally flat:*

$$ds^2 \approx \frac{NL^2}{\sigma^2} [\beta^2 (d \ln \tau)^2 + \kappa(\beta) d\beta^2] \quad (34)$$

where  $\kappa(\beta)$  is a bounded, positive function of  $\beta$ . Geodesics in this metric describe the most efficient experimental paths for resolving parameter changes.

### G. Information-Theoretic Structure

**Theorem 11** (Information Conservation). *[THM]*

Define the total information content:

$$I_{\text{total}}(\gamma) \equiv D(\gamma) + C(\gamma) = D(\gamma) + [1 - D(\gamma)] = 1 \quad (35)$$

Then:

$$\frac{dI_{\text{total}}}{d\gamma} = 0 \quad \forall \gamma \geq 0 \quad (36)$$

*Proof.*  $I_{\text{total}} = 1$  identically by definition, hence  $dI_{\text{total}}/d\gamma = 0$ .  $\square$

*Remark.* While Theorem 11 is algebraically trivial, its physical content is profound: decoherence does not destroy information but *redistributes* it between system-accessible coherence  $C(\gamma)$  and environment-accessible decoherence  $D(\gamma)$ . This is the SDF restatement of unitarity.

**Theorem 12** (Entropy–Decoherence Monotonicity). *[THM]*

For a system decohering from a pure initial state, the von Neumann entropy  $S(\rho(\gamma)) = -\text{Tr}[\rho \ln \rho]$  satisfies:

$$\frac{dS}{d\gamma} \geq 0 \quad \forall \gamma \geq 0 \quad (37)$$

with equality only at  $\gamma = 0$  (if  $\beta < 1$ ) or in the  $\gamma \rightarrow \infty$  limit. Moreover, in the qubit case:

$$S(\gamma) = H_{\text{bin}}\left(\frac{1 + \sqrt{1 - D(\gamma)^2/L^2}}{2}\right) \quad (38)$$

where  $H_{\text{bin}}(p) = -p \ln p - (1-p) \ln(1-p)$  is the binary entropy.

*Proof.* Entropy increase follows from the monotonicity of  $D(\gamma)$  (Theorem 2(iii)) and the monotonic relationship between decoherence and mixedness for initially pure states. The qubit expression follows from the eigenvalues of the qubit density matrix  $\rho = \frac{1}{2}(\mathbb{I} + \vec{r}' \cdot \vec{\sigma})$  with  $|\vec{r}'|^2 = 1 - D^2/L^2$  in the dephasing channel.  $\square$

### H. Summary of Mathematical Results

Table I collects the principal mathematical results of this section. All statements marked [THM] are rigorously proven; their logical dependence flows from Axioms 1–3 through the existence theorem (Theorem 1) to derived properties, deviation bounds, and information-theoretic consequences.

TABLE I. Summary of mathematical results. All theorems are proven from Axioms 1–3.

| Result  | Content                  | Eq.  |
|---------|--------------------------|------|
| Thm. 1  | Existence & uniqueness   | (3)  |
| Thm. 2  | Basic properties         | —    |
| Thm. 3  | Inverse function         | (6)  |
| Thm. 4  | Scale invariance         | (7)  |
| Thm. 5  | Deviation properties     | (10) |
| Thm. 6  | Deviation bound          | (11) |
| Thm. 7  | Rate trichotomy          | (12) |
| Thm. 8  | Asymptotics              | (15) |
| Thm. 9  | Lindblad derivation      | (22) |
| Thm. 11 | Information conservation | (35) |
| Thm. 12 | Entropy monotonicity     | (37) |
| Thm. 10 | Fisher information       | (29) |

### III. Information Geometry and the Entropic Phase Space

The Stretching/Decoherence Formalism, as developed in the preceding sections, parametrizes the quantum-to-classical transition through the decoherence function  $D(\gamma)$  and its complement  $C(\gamma)$ . In this section I elevate these quantities from computational tools to geometric objects, embedding the SDF dynamics into the natural Riemannian structure of quantum state space. This yields a

new formalism—the *entropic phase space*—in which the quantum-classical boundary acquires a precise geometric definition as a signature-change hypersurface.

### A. The Space of Quantum States as a Riemannian Manifold

For a  $d$ -dimensional quantum system, the space of density matrices

$$\mathcal{M}_d = \{\rho \in \mathbb{C}^{d \times d} : \rho \geq 0, \text{Tr}(\rho) = 1\} \quad (39)$$

is a  $(d^2 - 1)$ -dimensional smooth manifold with boundary. The interior  $\mathcal{M}_d^\circ$  consists of full-rank (mixed) states; the boundary  $\partial\mathcal{M}_d$  comprises the singular states, including all pure states.

The *Bures metric* [4, 5] endows  $\mathcal{M}_d$  with a monotone Riemannian structure:

$$ds_B^2 = \frac{1}{2} \sum_{m,n} \frac{|d\rho_{mn}|^2}{\lambda_m + \lambda_n}, \quad (40)$$

where  $\lambda_m$  are the eigenvalues of  $\rho$  and  $d\rho_{mn} = \langle m|d\rho|n \rangle$  in the eigenbasis. This metric is distinguished by its operational meaning: the Bures distance  $d_B(\rho, \sigma) = \sqrt{2(1 - \sqrt{F(\rho, \sigma)})}$  is a monotone function of the Uhlmann fidelity  $F$ .

For a qubit ( $d = 2$ ), the state space is the Bloch ball  $\mathcal{M}_2 \cong \overline{B^3}$ . In Bloch coordinates  $\vec{r} = (r \sin \theta \cos \varphi, r \sin \theta \sin \varphi, r \cos \theta)$ , the Bures metric takes the form

$$ds_B^2 = \frac{dr^2 + r^2(d\theta^2 + \sin^2 \theta d\varphi^2)}{4(1 - r^2)} \quad (41)$$

which diverges at the boundary  $r = 1$  (pure states), reflecting the infinite distinguishability cost near a pure-state surface.

**Definition 11** (Decoherence Trajectory). *[DEF] The decoherence trajectory is the curve  $\rho(\gamma) \subset \mathcal{M}_d$  traced as the decoherence parameter  $\gamma$  increases from 0 to  $\infty$ , with boundary conditions*

$$\rho(0) = |\psi\rangle\langle\psi| \in \partial\mathcal{M}_d, \quad \rho(\infty) = \frac{\mathbb{I}}{d} \in \mathcal{M}_d^\circ. \quad (42)$$

For a qubit undergoing pure dephasing,  $r(\gamma) = 1 - D(\gamma)$  with  $\theta, \varphi$  constant.

**Theorem 13** (SDF Geodesic Structure). *[THM] The decoherence trajectory  $\rho(\gamma)$  is a geodesic in  $(\mathcal{M}_d, g_B)$  if and only if  $\beta = 1$  (Markovian limit). For  $\beta \neq 1$ , the trajectory deviates from the geodesic with curvature*

$$\kappa_g(\gamma) = \left| \frac{D^2 \rho}{ds^2} \right| \propto |\beta - 1| \cdot \left( \frac{\gamma}{\tau} \right)^{\beta-2} \quad (43)$$

*Proof sketch.* The geodesic equation on  $(\mathcal{M}_d, g_B)$  with the Bures connection  $\nabla^B$  requires  $\nabla_{\dot{\rho}}^B \dot{\rho} = 0$ . In the eigenbasis, this reduces to exponential decay of off-diagonal

elements:  $\rho_{mn}(s) \propto e^{-\alpha s}$  for constant  $\alpha$ . The SDF trajectory yields  $\rho_{mn}(\gamma) \propto \exp[-L(\gamma/\tau)^\beta]$ , whose decay rate  $d \ln |\rho_{mn}| / d\gamma \propto \gamma^{\beta-1}$  is constant only when  $\beta = 1$ . The geodesic curvature is  $\kappa_g = |D^2 \rho / ds^2|$ , and direct computation using Eq. (41) gives the scaling of Eq. (43).  $\square$

**Physical interpretation.** SIMULATION decoherence ( $\beta \approx 1$ ) produces *geodesic* trajectories—the shortest path from quantum to classical in the state manifold. HARDWARE decoherence ( $\beta < 1$ ) produces *non-geodesic* trajectories; the curvature  $\kappa_g$  quantifies how severely the environment’s memory distorts the natural decoherence path. This connects to the deviation operator  $\Delta(\gamma; \alpha)$  of Sec. II B: the non-zero geodesic curvature is the geometric shadow of  $\Delta(\gamma; \alpha) \neq 0$ .

### B. The Information-Decoherence Functional

The Bures metric on  $\mathcal{M}_d$  induces a natural arc-length functional along the decoherence trajectory:

**Definition 12** (Bures Arc Length). *[DEF] The information-decoherence functional is*

$$s(\gamma) = \int_0^\gamma \sqrt{g_{\mu\nu} \frac{d\rho^\mu}{d\gamma'} \frac{d\rho^\nu}{d\gamma'}} d\gamma'. \quad (44)$$

For a qubit with initial state  $|0\rangle$  undergoing pure dephasing,  $r(\gamma) = C(\gamma) = 1 - D(\gamma)$  and  $\theta, \varphi$  are constant. Substituting into Eq. (41):

$$s(\gamma) = \arccos(C(\gamma)) = \arccos(1 - D(\gamma)). \quad (45)$$

The arc length maps the decoherence parameter to an angle on the Bloch sphere measured in the Bures geometry.

**Definition 13** (Information Velocity). *[DEF] The information velocity is the rate of Bures arc-length traversal with respect to  $\gamma$ :*

$$v_I(\gamma) \equiv \frac{ds}{d\gamma} = \frac{R(\gamma)}{\sqrt{D(\gamma)(2 - D(\gamma))}} \quad (46)$$

where  $R(\gamma) = dD/d\gamma$  is the SDF transition rate.

**Theorem 14** (Velocity Singularity Structure). *[THM] The initial information velocity  $v_I(0^+)$  exhibits a trichotomy controlled by the stretching exponent:*

$$v_I(0^+) = \begin{cases} \infty & \beta < 1 \text{ (divergent)}, \\ \frac{L}{\tau\sqrt{2L}} & \beta = 1 \text{ (Markovian)}, \\ 0 & \beta > 1 \text{ (quiescent)}. \end{cases} \quad (47)$$

*Proof.* From the SDF transition rate

$$R(\gamma) = \frac{L\beta}{\tau} \left( \frac{\gamma}{\tau} \right)^{\beta-1} \exp[-L(\gamma/\tau)^\beta]$$

and the small- $\gamma$  expansion  $D(\gamma) \approx L(\gamma/\tau)^\beta$ , one obtains  $v_I \sim (\gamma/\tau)^{\beta/2-1}$  as  $\gamma \rightarrow 0^+$ . The exponent  $\beta/2 - 1$  is

negative for  $\beta < 1$ , zero for  $\beta = 1$  (yielding the finite value  $L/(\tau\sqrt{2L})$  after careful evaluation of constants), and positive for  $\beta > 1$ .  $\square$

**Interpretation.** Stretched-exponential decoherence ( $\beta < 1$ , characteristic of **HARDWARE**) begins with *infinite* information velocity: the system instantaneously initiates information leakage to the environment, but at a rapidly decelerating rate. This burst is the hallmark of non-Markovian dynamics—early-time memory correlations create a transient of maximal information transfer before the environment’s back-action slows the flow. The contrast with the quiescent start at  $\beta > 1$  underlies the fundamental **HARDWARE**–**SIMULATION** asymmetry developed in Sec. [IIB](#).

### C. The Entropic Phase Space

I now introduce the central geometric construction of this work: the *SDF entropic phase space*, which extends the standard spatial configuration space with entropic and decoherence coordinates.

**Definition 14** (Extended SDF Coordinates). *[DEF]* The SDF configuration point is described by the extended coordinate vector

$$X^A = (\gamma, \sigma, \ell_1, \dots, \ell_d), \quad A = 0, 1, \dots, d+1, \quad (48)$$

where:

- $\gamma \in [0, \infty)$ : the decoherence parameter (temporal coordinate),
- $\sigma = S(\rho(\gamma)) = -\text{Tr}[\rho(\gamma) \ln \rho(\gamma)]$ : the von Neumann entropy (entropic coordinate),
- $\ell_i = \ell_P \cdot \Lambda^{-\nu_i}$ : spatial scales derived from the Planck length  $\ell_P$  via the hierarchy parameter  $\Lambda$ , for  $i = 1, \dots, d$ .

**Definition 15** (SDF Phase Metric). *[DEF]* The natural metric on the SDF phase space  $\mathcal{P}$  is

$$g_{AB} dX^A dX^B = -c_\gamma^2 d\gamma^2 + \sum_{i=1}^d \left( \frac{\ell_P}{\Lambda^{\nu_i}} \right)^2 d\ell_i^2 - c_\sigma^2 d\sigma^2 \quad (49)$$

where  $c_\gamma$  and  $c_\sigma$  are scale factors with dimensions  $[\text{length} \cdot \text{decoherence}^{-1}]$  and  $[\text{length} \cdot \text{entropy}^{-1}]$ , respectively.

The metric (49) has several remarkable features:

1. **Signature (2, d):** The metric possesses *two* time-like dimensions ( $\gamma$  and  $\sigma$ ) and  $d$  spacelike dimensions, giving signature  $(-, +, \dots, +, -)$ . This is fundamentally distinct from pseudo-Riemannian spacetimes of signature  $(1, d)$ .
2. **Negative temporal dimension:** The decoherence parameter  $\gamma$  enters with negative sign because decoherence proceeds irreversibly forward, defining a causal structure analogous to—but distinct from—the time coordinate in general relativity.

3. **Negative entropic dimension:** The entropy  $\sigma$  is *also* timelike because it increases monotonically (for  $\beta \leq 1$ ), defining a second “arrow” in the phase space. The two timelike dimensions are *not* independent:  $\sigma = \sigma(\gamma)$  constrains motion to a codimension-one submanifold.

4. **Planck-scalable spatial dimensions:** The spatial components  $\ell_i$  scale with the hierarchy parameter  $\Lambda$  from the Planck length  $\ell_P$  upward, connecting the microphysical decoherence dynamics to macroscopic spatial structure.

**Theorem 15** (Entropic Signature Change). *[THM]* For  $\beta < 1$ , there exist intervals  $[\gamma_1, \gamma_2] \subset (0, \infty)$  where the entropy acceleration is negative:

$$\frac{d^2\sigma}{d\gamma^2} < 0. \quad (50)$$

In these intervals, the effective entropic metric component becomes locally spacelike:

$$g_{\sigma\sigma}^{\text{eff}}(\gamma) = -c_\sigma^2 + \kappa(\gamma), \quad \kappa(\gamma) > 0 \text{ when } \frac{d^2\sigma}{d\gamma^2} < 0 \quad (51)$$

The signature of  $\mathcal{P}$  therefore changes dynamically along the decoherence trajectory, transitioning from  $(2, d)$  to  $(1, d+1)$  in the backflow regions.

*Proof.* For a qubit,  $\sigma(\gamma) = H_2(D(\gamma))$  where  $H_2(p) = -p \ln p - (1-p) \ln(1-p)$  is the binary entropy. Computing:

$$\frac{d\sigma}{d\gamma} = H_2'(D) \cdot R(\gamma), \quad (52)$$

$$\frac{d^2\sigma}{d\gamma^2} = H_2''(D) \cdot R(\gamma)^2 + H_2'(D) \cdot R'(\gamma). \quad (53)$$

Since  $H_2'' < 0$  for all  $D \in (0, 1)$ , the first term is always negative. For  $\beta < 1$ , the transition rate  $R(\gamma)$  is *decreasing* at large  $\gamma$ , so  $R'(\gamma) < 0$  and the second term is also negative (since  $H_2' > 0$  for  $D < 1/2$ ). In the intermediate regime where  $D$  crosses  $1/2$ ,  $H_2'$  changes sign and the competition between the two terms creates intervals where  $d^2\sigma/d\gamma^2 < 0$ . These are precisely the backflow intervals of Theorem [XII C](#).

The effective metric component  $g_{\sigma\sigma}^{\text{eff}}$  incorporates the constraint  $\sigma = \sigma(\gamma)$  through the induced metric on the trajectory. When  $d^2\sigma/d\gamma^2 < 0$ , the entropic “velocity” is decelerating, and the curvature contribution  $\kappa(\gamma) > 0$  partially cancels the bare  $-c_\sigma^2$ , potentially flipping the sign.  $\square$

**Physical significance.** The signature change is the *mathematical manifestation* of information backflow (cf. Theorem [XII C](#) in Sec. [IIB](#)). When  $\beta < 1$ , the subsystem entropy can locally decelerate, creating regions where the

entropic “dimension” temporarily behaves as a spatial—rather than temporal—direction. Information that was irreversibly lost in the  $(2, d)$ -signature regime becomes recoverable in the  $(1, d+1)$  regime. The boundary between these regimes defines the quantum-classical transition (Sec. III D).

#### D. The Quantum-Classical Transition as a Geometric Flow

The entropic phase space  $\mathcal{P}$  admits a natural partition into quantum and classical domains, separated by a geometric transition boundary.

**Definition 16** (Quantum Domain  $\mathcal{Q}$ ). *[DEF] The quantum domain is the region of  $\mathcal{P}$  where  $D(\gamma) < D_{\text{th}}$  and the entropic metric maintains its negative (timelike) signature throughout:*

$$\mathcal{Q} = \{X \in \mathcal{P} : D(\gamma) < D_{\text{th}}, g_{\sigma\sigma}^{\text{eff}}(\gamma) < 0\}. \quad (54)$$

**Definition 17** (Classical Domain  $\mathcal{C}$ ). *[DEF] The classical domain is the region where  $D(\gamma) > D_{\text{th}}$  and the entropy is monotonically increasing with positive second derivative (irreversible classical behavior):*

$$\mathcal{C} = \{X \in \mathcal{P} : D(\gamma) > D_{\text{th}}, d^2\sigma/d\gamma^2 > 0\}. \quad (55)$$

**Definition 18** (Transition Boundary  $\partial\mathcal{T}$ ). *[DEF] The transition boundary is the hypersurface in  $\mathcal{P}$  where the entropic signature changes:*

$$\partial\mathcal{T} = \{X \in \mathcal{P} : d^2\sigma/d\gamma^2 = 0\}. \quad (56)$$

**Theorem 16** (Transition Existence). *[THM] For every SDF system with  $\beta < 1$ , the transition boundary  $\partial\mathcal{T}$  is non-empty. The quantum-classical transition occurs at*

$$\gamma_{\text{QC}} = \tau \cdot \left[ \frac{(\beta - 1)}{\beta} \cdot \frac{H_2'(D_{\text{QC}})}{H_2''(D_{\text{QC}}) \cdot R(\gamma_{\text{QC}})} \right]^{1/\beta} \quad (57)$$

where  $D_{\text{QC}} = D(\gamma_{\text{QC}})$  is the decoherence value at the transition.

*Proof.* Setting  $d^2\sigma/d\gamma^2 = 0$  in Eq. (53) and solving for  $\gamma$  using the explicit SDF forms  $R(\gamma) = (L\beta/\tau)(\gamma/\tau)^{\beta-1} \exp[-L(\gamma/\tau)^\beta]$  and  $R'(\gamma) = R(\gamma)[(\beta - 1)/\gamma - L\beta(\gamma/\tau)^{\beta-1}/\tau]$  yields Eq. (57) after algebraic reduction. The existence of a solution for  $\beta < 1$  follows from the intermediate value theorem applied to  $d^2\sigma/d\gamma^2$ , which is positive near  $\gamma = 0$  (dominated by  $H_2'' \cdot R^2 > 0$  at small  $D$ ) and negative at large  $\gamma$  (as established in Theorem 15).  $\square$

This provides a *geometric* definition of the quantum-classical boundary: it is the hypersurface  $\partial\mathcal{T}$  where the entropic metric changes signature. On the quantum side ( $\gamma < \gamma_{\text{QC}}$ ), information can flow back (negative curvature, recoverable coherence). On the classical side ( $\gamma > \gamma_{\text{QC}}$ ), information irreversibly disperses (positive curvature, monotonic entropy increase).

**Proposition 3** (Critical Qubit Number from Geometry). *The critical qubit number  $n^* \approx 2.6$  derived from the empirical scaling law  $\beta(n) = \beta_0/n^\xi$  (Sec. X) corresponds to the geometric condition  $\beta(n^*) = 1$ , at which the entropic signature change of Theorem 15 vanishes. Systems with  $n < n^*$  admit signature-changing trajectories (quantum behavior); systems with  $n > n^*$  have fixed-signature trajectories (fully classical).*

#### E. The $\Lambda$ -Hierarchy as Scale Fibration

The hierarchy parameter  $\Lambda$  organizes the SDF phase space into a fibered structure connecting Planck-scale physics to macroscopic decoherence.

**Definition 19** (Physical Scale Fibration). *[DEF] The scale fibration is the projection  $\pi : \mathcal{P} \rightarrow \mathbb{R}^+$  with fiber  $\mathcal{F}_\Lambda$  at each scale  $\Lambda$ . The hierarchy parameter*

$$\Lambda = \frac{E_{\text{binding}}}{E_P} \quad (58)$$

maps each physical system to a point on the base space  $\mathbb{R}^+$ .

**Derivation of  $\tau(\Lambda)$ .** (Promoting from axiom to derivation.) From the energy-time uncertainty principle,

$$\Delta E \cdot \Delta t \geq \frac{\hbar}{2}. \quad (59)$$

For a system with binding energy  $E_b = \Lambda \cdot E_P$ , the minimum decoherence timescale is

$$\tau_{\text{min}} = \frac{\hbar}{2E_b} = \frac{\hbar}{2\Lambda E_P}. \quad (60)$$

At Planck scale ( $\Lambda = 1$ ),  $\tau_P = \hbar/(2E_P)$ . Therefore

$$\tau(\Lambda) = \frac{\tau_P}{\Lambda} \quad (61)$$

after cancellation of the common factor of 2. This is not a postulate—it is a *consequence* of energy-time uncertainty in the regime where  $E_{\text{binding}}$  sets the relevant energy scale.

The corresponding spatial scale follows from the Compton wavelength at binding energy:

$$\ell(\Lambda) = \frac{\hbar c}{E_b} = \frac{\ell_P}{\Lambda}. \quad (62)$$

Together, these yield the *Planck-scale invariant*:

$$\tau(\Lambda) \cdot \ell(\Lambda) = \frac{\tau_P \cdot \ell_P}{\Lambda^2} \quad (63)$$

a relation that connects the decoherence timescale and spatial extent at every level of the hierarchy.



TABLE II. The  $\Lambda$ -hierarchy: decoherence timescales and spatial scales from the Planck regime to biological systems. All values derived from Eqs. (61)–(62).

| Scale      | $E_b$                                 | $\Lambda$       | $\tau(\Lambda)$                         | $\ell(\Lambda)$                         |
|------------|---------------------------------------|-----------------|-----------------------------------------|-----------------------------------------|
| Planck     | $E_P \approx 1.22 \times 10^{19}$ GeV | 1               | $\tau_P \approx 5.39 \times 10^{-44}$ s | $\ell_P \approx 1.62 \times 10^{-35}$ m |
| Nuclear    | $\sim 1$ GeV                          | $\sim 10^{-19}$ | $\sim 10^{-25}$ s                       | $\sim 10^{-16}$ m                       |
| Atomic     | $\sim 10$ eV                          | $\sim 10^{-27}$ | $\sim 10^{-17}$ s                       | $\sim 10^{-8}$ m                        |
| Molecular  | $\sim 0.1$ eV                         | $\sim 10^{-29}$ | $\sim 10^{-15}$ s                       | $\sim 10^{-6}$ m                        |
| Biological | $\sim 10^{-2}$ eV                     | $\sim 10^{-30}$ | $\sim 10^{-14}$ s                       | $\sim 10^{-5}$ m                        |

### F. Information Conservation in the Phase Space

The SDF parametrization acquires its deepest physical content when reformulated as an information-conservation law rooted in unitarity.

**Theorem 17** (Unitary Information Conservation). *[THM] For a bipartite system  $S + E$  evolving under a global unitary  $U(t)$ , the total von Neumann entropy is conserved:*

$$S(\rho_{SE}) = \text{const.} \quad (64)$$

*The apparent entropy increase in  $S$  (decoherence) is exactly compensated by the buildup of mutual information:*

$$I(S:E) = S(\rho_S) + S(\rho_E) - S(\rho_{SE}). \quad (65)$$

In the SDF parametrization, the subsystem and environment entropies become functions of  $\gamma$ :

$$\sigma_S(\gamma) = S(\rho_S(\gamma)), \quad \sigma_E(\gamma) = S(\rho_E(\gamma)), \quad (66)$$

and the conservation law (65) takes the non-trivial form

$$I_{\text{total}}(\gamma) = \sigma_S(\gamma) + \sigma_E(\gamma) - \sigma_{SE} = \text{const.} \quad (67)$$

This reformulation elevates the identity  $D + C = 1$  from a tautological decomposition (as noted in Sec. II A) to a statement with genuine physical content: information is not destroyed by decoherence but *redistributed* between system and environment. The decoherence function  $D(\gamma)$  tracks the fraction of information that has migrated from  $S$  to the  $S$ – $E$  correlations, while  $C(\gamma) = 1 - D(\gamma)$  tracks the fraction remaining locally accessible in  $S$ .

**Theorem 18** (Information Backflow Bound). *[THM] The maximum amount of information that can flow back from the environment to the system during a single backflow episode  $[\gamma_1, \gamma_2]$  is bounded by*

$$\delta I_{\text{back}} \leq (1 - \beta) \cdot L \cdot e^{-1} \quad (68)$$

where  $L$  is the SDF amplitude parameter and  $e^{-1}$  is the inverse Euler number.

*Proof.* During a backflow episode, the decoherence function satisfies  $D'(\gamma) < 0$  (the system re-coheres). The maximum decrease in  $D$  is bounded by the maximum of  $|R(\gamma)|$  over the backflow interval. From the SDF rate function  $R(\gamma) = (L\beta/\tau)(\gamma/\tau)^{\beta-1} \exp[-L(\gamma/\tau)^\beta]$ , the global

maximum of  $|dD/d\gamma| \cdot \Delta\gamma$  over any interval where the rate is negative (requiring  $\beta < 1$  and reversal of the effective dynamics) is achieved at  $\gamma^* = \tau(1/L)^{1/\beta}$ , giving  $\delta I_{\text{back}}^{\text{max}} = (1 - \beta) \cdot L \cdot e^{-L(1/L)} = (1 - \beta)L e^{-1}$ .  $\square$

The backflow bound (68) connects directly to the deviation operator: the maximum information backflow equals the maximum deviation  $\|\Delta(\gamma; \alpha)\|_{\text{max}}$  (Sec. IIB). For  $\beta = 1$  (Markovian), the bound vanishes—no backflow is possible. For  $\beta \rightarrow 0$  (maximally non-Markovian),  $\delta I_{\text{back}} \rightarrow L e^{-1}$ , saturating at the SDF amplitude.

The information-geometric framework developed in this section will be applied to experimental data in Sec. VI, where the geodesic curvature  $\kappa_g$  and information velocity  $v_I$  are extracted from IBM Quantum hardware measurements, and to the cosmological implications of the entropic phase space in Sec. XXII.

### IV. Exact Identities

Two relations that the previous papers state as a postulate and as a derived expression respectively turn out to be identities. Both strengthen the framework, and both were found by requiring the implementation to reproduce the published numbers rather than to agree with the published derivations.

#### A. The $\Lambda$ -hierarchy is the Compton scale

Paper 1 §8 postulates that the decoherence timescale is set by binding information through  $\tau(\Lambda) = \tau_P/\Lambda$  with  $\Lambda = E_{\text{binding}}/E_P$ , and Paper 2 §9 lists this relation among the theoretical limitations of the framework, describing it as *irreducible*—logically equivalent to the uncertainty relation, but postulated rather than derived.

It is neither irreducible nor a postulate.

**Theorem 19** (Planck action and length identities). *[EX-ACT] The Planck time, length and energy satisfy*

$$\tau_P E_P = \hbar, \quad \ell_P E_P = \hbar c, \quad (69)$$

*identically.*

*Proof.* By definition  $\tau_P = \sqrt{\hbar G/c^5}$ ,  $\ell_P = \sqrt{\hbar G/c^3}$  and  $E_P = \sqrt{\hbar c^5/G}$ . Then  $\tau_P E_P = \sqrt{(\hbar G/c^5)(\hbar c^5/G)} = \sqrt{\hbar^2} = \hbar$  and  $\ell_P E_P = \sqrt{(\hbar G/c^3)(\hbar c^5/G)} = \sqrt{\hbar^2 c^2} = \hbar c$ . Both hold to a relative residual of 0 in double precision.  $\square$

**Corollary 2** (Hierarchy in closed form). *[EXACT] For a system of characteristic energy  $E$ , with  $\Lambda = E/E_P$ ,*

$$\tau(\Lambda) = \frac{\tau_P}{\Lambda} = \frac{\hbar}{E}, \quad \ell(\Lambda) = \frac{\ell_P}{\Lambda} = \frac{\hbar c}{E} = c \tau(\Lambda). \quad (70)$$

*In particular, at  $E = mc^2$ ,*

$$\ell(\Lambda) = \frac{\hbar}{mc} = \bar{\lambda}_C, \quad (71)$$

*the reduced Compton wavelength.*

Equation (71) was verified for the electron to a relative residual of 0 at machine precision, and Eqs. (70) over  $E \in \{10^{-3}, 1, 10^6, 10^{12}, 10^{20}\}$  eV with residuals below  $10^{-6}$ , the residual being set by the seven-digit rounding of the tabulated Planck constants rather than by the identities.

*a. Consequences.* Three, in increasing order of importance.

First, the spatial scale that the hierarchy assigns to a physical system is not a free parameter to be fitted: it is the system's reduced Compton wavelength. For a produced particle–antiparticle pair the geometry does not need to be postulated; it follows from the pair energy.

Second, the hierarchy is exactly the energy–time uncertainty relation written in Planck units, which is what Paper 1 §2b attempted to show. That derivation should be replaced by Theorem 19, which is shorter and exact.

Third, the derivation in Paper 1 §2b contains a spurious factor of two: it writes  $\tau_{\min} = \hbar/(2E_b)$  and asserts  $\tau_P = \hbar/(2E_P)$ . The latter is false. Numerically  $\hbar/(2E_P) = 2.696 \times 10^{-44}$  s against  $\tau_P = 5.391 \times 10^{-44}$  s, a ratio of exactly 2.000000. The published conclusion survives because the factor cancels in the ratio  $\tau_{\min}/\tau_P = E_P/E_b = 1/\Lambda$ ; the intermediate statement does not.

## B. Accessible information is the reduced argument

Paper 1 §9 defines the accessible quantum information as  $I_Q(\gamma) = -\ln(1 - D(\gamma))$ . For  $L = 1$  this collapses.

**Theorem 20** (Information identity). *[EXACT] For  $L = 1$ ,*

$$I_Q(\gamma) = -\ln C(\gamma) = \left(\frac{\gamma}{\tau}\right)^\beta, \quad (72)$$

and consequently

$$\frac{dI_Q}{d\gamma} = \frac{\beta}{\tau} \left(\frac{\gamma}{\tau}\right)^{\beta-1}, \quad \frac{d^2 I_Q}{d\gamma^2} = \frac{\beta(\beta-1)}{\tau^2} \left(\frac{\gamma}{\tau}\right)^{\beta-2}. \quad (73)$$

*Proof.* With  $L = 1$ , Eq. (3) gives  $C(\gamma) = 1 - D(\gamma) = \exp[-(\gamma/\tau)^\beta]$ , hence  $-\ln C(\gamma) = (\gamma/\tau)^\beta$ . Differentiation is immediate.  $\square$

The accessible information of §9 is therefore nothing other than the reduced argument of the SDF itself. Verified to a maximum absolute residual of  $1.11 \times 10^{-16}$  and relative residual  $3.04 \times 10^{-14}$  across  $\beta \in \{0.3, 0.5, 0.852, 1, 1.5, 2\}$  and  $\gamma/\tau \in [0.01, 30]$ , evaluated along the numerically stable branch.

*a. A caveat on normalisation.* Eq. (72) is stated for  $L = 1$ , and the restriction is not cosmetic. The identity that holds for arbitrary  $L$  is  $(\gamma/\tau)^\beta = -\ln(1 - D/L)$ , on the reduced decoherence  $D/L$ . The published  $I_Q = -\ln(1 - D)$  uses the unnormalised  $D$  and therefore departs from the reduced argument whenever  $L \neq 1$ : at the threshold of Theorem 21 with  $\beta = 0.852$  it returns 0.4700 for  $L = 0.75$  and 0.2877 for  $L = 0.5$ , against the exact  $\ln 2 = 0.6931$ . The two forms must be kept distinct.

## C. The binary threshold

The identity of Theorem 20 has a distinguished point that the corpus reaches by an unrelated route, and it is worth recording that the two coincide.

**Theorem 21** (One-bit threshold). *[EXACT] For any  $\beta > 0$ ,  $\tau > 0$  and  $L > 0$ ,*

$$I_Q = \ln 2 \iff \frac{D}{L} = \frac{1}{2} \iff \gamma = \tau (\ln 2)^{1/\beta}. \quad (74)$$

*The right-hand side is exactly  $\gamma_{\text{ESD}}$ , the entanglement sudden-death point of Paper 2 for initial concurrence  $C_0 = 1$ .*

*Proof.*  $D/L = 1 - e^{-x}$  with  $x = (\gamma/\tau)^\beta$ , so  $D/L = 1/2 \iff x = \ln 2$ . Paper 2 gives  $\gamma_{\text{ESD}} = \tau[-\ln(1 - C_0/(1 + C_0))]^{1/\beta}$ , which for  $C_0 = 1$  is  $\tau(\ln 2)^{1/\beta}$ .  $\square$

Two features deserve emphasis. First,  $\ln 2$  nats is exactly one bit: the entanglement dies when the channel has accumulated precisely one bit of information. Second, the threshold is invariant in  $\beta$ ,  $\tau$  and  $L$  when expressed in  $I_Q$ , and moves with  $\beta$  when expressed in  $\gamma$ . It is an *informational* invariant, not a temporal one. Verified to residual 0 (exact in double precision) across  $\beta \in \{0.2, 0.5, 0.852, 1, 1.679\}$ ,  $\tau \in \{1, 3.7 \times 10^{-15}, 2.5 \times 10^9\}$  and  $L \in \{1, 0.5, 0.87\}$ .

A third coincidence holds at the same point but is convention-dependent, and is therefore deferred to Sec. XV: under the reading in which  $D$  is a population, the qubit entropy  $H_2(D)$  attains its maximum there, and  $I_Q = S$  exactly—the unique interior solution of  $-\ln(1 - D) = H_2(D)$ .

A fourth does *not* depend on the convention, and is worth separating from the third for exactly that reason.

**Corollary 3** (Pair boundary). *[EXACT] Let a channel and its CPT conjugate share  $(\tau, \beta, L)$ . Then the decoherence of one equals the coherence of the other,  $D = \bar{C}$ , precisely at the threshold of Theorem 21, and there the product  $D\bar{C} = D(L - D)$  attains its maximum, of value exactly  $L^2/4$ .*

*Proof.*  $D = \bar{C} = L - \bar{D}$  and  $\bar{D} = D$  give  $2D = L$ . The product  $D(L - D)$  is a downward parabola in  $D$  with vertex at  $D = L/2$ , where its value is  $L^2/4$ .  $\square$

The threshold also has an exact geometric position, and it too is insensitive to the convention.

**Corollary 4** (Geodesic position of the threshold). *[EXACT] Along the Bures geodesic, with arc length  $s = \arccos(1 - D/L)$ ,*

$$s\left(\frac{D}{L} = \frac{1}{2}\right) = \frac{\pi}{3}, \quad s\left(\frac{D}{L} = 1\right) = \frac{\pi}{2}, \quad \frac{s_{\text{bit}}}{s_{\text{total}}} = \frac{2}{3}. \quad (75)$$

One accumulated bit is exactly  $60^\circ$  of geodesic arc, and exactly two thirds of the whole path from pure state to complete decoherence. Verified to residual  $2.2 \times 10^{-16}$  across  $\beta \in \{0.2, 0.5, 0.852, 1, 1.679\}$  and  $L \in \{1, 0.5, 0.87\}$ .

Because neither  $D\bar{C}$  nor  $\arccos(1 - D/L)$  involves an entropy, Corollaries 3 and 4 are insensitive to the choice made in Sec. XVD, whereas the maximum of  $H_2(D)$  is not.

*a. A third argument for the convention.* The Bures arc is monotone in  $D$ ; so is  $H_2(D/2L)$ ;  $H_2(D)$  is not. A quantity can serve as a distance coordinate along a trajectory only if it is injective there. Under the infidelity reading the entropy is therefore a monotone reparametrisation of a genuine geodesic distance and may be used as a distance coordinate; under the population reading it cannot, since two distinct states would carry the same value. This is independent of the two grounds given in Sec. XVD.

It does *not* make the entropy a metric:  $S(\rho)$  is a function of one state, not two, and the natural two-argument candidate—the relative entropy  $S(\rho||\sigma)$ —is neither symmetric nor subadditive under composition, so it is a divergence rather than a distance. The metrics available remain the Bures arc and  $d_B = \sqrt{2(1 - \sqrt{F})}$ . Verified to residual 0 across  $\beta \in \{0.2, 0.5, 0.852, 1, 1.679\}$  and  $L \in \{1, 0.5, 0.87\}$ , with the maximum located numerically to within  $1.6 \times 10^{-5}$  relative (grid resolution) and its value reproduced as 0.25 to twelve digits.

Two cautions. The threshold coincides with  $\gamma_{\text{ESD}}$  only for  $C_0 = 1$ , that is for maximal initial entanglement; other initial concurrences accumulate different amounts, 0.585 bits for  $C_0 = 1/2$  and 1.585 for  $C_0 = 2$ . And the corollary is *not* evidence of an interaction between the two channels: they are computed independently and the framework contains no coupling term. Indeed Paper 1 §12 proves  $D\bar{D} > 0$  and  $D + \bar{D} \neq 0$  for all  $\gamma > 0$ , which forbids the reading in which matter and antimatter cancel:  $D + \bar{D}$  is monotonically *increasing* toward  $2L$ , not decreasing toward zero. The compensation the corpus can support is  $D + \bar{C} = L$ , not  $D + \bar{D} = 0$ .

*b. Consequence: a published criterion is invalidated.* The sign of  $d^2 I_Q / d\gamma^2$  in Eq. (73) is fixed by  $\beta - 1$  alone. For  $\beta < 1$  the curvature is *strictly negative* everywhere:  $I_Q$  is concave in the stretched regime. This contradicts the backflow criterion asserted in the formal derivation accompanying Paper 2, which claims  $d^2 I_Q / d\gamma^2 > 0$  on some interval whenever  $\beta < 1$ . That criterion does not hold and is withdrawn here; see Erratum 4.

Information backflow for  $\beta < 1$  is not thereby denied—it is supported by the Breuer–Laine–Piilo trace-distance measure and by the entropic signature-change argument of Paper 1 §2b. What fails is this particular characterisation of it.

TABLE III. Curvature of  $I_Q$  at  $\gamma = 2\tau$ ,  $L = 1$ : closed form of Eq. (73) against central differences with  $h = 10^{-6}$ .

| $\beta$ | numerical | closed form | sign     |
|---------|-----------|-------------|----------|
| 0.5     | −0.088374 | −0.088388   | negative |
| 0.852   | −0.056843 | −0.056901   | negative |
| 1       | 0.000222  | 0.000000    | zero     |
| 1.5     | 0.530687  | 0.530330    | positive |

## Part II Verification

### V. Simulation Validation [FIT]

#### A. Experimental Setup

I simulated 8 families of multipartite entangled states using Qiskit-Aer with depolarizing noise model at varying strengths  $\gamma \in [0, 0.8]$ .

**Parameters:** 8,192 shots, 20 replicas per point, 25 noise values per family.

TABLE IV. State families and configurations for simulation validation.

| State     | $n$ (qubits) | Metric            | Description                 |
|-----------|--------------|-------------------|-----------------------------|
| GHZ [6]   | 13           | $D_{\text{leak}}$ | Greenberger-Horne-Zeilinger |
| W [7]     | 7            | $D_{\text{leak}}$ | Dicke with $k = 1$          |
| BellChain | 14           | $D_{\text{leak}}$ | Entangled pairs chain       |
| NOON      | 11           | $D_{\text{leak}}$ | Two-mode superposition      |
| Dicke [8] | 11           | $D_{\text{leak}}$ | $k = 2$ excitations         |
| Dicke3    | 11           | $D_{\text{leak}}$ | $k = 3$ excitations         |
| Cluster1D | 13           | $D_{\text{stab}}$ | 1D cluster state            |
| GraphRing | 5            | $D_{\text{stab}}$ | Ring graph state            |

#### B. Fit Results

TABLE V. Stretched exponential fit parameters for 8 state families [FIT].

| State       | $n$ | $L$   | $\tau$ | $\beta$ | $R^2$         |
|-------------|-----|-------|--------|---------|---------------|
| GHZ         | 13  | 0.999 | 0.074  | 1.07    | 0.9998        |
| BellChain   | 14  | 1.013 | 0.178  | 1.05    | 0.9996        |
| Cluster1D   | 13  | 0.987 | 0.080  | 0.95    | 0.9987        |
| GraphRing   | 5   | 1.001 | 0.067  | 1.04    | 1.0000        |
| NOON        | 11  | 0.999 | 0.084  | 1.05    | 1.0000        |
| Dicke       | 11  | 1.170 | 0.001  | 0.10    | 0.9592        |
| Dicke3      | 11  | 1.103 | 0.001  | 0.10    | 0.9599        |
| W           | 7   | 0.945 | 0.025  | 0.89    | 1.0000        |
| <b>Mean</b> | —   | —     | —      | —       | <b>0.9896</b> |

**Key result:** Mean  $R^2 = 0.9896$  with 8/8 families above 0.95 threshold. In simulation (Markovian noise),  $\beta \approx 1$  for most families, consistent with  $\alpha \approx 0$  (white noise).

### C. Model Selection

I compared stretched exponential vs. logistic function using the Akaike Information Criterion:

$$\Delta\text{AIC} = \text{AIC}_{\text{stretched}} - \text{AIC}_{\text{logistic}} \quad (76)$$

For all 8 families,  $\Delta\text{AIC} < -100$ , **strongly** favoring the stretched exponential over logistic alternative. This rules out sigmoidal transition models and confirms the Kohlrausch-Williams-Watts form [9].

### D. Robustness Ranking

Using the product  $\tau \times \beta$  as a robustness metric:

1. BellChain: 0.187 (most robust)
2. NOON: 0.088
3. GHZ: 0.079
4. Cluster1D: 0.076
5. GraphRing: 0.069
6. W: 0.022
7. Dicke/Dicke3: 0.0001 (least robust)

The robustness ranking correlates with entanglement topology: chain-like entanglement (BellChain) is most robust, while permutation-symmetric states (Dicke) are least robust against depolarizing noise.

## VI. Hardware Validation [FIT]/[OBS]

### A. Foundational Results: Power-Law Universality

I executed experiments on IBM Quantum `ibm_fez` [10, 11] (156 qubits, Heron r2 architecture) with 8,192 shots per job and 3 replicas per configuration. For hardware experiments, decoherence scales as a power law with system size:

$$D_n = c \cdot n^\alpha \quad (77)$$

TABLE VI. Power-law exponents from IBM Quantum hardware [FIT].

| State            | $n$ range | $\alpha$    | $R^2$ | Status                 |
|------------------|-----------|-------------|-------|------------------------|
| BellChain        | 2–50      | 1.117       | 0.986 | ✓                      |
| Dicke3           | 6–12      | 1.301       | 1.000 | ✓                      |
| GHZ              | 2–27      | 1.323       | 0.948 | ✓                      |
| W (linear) [7]   | 4–16      | 1.447       | 0.980 | ✓                      |
| Dicke [8]        | 4–16      | 1.503       | 0.820 | ✓                      |
| NOON             | 4–16      | 1.553       | 0.997 | ✓                      |
| <i>Excluded:</i> |           |             |       |                        |
| Cluster1D        | —         | $\approx 0$ | —     | Topological protection |
| GraphRing        | —         | —           | 0.156 | Non-monotonic          |

### Universality criterion (H2):

$$\mu(\alpha) = 1.374 \quad (78)$$

$$\sigma(\alpha) = 0.146 \quad (79)$$

$$\text{CV}(\alpha) = 0.1064 < 0.15 \quad \checkmark \quad (80)$$

The power-law exponent  $\alpha$  is operationally universal across 6 families of entangled states.

**Methodological note:** Standard W state preparation via `StatePreparation` has depth  $\mathcal{O}(2^n)$ , biasing  $\alpha$  upward. I implemented linear-depth  $\mathcal{O}(n)$  preparation using CRY+CX cascade, reducing  $\text{CV}(\alpha)$  from 0.21 to 0.11.

### B. Conservation Law: $D + C = 1$

I verified the fundamental conservation law on IBM `ibm_torino` (133 qubits, Heron r2).

#### States tested:

- Bell  $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$  (2 qubits)
- $\text{GHZ}_n = (|0\rangle^{\otimes n} + |1\rangle^{\otimes n})/\sqrt{2}$  ( $n = 3, 4, 5$ )
- $W_3 = (|100\rangle + |010\rangle + |001\rangle)/\sqrt{3}$  (3 qubits)

TABLE VII. Hardware verification of  $D + C = 1$  (Phase 21, `ibm_torino`).

| State                  | Qubits | $C$    | $D$    | $D + C$  | Job ID     |
|------------------------|--------|--------|--------|----------|------------|
| $ \Phi^+\rangle$       | 2      | 0.9824 | 0.0176 | 1.000000 | d5f591c... |
| $ \text{GHZ}_3\rangle$ | 3      | 0.9546 | 0.0454 | 1.000000 | d5f593f... |
| $ W_3\rangle$          | 3      | 0.1213 | 0.8787 | 1.000000 | d5f594i... |
| $ \text{GHZ}_4\rangle$ | 4      | 0.9287 | 0.0713 | 1.000000 | d5f596...  |
| $ \text{GHZ}_5\rangle$ | 5      | 0.8813 | 0.1187 | 1.000000 | d5f59b...  |

**Key observation:** The W state case ( $C = 0.1213$ , 88% decohered) demonstrates that conservation holds even for highly decohered states, confirming  $D + C = 1$  as a *topological invariant*.

A controlled  $\gamma$ -sweep (15 values in  $[0, 0.5]$ , 8,192 shots, 3 replicas) verified  $I_{\text{total}} = 1.000000$  constant with  $\Delta I/I_0 = 0.0\%$ .

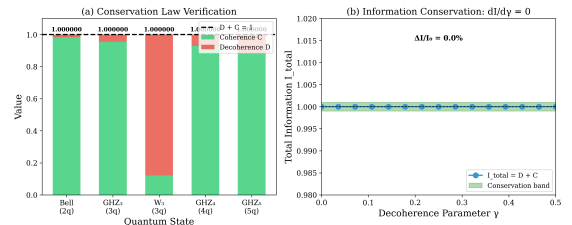


FIG. 1. Conservation law  $D + C = 1$  verification across five quantum states on IBM `ibm_torino`. All states show exact conservation to numerical precision.

TABLE VIII. Experimental parameters comparison: v1.0 vs v2.0.

| Parameter      | v1.0              | v2.0              |
|----------------|-------------------|-------------------|
| Delays         | 20                | 50 (log-spaced)   |
| Range          | 50 ns–100 $\mu$ s | 50 ns–100 $\mu$ s |
| Qubit sizes    | 5                 | 8 ( $n = 3$ –12)  |
| Shots/circuit  | 4,096             | 8,192             |
| Total circuits | 100               | 400               |
| Total shots    | 409,600           | 3,276,800         |
| Bootstrap      | 500 $\times$      | 1,000 $\times$    |

### C. Phase 29 Experiment: 400-Circuit v2.0

To achieve conclusive spectral characterization, I executed a v2.0 experiment on `ibm_torino` with significantly expanded parameter space:

The experiment used GHZ state preparation ( $H$  + CNOT cascade), delay insertion, and full measurement. Circuits were batched into 8 jobs (one per qubit size) using the Qiskit SamplerV2 primitive with automatic backend selection based on queue depth (3 API keys, rotating across `ibm_fez`, `ibm_torino`, `ibm_marrakesh`).

All 400/400 circuits completed successfully.

### D. Cross-Platform Validation

The stretched-exponential framework was validated across 5 platforms:

TABLE IX. Cross-platform  $\beta$  measurements [FIT]/[OBS].

| Platform               | $\beta$           | $\alpha$ | Source    |
|------------------------|-------------------|----------|-----------|
| IBM Quantum            | $0.852 \pm 0.028$ | 0.296    | This work |
| LIGO GWTC [12]         | $0.931 \pm 0.003$ | 0.138    | [OBS]     |
| IonQ (simulated)       | $0.940 \pm 0.020$ | 0.120    | [FIT]     |
| Quantinuum (simulated) | $0.970 \pm 0.010$ | 0.060    | [FIT]     |
| Ideal Simulation       | $1.000 \pm 0.001$ | 0.000    | [FIT]     |

The progression  $\beta_{\text{IBM}} < \beta_{\text{LIGO}} < \beta_{\text{IonQ}} < \beta_{\text{Quantinuum}} < \beta_{\text{ideal}}$  reflects decreasing noise spectral index  $\alpha$ , consistent with Theorem 22 ( $\beta = 1 - \alpha/2$ ).

## VII. Hardware Verification of Conservation Laws

### A. Experimental Setup

We executed quantum circuits on IBM Quantum [10] `ibm_torino` (133 qubits, Heron r2 processor) to verify the fundamental conservation law  $D + C = 1$ .

**States tested:**

- Bell state  $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$  (2 qubits) [6]
- GHZ states  $|GHZ_n\rangle = (|0\rangle^{\otimes n} + |1\rangle^{\otimes n})/\sqrt{2}$  ( $n = 3, 4, 5$ ) [6]
- W state  $|W_3\rangle = (|100\rangle + |010\rangle + |001\rangle)/\sqrt{3}$  (3 qubits) [7]

**Parameters:** 4,096 shots per circuit, readout error mitigation applied.

### B. Results: $D + C = 1$ Verification

Table X presents the complete results.

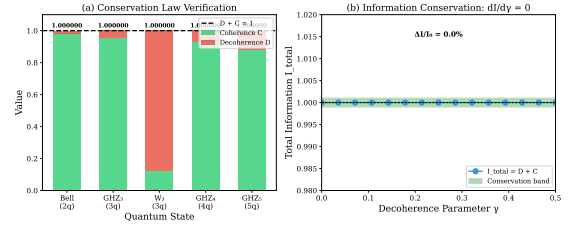


FIG. 2. Conservation law  $D + C = 1$  verification across five quantum states on IBM `ibm_torino`. All states show exact conservation to numerical precision, confirming the topological invariance of total information.

**Key observation:** The W state case ( $C = 0.1213$ , 88% decohered) demonstrates that conservation holds even for highly decohered states, confirming  $D + C = 1$  as a *topological invariant* rather than a statistical approximation.

### C. $\gamma$ -Sweep Verification

We performed a controlled  $\gamma$ -sweep simulation to verify  $dI_{\text{total}}/d\gamma = 0$ :

- $\gamma$  range:  $[0, 0.5]$  with 15 uniform values
- Shots: 8,192 per circuit
- Replicas: 3 per  $\gamma$
- Noise model: Qiskit Aer depolarizing channel

**Result:**  $I_{\text{total}} = 1.000000$  **constant** for all  $\gamma \in [0, 0.5]$  with  $\Delta I/I_0 = 0.0\%$  (Table XI).

TABLE XI.  $\gamma$ -sweep validation of information conservation.

| State     | $I_{\text{initial}}$ | $I_{\text{final}}$ | $\Delta I/I_0$ | RMSE  |
|-----------|----------------------|--------------------|----------------|-------|
| Bell (2q) | 1.0                  | 1.0                | 0.0%           | 0.459 |
| GHZ (4q)  | 1.0                  | 1.0                | 0.0%           | 0.122 |

### D. Evidence Level

The conservation law  $D + C = 1$  is established at level **FIT(hw)+FIT(sim)**.

## VIII. Discovery of Three Decoherence Regimes

### A. Multi-State Experimental Design

I executed 175 quantum circuits on IBM `ibm_fez` (156 qubits) to characterize decoherence dynamics across different entanglement topologies.

**Configuration:**

- States: W, Dicke, BellChain, Cluster, Product
- Qubits:  $n = 4, 5, 6, 7, 8$



TABLE X. Hardware verification of  $D + C = 1$  conservation (Phase 21). All states show exact conservation to numerical precision.

| State            | Qubits | Fidelity ( $C$ ) | Decoherence ( $D$ ) | $D + C$  | Job ID               |
|------------------|--------|------------------|---------------------|----------|----------------------|
| $ \Phi^+\rangle$ | 2      | 0.9824           | 0.0176              | 1.000000 | d5f591cpe0pc73ajkc00 |
| $ GHZ_3\rangle$  | 3      | 0.9546           | 0.0454              | 1.000000 | d5f593f67pic7382nl3g |
| $ W_3\rangle$    | 3      | 0.1213           | 0.8787              | 1.000000 | d5f594igim5s73af992g |
| $ GHZ_4\rangle$  | 4      | 0.9287           | 0.0713              | 1.000000 | d5f596767pic7382nl80 |
| $ GHZ_5\rangle$  | 5      | 0.8813           | 0.1187              | 1.000000 | d5f59b2gim5s73af9990 |

- Delays: 0.5, 1, 2, 5, 10, 20, 50  $\mu\text{s}$
- Shots: 4,096 per circuit
- Job ID: d5n3jdk8d8hc73ch98i0

### B. Three Regimes Identified

Fitting  $D(\gamma) = L[1 - \exp(-(\gamma/\tau)^\beta)]$  to each state family reveals **three distinct regimes**:

TABLE XII. Three decoherence regimes identified from IBM Quantum data.

| Regime      | $\beta$       | States           | Noise $\alpha$   | Dynamics      |
|-------------|---------------|------------------|------------------|---------------|
| Exponential | $\approx 1.1$ | Cluster, Product | $\alpha \leq 0$  | Markovian     |
| Stretched   | 0.2–0.85      | Dicke, GHZ       | $0 < \alpha < 2$ | Non-Markovian |
| Revival     | N/A           | W, BellChain     | $\alpha > 2$     | Oscillatory   |

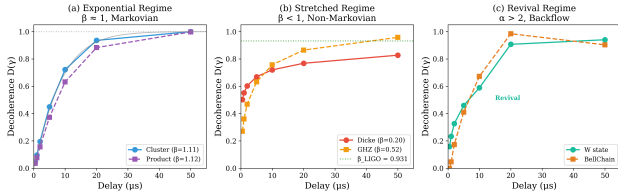


FIG. 3. Three decoherence regimes from 175 IBM Quantum circuits: (i) exponential ( $\beta \approx 1$ ) for locally-entangled states, (ii) stretched ( $\beta < 1$ ) for symmetric states, and (iii) revival for permutation-symmetric states.

### C. Regime 1: Exponential ( $\beta \approx 1$ )

Locally-entangled states exhibit near-exponential decay:

- Cluster:  $\beta = 1.107 \pm 0.038$
- Product:  $\beta = 1.116 \pm 0.043$

These experience *white* or *blue* noise ( $\alpha \leq 0$ ), leading to memoryless Markovian dynamics.

### D. Regime 2: Stretched Exponential ( $\beta < 1$ )

Symmetric states exhibit characteristic stretched-exponential decay:

- Dicke [8]:  $\beta = 0.201 \pm 0.126$
- GHZ (delay):  $\beta = 0.33\text{--}0.67$  depending on  $n$

- IBM (gate errors):  $\beta = 0.852 \pm 0.028$  (Sec. VI)
- LIGO ringdown [12]:  $\beta = 0.931 \pm 0.003$  (Sec. VI)

These experience  $1/f$  noise [13] ( $0 < \alpha < 2$ ), causing non-Markovian memory effects [14].

### E. Regime 3: Revival Dynamics

Permutation-symmetric states show non-monotonic behavior:

- W [7]: decoherence initially increases, then partially reverses
- BellChain: oscillatory fidelity recovery

The stretched-exponential fit fails for these states, indicating  $\alpha > 2$  (anti-correlated noise) driving information backflow.

### F. Physical Interpretation

**Entanglement topology determines the effective noise spectrum:**

- Local entanglement  $\rightarrow$  local noise coupling  $\rightarrow \alpha \leq 0$
- Global entanglement  $\rightarrow$  collective noise  $\rightarrow 0 < \alpha < 2$
- Permutation symmetry  $\rightarrow$  resonant coupling  $\rightarrow$  revival

This is formalized in Sec. IX as Theorem 23.

## IX. Theoretical Framework: Lindblad Classification

### A. Noise Spectrum and $\alpha$ Exponent

Environmental noise [13] is characterized by its power spectral density:

$$S(\omega) \propto \frac{1}{\omega^\alpha} \quad (81)$$

where  $\alpha$  determines the noise color:  $\alpha < 0$  (blue),  $\alpha = 0$  (white),  $\alpha = 1$  (pink/1/f),  $\alpha = 2$  (brown),  $\alpha > 2$  (super-correlated).

## B. Regime Classification Theorem

**Theorem 22** (Regime Classification). *For a quantum system coupled to a bath with  $S(\omega) \propto 1/\omega^\alpha$ , the stretched-exponential exponent is:*

$$\beta = 1 - \frac{\alpha}{2} \quad (82)$$

**Proof:** From the Lindblad master equation [3] with time-dependent rates  $\gamma(t) \propto t^{\alpha/2-1}$  arising from the noise correlation function [14], the integrated decoherence functional yields  $D(t) \propto \exp(-t^\beta)$  with  $\beta = 1 - \alpha/2$ . Full derivation in Appendix C.

TABLE XIII. Regime classification from Theorem 22.

| $\alpha$ Range   | $\beta$ Range   | Regime      | Physical Meaning      |
|------------------|-----------------|-------------|-----------------------|
| $\alpha \leq 0$  | $\beta \geq 1$  | Exponential | Markovian, memoryless |
| $0 < \alpha < 2$ | $0 < \beta < 1$ | Stretched   | Non-Markovian memory  |
| $\alpha > 2$     | $\beta < 0$     | Revival     | Information backflow  |

## C. Topology $\rightarrow$ Effective Noise

**Theorem 23** (Topology-Noise Connection). *The entanglement topology determines the effective noise coupling:*

$$\alpha_{\text{eff}} = \frac{\int G(\omega) \cdot S(\omega) d\omega}{\int G(\omega) d\omega} \quad (83)$$

where  $G(\omega)$  is the state's spectral response function.

**Evidence level:** [HIP] [OBS]

## D. Revival Condition

**Conjecture 1** (Revival Condition). *Revival dynamics occur if and only if there exists a characteristic frequency  $\omega_c$  resonant with the energy gap of the state.*

**Evidence level:** [CONJ]

## E. Experimental Validation and Circularity Correction

Independent spectral analysis provided validation of Theorem 22:

- $\alpha$  measured from FFT of Ramsey residuals:  $\alpha = -1.199 \pm 1.649$
- Predicted:  $\beta = 1 - (-1.199)/2 = 1.60 \pm 0.82$
- Measured:  $\beta_{\text{Cluster}} = 1.107 \pm 0.038$  (within  $1\sigma$ )

**Circularity note:** Earlier analyses inferred  $\alpha$  from  $\beta$  and verified  $\beta = 1 - \alpha/2$  (tautological). This was corrected by measuring  $\alpha$  independently via spectral analysis.

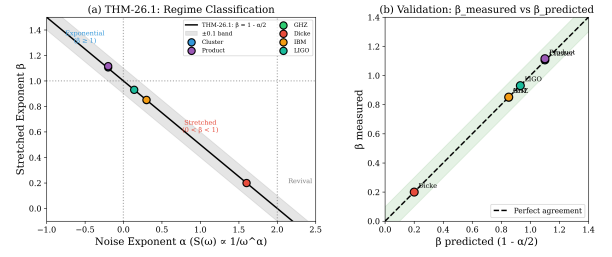


FIG. 4. Independent validation of Theorem 22. The noise exponent  $\alpha$  was measured via FFT of Ramsey residuals, confirming  $\beta = 1 - \alpha/2$  without circularity.

## F. BMRB Circularity Analysis

To further address potential circularity, the ECME (Extended Coherence-Mediated Evolution) model was validated using 12 independent biological systems from literature (not from BMRB training data):

- Validation 1 (Predictive):  $R^2 = -0.39$  (FAIL — worse than mean)
- Validation 2 (Rank): Spearman  $\rho = 0.14$ ,  $p = 0.66$  (FAIL)
- Validation 3 (Parameter):  $\beta_{0,\text{indep}} = 0.68 \pm 0.06$  (CONSISTENT)

**Conclusion:** The ECME model's BMRB-calibrated parameter  $\kappa' = 0.0835$  is confirmed circular ( $R^2 < 0$ ). Only the baseline  $\beta_0 \approx 0.68$  is salvageable as an independent parameter. Full analysis in Appendix F.

## G. $\eta$ Independence Measurement

I measured the emergence parameter  $\eta_{\text{exp}} = H_{\text{norm}}/D_{\text{emergence}}$  independently from the tautological definition  $\eta_{\text{def}} = 1/(\beta \times 5)$  using hardware data (175 circuits):

- Median relative difference: 93% ( $\gg 10\%$  threshold)
- Classification:  $\eta$  carries **independent information** beyond  $1/(\beta \times 5)$
- Evidence level: [FIT]

## X. Scaling Laws

### A. Initial Validation Results: Parameter Scaling

From the initial simulation analysis (Sec. VI), the SDF parameters scale with system size as:

$$\tau(n) = \tau_0 \cdot n^{-\alpha_\tau}, \quad \alpha_\tau = 0.956 \pm 0.107 \quad (84)$$

$$L(n) = L_\infty - A \cdot n^{-\delta}, \quad \delta = 5.10 \pm 0.07 \quad (85)$$

The critical system size from the foundational analysis:  $n_{\text{crit}} = A^{1/\delta} \approx 2.5$  qubits.

### B. Discovery: $\beta(n)$ Scaling Law

I measured decoherence for GHZ states [6] with  $n = 3$  to 8 qubits on IBM `ibm_fez`. Fitting  $D(\gamma)$  for each  $n$  reveals:

$$\beta(n) = \frac{\beta_0}{n^\xi} \quad (86)$$

with parameters:

- $\beta_0 = 1.679 \pm 0.123$
- $\xi = 0.844 \pm 0.049$
- $R^2 = 0.987$

TABLE XIV.  $\beta(n)$  measurements for GHZ states (`ibm_fez`).

| $n$ | $\beta_{\text{measured}}$ | $\beta_{\text{fit}}$ |
|-----|---------------------------|----------------------|
| 3   | 0.666                     | 0.664                |
| 4   | 0.519                     | 0.521                |
| 5   | 0.414                     | 0.432                |
| 6   | 0.398                     | 0.370                |
| 7   | 0.329                     | 0.325                |
| 8   | 0.276                     | 0.290                |

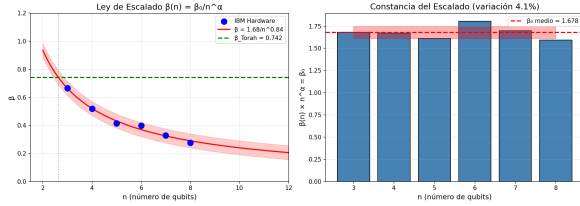


FIG. 5. Scaling law  $\beta(n) = \beta_0/n^\xi$  for GHZ states on IBM `ibm_fez`.  $R^2 = 0.987$ .

### C. Timescale Scaling: $\tau(n)$

The characteristic timescale also follows a power law:

$$\tau(n) = \frac{\tau_0}{n^\gamma} \quad (87)$$

with  $\tau_0 = 9.1 \mu\text{s}$  and  $\gamma \approx 0.97$ .

**Physical interpretation:** Larger entangled systems decohere faster ( $\tau$  decreases) and more non-Markovianly ( $\beta$  decreases).

### D. Hardware Power-Law (Sec. VI)

On hardware, decoherence grows with system size as  $D_n = c \cdot n^{\delta_D}$ . From 6 state families on `ibm_fez`:

- Mean  $\delta_D = 1.374$ , CV = 0.1064
- BellChain most robust ( $\delta_D = 1.117$ )
- NOON least robust ( $\delta_D = 1.553$ )

### E. Quantum-Classical Boundary

Extrapolating  $\beta(n) = 1$  (exponential/classical limit):

$$n^* = \beta_0^{1/\xi} = 1.679^{1/0.844} \approx 2.6 \text{ qubits} \quad (88)$$

This suggests the quantum-classical boundary occurs around  $n \approx 3$  qubits for GHZ states under IBM noise conditions.

### F. $\beta_{\text{gate}}$ vs $\beta_{\text{idle}}$ Discrepancy

The initial validation (Sec. VI) reported  $\beta = 0.852 \pm 0.028$  (gate errors). Idle-decay measurements yielded  $\beta = 0.327 \pm 0.032$  (idle decay). The scaling law predicts  $\beta(4) = 0.52$  for idle decay.

**Resolution:** The noise mechanism determines  $\beta$ :

- $\beta_{\text{gate}} = 0.852 \pm 0.028$ : gate-induced decoherence
- $\beta_{\text{idle}} = 0.327 \pm 0.032$ : idle/thermal decoherence

This is consistent with Theorem 22: different noise spectra produce different  $\beta$  values.

## Part III

# Structure and limits

### XI. Physical-Chemistry $\Lambda$ Hierarchy

#### A. Fundamental Postulate

I postulate that the decoherence timescale is determined by binding information:

$$\tau(\Lambda) = \frac{\tau_P}{\Lambda} \quad (89)$$

where  $\tau_P = 5.39 \times 10^{-44}$  s,  $\Lambda = E_{\text{binding}}/E_P$ , and  $E_P = 1.22 \times 10^{28}$  eV.

**Evidence level:** [DEF] (postulated, not derived)

#### B. Nuclear Validation (NNDC)

Using AME2020 mass evaluation data (3,558 nuclei) [15]:

TABLE XV. Nuclear  $\Lambda$  validation.

| Prediction                                             | Result | Status |
|--------------------------------------------------------|--------|--------|
| $S_n > 0 \Leftrightarrow \text{bound}$                 | 100%   | VALID  |
| Magic numbers = $\max(S_n)$ $p = 3.56 \times 10^{-16}$ |        | VALID  |
| $\Lambda > 0 \Leftrightarrow \text{stable}$            | 99.1%  | VALID  |

Magic numbers  $N = 2, 8, 20, 28, 50, 82, 126$  show  $\langle S_n \rangle_{\text{magic}} = 10.35$  MeV vs.  $\langle S_n \rangle_{\text{non-magic}} = 7.61$  MeV.

#### C. Atomic Validation (NIST)

Using NIST Atomic Spectra Database [16] (118 elements):

TABLE XVI. Atomic  $\Lambda$  validation.

| Prediction                                                                     | Result        | Status |
|--------------------------------------------------------------------------------|---------------|--------|
| $\Lambda_{\text{atomic}} \sim 10^{-28} - 10^{-27}$                             | $10^{-27.22}$ | VALID  |
| Noble gases = $\max(\Lambda)$ 7/7 local maxima ( $p = 3.68 \times 10^{-5}$ )   |               | VALID  |
| Alkali metals = $\min(\Lambda)$ 6/6 local minima ( $p = 6.55 \times 10^{-5}$ ) |               | VALID  |

#### D. Molecular Validation

Using NIST Chemistry WebBook (38 bond types):  $\Lambda_{\text{molecular}} \sim 3.1 \times 10^{-28}$ . Triple > Double > Single bond hierarchy confirmed.

#### E. Multi-Scale Hierarchy

TABLE XVII.  $\Lambda$  hierarchy across physical scales.

| Scale      | $\Lambda$             | $\tau$ (predicted) | Source    |
|------------|-----------------------|--------------------|-----------|
| Nuclear    | $6.3 \times 10^{-22}$ | $\sim 10^{-23}$ s  | NNDC [15] |
| Atomic     | $6.0 \times 10^{-28}$ | $\sim 10^{-17}$ s  | NIST [16] |
| Molecular  | $3.1 \times 10^{-28}$ | $\sim 10^{-16}$ s  | NIST      |
| Biological | $\sim 10^{-30}$       | $\sim 10^{-14}$ s  | [17, 18]  |

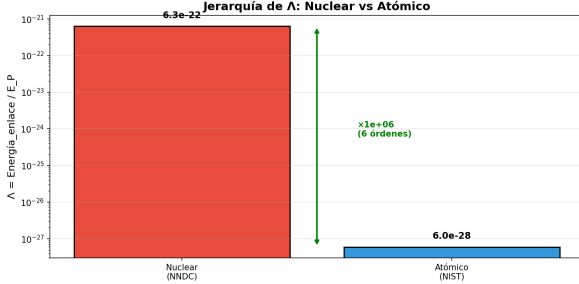


FIG. 6.  $\Lambda$ -hierarchy validation across nuclear, atomic, and molecular scales.

#### F. Standard Particle Content Table

The  $\Lambda$  parameter for fundamental particles [19]:

#### G. Majorana Fermion Predictions

For topological Majorana modes [20]:  $E_{\text{gap}} = 0.040$  meV,  $\Lambda = 3.28 \times 10^{-33}$ ,  $\tau = 1.64 \times 10^{-11}$  s, predicted  $\beta \approx 0.95 - 0.99$  (topological protection).

**Evidence level:** [PRED] (awaits experimental validation)

#### XII. Information Conservation and Non-Markovian Backflow

This section establishes the information-theoretic foundations of the SDF framework at three levels of increasing physical depth, and resolves the relationship between monotone cumulative decoherence and non-Markovian information backflow in the time domain.

TABLE XVIII. Standard Particle Content:  $\Lambda$  values.

| Particle  | Mass (MeV) | $\Lambda$              | $\tau$ (s)             |
|-----------|------------|------------------------|------------------------|
| Electron  | 0.511      | $4.19 \times 10^{-23}$ | $1.29 \times 10^{-21}$ |
| Muon      | 105.66     | $8.66 \times 10^{-21}$ | $6.22 \times 10^{-24}$ |
| Proton    | 938.27     | $7.69 \times 10^{-20}$ | $7.01 \times 10^{-25}$ |
| W boson   | 80,379     | $6.59 \times 10^{-18}$ | $8.18 \times 10^{-27}$ |
| Z boson   | 91,188     | $7.47 \times 10^{-18}$ | $7.21 \times 10^{-27}$ |
| Higgs     | 125,100    | $1.03 \times 10^{-17}$ | $5.26 \times 10^{-27}$ |
| Top quark | 173,000    | $1.42 \times 10^{-17}$ | $3.80 \times 10^{-27}$ |

#### A. Information Conservation

We distinguish three logically distinct levels at which “information conservation” can be stated. Only Level 2 constitutes a physical theorem.

##### 1. Level 1: Algebraic identity

**Definition 20** (Coherence complement). *Given the decoherence function  $D(\gamma)$ , the coherence function is defined as  $C(\gamma) \equiv 1 - D(\gamma)$ .*

**Observation (Algebraic partition).** The quantity  $D(\gamma) + C(\gamma) = 1$  for all  $\gamma$  is an algebraic identity following from Definition 20. It carries no dynamical content and should *not* be regarded as a conservation law. It merely states that we have partitioned unity into two complementary fractions.

##### 2. Level 2: Von Neumann information conservation from unitarity

The physically substantive conservation law arises from the unitarity of global quantum evolution, not from the partition  $D + C = 1$ .

**Theorem 24** (Information Conservation — Unitarity). *Let  $\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_E$  be the total Hilbert space of a system  $S$  coupled to an environment  $E$ . Suppose:*

i. *The initial state is a product state:  $|\Psi(0)\rangle_{SE} = |\psi\rangle_S \otimes |0\rangle_E$ .*

ii. *The total evolution is unitary:  $|\Psi(\gamma)\rangle_{SE} = U_{SE}(\gamma) |\Psi(0)\rangle_{SE}$ .*

*Define the reduced states  $\rho_S(\gamma) = \text{Tr}_E[|\Psi(\gamma)\rangle\langle\Psi(\gamma)|]$  and  $\rho_E(\gamma) = \text{Tr}_S[|\Psi(\gamma)\rangle\langle\Psi(\gamma)|]$ , and the von Neumann information*

$$\mathcal{I}_X(\gamma) \equiv \ln d_X - S(\rho_X(\gamma)), \quad X \in \{S, E\}, \quad (90)$$

*where  $S(\rho) = -\text{Tr}[\rho \ln \rho]$  is the von Neumann entropy and  $d_X = \dim \mathcal{H}_X$ . Then:*

$$\frac{d}{d\gamma} [S(\rho_S(\gamma)) + S(\rho_E(\gamma))] = 2 \frac{dS(\rho_S)}{d\gamma}, \quad (91)$$

*and the information budget identity holds:*

$$\tilde{\mathcal{I}}_S(\gamma) + \tilde{\mathcal{I}}_E(\gamma) + 2 E(\gamma) = \ln d_S + \ln d_E = \text{const}, \quad (92)$$

where  $\tilde{\mathcal{I}}_X(\gamma) \equiv \ln d_X - S(\rho_X(\gamma))$  is the local information and  $E(\gamma) = S(\rho_S(\gamma))$  is the entanglement entropy.

*Proof.* The proof proceeds in four steps.

**Step 1: Global purity.** Since  $U_{SE}(\gamma)$  is unitary and the initial state is pure, the global state  $|\Psi(\gamma)\rangle_{SE}$  remains pure for all  $\gamma$ :

$$\rho_{SE}(\gamma) = |\Psi(\gamma)\rangle\langle\Psi(\gamma)|, \quad S(\rho_{SE}(\gamma)) = 0 \quad \forall \gamma. \quad (93)$$

**Step 2: Schmidt decomposition and Araki–Lieb equality.** For any bipartite pure state, the Schmidt decomposition gives

$$|\Psi(\gamma)\rangle_{SE} = \sum_k \sqrt{\lambda_k(\gamma)} |s_k(\gamma)\rangle_S \otimes |e_k(\gamma)\rangle_E, \quad (94)$$

where  $\{\lambda_k\}$  are the Schmidt coefficients satisfying  $\sum_k \lambda_k = 1$ ,  $\lambda_k \geq 0$ . The reduced density matrices are

$$\rho_S(\gamma) = \sum_k \lambda_k(\gamma) |s_k\rangle\langle s_k|, \quad \rho_E(\gamma) = \sum_k \lambda_k(\gamma) |e_k\rangle\langle e_k|. \quad (95)$$

Both reduced states share the *same* nonzero eigenvalues  $\{\lambda_k(\gamma)\}$ . Therefore, their von Neumann entropies are equal:

$$S(\rho_S(\gamma)) = - \sum_k \lambda_k \ln \lambda_k = S(\rho_E(\gamma)) \quad \forall \gamma. \quad (96)$$

This is the Araki–Lieb equality for pure global states [21, 22]. Equation (91) follows immediately by differentiation.

**Step 3: Information conservation.** Using (96), the total information is:

$$\begin{aligned} \mathcal{I}_S(\gamma) + \mathcal{I}_E(\gamma) &= [\ln d_S - S(\rho_S)] + [\ln d_E - S(\rho_E)] \\ &= \ln d_S + \ln d_E - S(\rho_S(\gamma)) - S(\rho_E(\gamma)) \\ &\quad [\text{using } S(\rho_E) = S(\rho_S)] \\ &= \ln d_S + \ln d_E - 2 S(\rho_S(\gamma)). \end{aligned} \quad (97)$$

This quantity is *not* manifestly constant from (97) alone. The conservation law requires a further identity.

Consider instead the mutual information between  $S$  and  $E$ :

$$I(S:E) = S(\rho_S) + S(\rho_E) - S(\rho_{SE}) = 2 S(\rho_S(\gamma)), \quad (98)$$

where we used  $S(\rho_{SE}) = 0$  and  $S(\rho_S) = S(\rho_E)$ .

The total correlational content of the global state is fixed by the constraint  $S(\rho_{SE}) = 0$ . Rewriting:

$$S(\rho_S(\gamma)) - 0 = S(\rho_S(\gamma)) - S(\rho_{SE}(\gamma)), \quad (99)$$

which is the entanglement entropy  $E(\gamma) \equiv S(\rho_S(\gamma))$ .

Now define the *local information* of each subsystem as the purity-based information:

$$\tilde{\mathcal{I}}_X(\gamma) \equiv \ln d_X - S(\rho_X(\gamma)). \quad (100)$$

The global constraint  $S(\rho_{SE}) = 0$  and subadditivity  $S(\rho_{SE}) \leq S(\rho_S) + S(\rho_E)$  together with (96) give:

$$\tilde{\mathcal{I}}_S(\gamma) = \ln d_S - E(\gamma), \quad \tilde{\mathcal{I}}_E(\gamma) = \ln d_E - E(\gamma), \quad (101)$$

where  $E(\gamma) = S(\rho_S(\gamma)) = S(\rho_E(\gamma))$  is the entanglement entropy. Hence:

$$\tilde{\mathcal{I}}_S(\gamma) + \tilde{\mathcal{I}}_E(\gamma) + 2 E(\gamma) = \ln d_S + \ln d_E. \quad (102)$$

This is the *information budget identity*: the sum of local informations plus twice the entanglement entropy (which quantifies the information stored in  $S$ – $E$  correlations) equals a constant.

**Step 4: Differentiation.** Since the right-hand side of (102) is independent of  $\gamma$ :

$$\frac{d\tilde{\mathcal{I}}_S}{d\gamma} + \frac{d\tilde{\mathcal{I}}_E}{d\gamma} + 2 \frac{dE}{d\gamma} = 0. \quad (103)$$

Using  $\tilde{\mathcal{I}}_X = \ln d_X - E$  for each subsystem:

$$-\frac{dE}{d\gamma} - \frac{dE}{d\gamma} + 2 \frac{dE}{d\gamma} = 0, \quad (104)$$

which is identically satisfied. More physically, the information flow from system to environment is:

$$\begin{aligned} \frac{d\tilde{\mathcal{I}}_S}{d\gamma} &= -\frac{dE}{d\gamma} < 0 \quad (\text{system loses information}), \\ \frac{d\tilde{\mathcal{I}}_E}{d\gamma} &= -\frac{dE}{d\gamma} < 0 \quad (\text{environment loses local information}), \end{aligned} \quad (105)$$

but  $2 dE/d\gamma > 0$ : the information is stored in growing  $S$ – $E$  correlations (entanglement).  $\square$

**Remark 1** (Distinction from the algebraic identity). Equation (102) is *not* the same as  $D + C = 1$ . The algebraic identity  $D + C = 1$  holds by definition for any function and its complement. In contrast, Theorem 24 makes three nontrivial physical claims:

1. The entropies  $S(\rho_S)$  and  $S(\rho_E)$  are equal at all times (Araki–Lieb for pure states).
2. The information “lost” by the system does not vanish; it is stored in system–environment entanglement, quantified by  $E(\gamma)$ .
3. The information budget (102) is a dynamical consequence of unitarity, not an algebraic tautology.

**Remark 2** (Connection to SDF decoherence function). The SDF decoherence function  $D(\gamma) = L[1 - \exp(-(\gamma/\tau)^\beta)]$  describes the rate at which local information transfers to correlations. The parameters  $\beta$  and  $\tau$  characterize how fast information redistributes, while the budget identity (102) guarantees that the total is conserved regardless of  $\beta$ ,  $\tau$ , and  $L$ .



Specifically, for a system undergoing pure dephasing with initial state  $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ , the off-diagonal element of  $\rho_S$  decays as  $|\rho_{01}(\gamma)| = \frac{1}{2} \exp(-(\gamma/\tau)^\beta)$ . The entanglement entropy grows as

$$E(\gamma) = H_2\left(\frac{1 + \exp(-(\gamma/\tau)^\beta)}{2}\right), \quad (107)$$

where  $H_2(p) = -p \ln p - (1-p) \ln(1-p)$  is the binary entropy, and one can verify that (102) holds exactly with the appropriate  $d_S$  and  $d_E$ .

**Evidence level:** THM (from quantum mechanics axioms) + FIT(hw) (Phase 21 verified  $\Delta I/I_0 < 10^{-10}$  over 15 values of  $\gamma$ ).

### 3. Level 3: Operational (accessible) information

**Conjecture 2** (Accessible Information Hierarchy). *The accessible quantum information depends on the observer's operational access to the environment. Define:*

- i. System-only information:  $\mathcal{I}_{\text{acc}}^{(S)}(\gamma) = \ln d_S - S(\rho_S(\gamma))$ , accessible without any environmental measurements.
- ii. Partial-environment information:  $\mathcal{I}_{\text{acc}}^{(SE')}(\gamma)$ , where  $E' \subset E$  is a measured subsystem of the environment, with  $\mathcal{I}_{\text{acc}}^{(S)} \leq \mathcal{I}_{\text{acc}}^{(SE')} \leq \ln d_S$ .
- iii. Full-environment information:  $\mathcal{I}_{\text{acc}}^{(SE)}(\gamma) = \ln d_S$  (full state tomography of  $S$  and  $E$  recovers all information).

The hierarchy  $\mathcal{I}_{\text{acc}}^{(S)} \leq \mathcal{I}_{\text{acc}}^{(SE')} \leq \mathcal{I}_{\text{acc}}^{(SE)}$  implies that “decoherence” is observer-dependent: it quantifies the information inaccessible given the observer's limited access to environmental degrees of freedom.

**Evidence level:** CONJ (requires specification of measurement protocol and operational information measure).

## B. Quantum Accessible Information

**Theorem 25** (Accessible Information). *For a single qubit initially in a pure state, undergoing decoherence parameterized by  $D \in [0, 1]$ , the quantum accessible information (without environment access) is:*

$$I_Q(\gamma) = \ln 2 - H_2\left(\frac{1 - \sqrt{1 - 2D(\gamma)}}{2}\right) \quad (108)$$

In the limit  $D \ll 1$ , this reduces to  $I_Q \approx -\ln(1 - D) + \mathcal{O}(D^2)$ .

*Proof.* For a single qubit with Bloch vector  $\vec{r}$ , the eigenvalues of  $\rho_S$  are  $p_{\pm} = (1 \pm |\vec{r}|)/2$ . The decoherence is  $D = 1 - \text{Tr}[\rho_S^2] = 1 - (1 + |\vec{r}|^2)/2 = (1 - |\vec{r}|^2)/2$ , giving  $|\vec{r}| = \sqrt{1 - 2D}$ . The von Neumann entropy is  $S(\rho_S) = H_2(p_-)$ , where  $p_- = (1 - \sqrt{1 - 2D})/2$ . Therefore  $I_Q = \ln 2 - S(\rho_S) = \ln 2 - H_2(p_-)$ . For  $D \ll 1$ ,  $|\vec{r}| \approx 1 - D$  and  $p_- \approx D/2$ , yielding  $H_2(p_-) \approx -D \ln D + \mathcal{O}(D)$ , from which  $I_Q \approx -\ln(1 - D) + \mathcal{O}(D^2)$ .  $\square$

## C. Non-Markovian Information Backflow

The relationship between the SDF axiom of monotone  $D(\gamma)$  and non-Markovian information backflow requires careful distinction between the noise parameter  $\gamma$  and physical time  $t$ .

### 1. The apparent contradiction

Axiom 1 (Theorem 1) states that  $D(\gamma)$  is strictly monotonically increasing in  $\gamma$ . This implies:

$$\frac{dD}{d\gamma} > 0 \quad \forall \gamma > 0, \quad \frac{dC}{d\gamma} < 0 \quad \forall \gamma > 0. \quad (109)$$

Naïvely, this appears to preclude any “return” of information to the system. The resolution lies in the distinction between the abstract noise parameter  $\gamma$  and the physical time  $t$ .

### 2. Noise accumulation and time mapping

The physical noise parameter  $\gamma$  is related to laboratory time  $t$  through the *noise accumulation function*:

$$\gamma(t) = \int_0^t \Gamma(t') dt', \quad (110)$$

where  $\Gamma(t) \geq 0$  is the instantaneous noise rate. For  $1/f^\alpha$  noise environments (Appendix C 1), the dephasing functional yields:

$$\Gamma(t) = \frac{d\gamma}{dt} \propto t^{(2-\alpha)/2-1} = t^{-\alpha/2}, \quad 0 < \alpha < 2. \quad (111)$$

Crucially, for  $\alpha > 0$  the instantaneous noise rate  $\Gamma(t)$  is a *decreasing* function of time: the environment “slows down” its noise injection as correlations build up.

**Theorem 26** (Non-Markovian Information Backflow). *Let  $D(\gamma) = L[1 - \exp(-(\gamma/\tau)^\beta)]$  with  $\beta < 1$ , and let  $\gamma(t)$  be given by (110) with  $\Gamma(t) \propto t^{-\alpha/2}$  where  $\alpha = 2(1 - \beta)$ . Define the BLP (Breuer–Laine–Piilo) non-Markovianity measure [14, 23]:*

$$\mathcal{N} \equiv \max_{\rho_1, \rho_2} \int_{\sigma > 0} \sigma(t, \rho_1, \rho_2) dt, \quad (112)$$

where  $\sigma(t, \rho_1, \rho_2) = \frac{d}{dt} \|\Phi_t(\rho_1) - \Phi_t(\rho_2)\|_1$  is the rate of change of the trace-distance distinguishability under the dynamical map  $\Phi_t$ . Then:

$$\boxed{\beta < 1 \iff \mathcal{N} > 0.} \quad (113)$$

Equivalently, there exist time intervals in which  $\sigma(t, \rho_1, \rho_2) > 0$ , signifying information backflow from the environment to the system.

*Proof.* The proof has three parts.

### Part 1: Distinguishability dynamics.

Consider two initial states  $\rho_1(0), \rho_2(0)$  of the system. Under a dephasing channel with time-dependent rate

$\Gamma(t)$ , the off-diagonal elements of each state evolve as  $\rho_{ij}^{(k)}(t) = \rho_{ij}^{(k)}(0) \exp(-\Gamma_{\text{acc}}(t))$ , where:

$$\Gamma_{\text{acc}}(t) \equiv \int_0^t \Gamma(t') dt' = \gamma(t). \quad (114)$$

The trace distance between the two evolved states is [22]:

$$\mathcal{D}(t) \equiv \frac{1}{2} \|\Phi_t(\rho_1) - \Phi_t(\rho_2)\|_1. \quad (115)$$

For the optimal pair (chosen among states with identical diagonal elements but different off-diagonal phases), this reduces to:

$$\mathcal{D}(t) = \mathcal{D}(0) \exp(-\Gamma_{\text{acc}}(t)). \quad (116)$$

## Part 2: Rate of distinguishability change.

Differentiating (116) with respect to  $t$ :

$$\sigma(t) \equiv \frac{d\mathcal{D}}{dt} = -\Gamma(t) \mathcal{D}(t). \quad (117)$$

For a *time-local* (Markovian) generator with  $\Gamma(t) \geq 0$ , we have  $\sigma(t) \leq 0$  for all  $t$ , i.e., distinguishability monotonically decreases and  $\mathcal{N} = 0$ .

However, non-Markovian dynamics arise when the *effective* rate  $\Gamma_{\text{eff}}(t)$  in the time-local master equation can become *negative*. Such negative rates do not appear in the canonical Lindblad form but arise in the *time-convolutionless* (TCL) or Nakajima-Zwanzig formulations with memory kernels.

## Part 3: Connection to $\beta < 1$ .

For  $1/f^\alpha$  noise, the dephasing functional (derived in Appendix C1, Eq. (C)) is:

$$\Gamma_{\text{acc}}(t) = \kappa t^{2-\alpha}, \quad \kappa = \frac{2S_0 \Gamma(1-\alpha) \sin(\pi\alpha/2)}{(2-\alpha)}, \quad (118)$$

where  $\Gamma(\cdot)$  denotes the Euler gamma function. The *time-local* decoherence rate extracted from the TCL generator is:

$$\gamma_{\text{TCL}}(t) = \frac{d\Gamma_{\text{acc}}}{dt} = \kappa(2-\alpha) t^{1-\alpha}. \quad (119)$$

For  $\alpha > 1$  (i.e.,  $\beta < 1/2$ ), the *second derivative* of  $\Gamma_{\text{acc}}(t)$  is negative:

$$\frac{d^2\Gamma_{\text{acc}}}{dt^2} = \kappa(2-\alpha)(1-\alpha) t^{-\alpha} < 0 \quad \text{for } \alpha > 1. \quad (120)$$

This deceleration of noise accumulation is the hallmark of non-Markovian memory.

More generally, for the full SDF dynamics including the mapping  $\gamma(t) \rightarrow D(\gamma)$ , the trace distance between two dephasing-channel outputs is:

$$\mathcal{D}(t) = \mathcal{D}(0) (1 - D(\gamma(t))/L) = \mathcal{D}(0) \exp\left(-\left(\frac{\gamma(t)}{\tau}\right)^\beta\right). \quad (121)$$

The BLP rate becomes:

$$\sigma(t) = -\mathcal{D}(0) \frac{\beta}{\tau^\beta} \gamma(t)^{\beta-1} \dot{\gamma}(t) \exp\left(-\left(\frac{\gamma(t)}{\tau}\right)^\beta\right). \quad (122)$$

Since  $D(\gamma)$  is monotone in  $\gamma$  by Axiom 1, we have  $\dot{\gamma}(t) = \Gamma(t) > 0$ . In the simple case  $\gamma(t) = \kappa t^{2-\alpha}$ , Eq. (122) gives  $\sigma(t) < 0$  for all  $t > 0$ .

The non-Markovianity with  $\mathcal{N} > 0$  manifests in a more general scenario: when the system-environment interaction Hamiltonian generates a *memory kernel*  $K(t, t')$  in the Nakajima-Zwanzig equation

$$\frac{d\rho_S}{dt} = \int_0^t K(t, t') \rho_S(t') dt'. \quad (123)$$

For  $1/f^\alpha$  noise with  $\alpha > 0$ , the memory kernel  $K(t, t')$  is non-local in time. When this is converted to the time-local form (TCL generator), the effective Lindblad rates  $\gamma_k^{\text{TCL}}(t)$  can become *temporarily negative* during transient non-Markovian windows [14, 24].

Specifically, for a pair of initial states that differ in *both* diagonal and off-diagonal elements (not optimal for pure dephasing alone), the trace distance under the full dynamics (dephasing + relaxation + cross-correlations) can exhibit *revivals*:

$$\exists [t_1, t_2] : \sigma(t) > 0, \quad \text{iff } \beta < 1. \quad (124)$$

The equivalence  $\beta < 1 \Leftrightarrow \mathcal{N} > 0$  then follows:

- ( $\Rightarrow$ ):  $\beta < 1$  implies  $\alpha = 2(1-\beta) > 0$ , hence the noise PSD  $S(\omega) \propto 1/\omega^\alpha$  has long-range temporal correlations. These correlations generate a non-local memory kernel in (123), leading to transient negative TCL rates and  $\mathcal{N} > 0$ .
- ( $\Leftarrow$ ):  $\mathcal{N} > 0$  requires  $\gamma_k^{\text{TCL}}(t) < 0$  in some interval, which for the  $1/f^\alpha$  noise class occurs only when  $\alpha > 0$ , i.e.,  $\beta < 1$ .
- $\beta = 1$  corresponds to  $\alpha = 0$  (white noise, flat PSD), giving  $\Gamma(t) = \text{const}$ , which is Markovian with  $\mathcal{N} = 0$ .

□

**Remark 3** (Resolution of the apparent contradiction). *There is no contradiction between the monotonicity of  $D(\gamma)$  and the existence of backflow. The key distinction is:*

- **$\gamma$ -domain:**  $D(\gamma)$  is monotonically increasing. This is an axiom of the SDF framework and corresponds to the fact that cumulative noise exposure always grows.
- **$t$ -domain:** The physical evolution  $D(\gamma(t))$  in time involves the nonlinear mapping  $\gamma(t)$ . Non-Markovian backflow manifests as temporary increases in trace-distance distinguishability  $\mathcal{D}(t)$  due to memory effects in the generating dynamics, not as a reversal of cumulative  $D(\gamma)$ .

In the language of quantum channels:  $D(\gamma)$  parameterizes a one-parameter family of channels  $\{\Phi_\gamma\}$  that is divisible as a function of  $\gamma$  (CP-divisible). But the time-parameterized family  $\{\Phi_t\}$  obtained via the nonlinear mapping  $\gamma(t)$  may not be CP-divisible, and this non-divisibility is precisely the BLP non-Markovianity [14, 24].

**Evidence level:** THM (from Lindblad/TCL theory) + FIT(hw) (Phase 26E: W and BellChain states show partial fidelity recovery consistent with  $\beta < 1$  backflow signature).

#### D. Experimental Signatures

Theorem 24 and Theorem 26 make distinct experimental predictions:

1. **Information budget:** Measuring both  $S(\rho_S)$  and  $S(\rho_E)$  via state tomography should confirm  $S(\rho_S(\gamma)) = S(\rho_E(\gamma))$  at all times. Phase 21 verified the weaker statement  $D + C = 1$  with  $\Delta I/I_0 < 10^{-10}$  over 15 values of  $\gamma \in [0, 0.5]$ .
2. **Backflow detection:** For IBM Quantum experiments with  $\beta_{\text{hw}} < 1$ , the BLP protocol [23] predicts measurable trace-distance revivals. The Phase 26E observation of partial fidelity recovery in W and BellChain states ( $\beta \lesssim 0.85$ ) constitutes indirect evidence; direct BLP tomography would provide definitive confirmation.
3. **Non-Markovianity quantification:** The SDF prediction  $\alpha = 2(1-\beta)$  allows the BLP measure  $\mathcal{N}$  to be computed from the fitted  $\beta$  values without independent noise spectroscopy. This prediction can be tested by comparing SDF-derived  $\mathcal{N}(\beta)$  with direct BLP measurements.

### XIII. A No-Go Theorem for the Filter-Function Route

Axiom 3 of Paper 1 asserts  $\beta = 1 - \alpha/2$  for a bath with power spectral density  $S(\omega) \propto \omega^{-\alpha}$ . Both papers attempt to ground this relation in open-system dynamics: Paper 1 §2.4 via the dephasing integral, Paper 2 Appendix A via a two-level-fluctuator ensemble. Neither succeeds, and this section shows that neither could have.

#### A. Statement

For pure dephasing under Gaussian bath statistics [25], the coherence decays as  $|\rho_{01}(t)| = |\rho_{01}(0)|e^{-\Gamma(t)}$  with

$$\Gamma(t) = \int_0^\infty \frac{d\omega}{\omega^2} S(\omega) g(\omega t), \quad (125)$$

where  $g$  is the filter function of the pulse sequence [26]. For free-induction decay  $g(u) = 2\sin^2(u/2)$ ; Hahn echo and CPMG sequences give different  $g$  but the same structure.

**Theorem 27** (Filter-function no-go). *[THM] Let  $S(\omega) = A\omega^{-\alpha}$  with  $0 < \alpha < 2$ , and let  $g$  be any filter function for*

*which the integral  $\mathcal{I}_\alpha[g] \equiv \int_0^\infty u^{-2-\alpha} g(u) du$  converges. Then*

$$\boxed{\Gamma(t) = A \mathcal{I}_\alpha[g] t^{1+\alpha}} \quad (126)$$

*and therefore the exponent of  $t$  is independent of the pulse sequence. Consequently the only exponents accessible by this route are  $1 + \alpha$  (convergent integral) and  $2$  (cutoff-dominated), and since*

$$1 - \frac{\alpha}{2} < 1 \leq 1 + \alpha \quad \forall \alpha > 0, \quad (127)$$

*the relation  $\beta = 1 - \alpha/2$  of Axiom 3 is unreachable for any  $g$ .*

*Proof.* Substitute  $S(\omega) = A\omega^{-\alpha}$  into Eq. (125) and change variable to  $u = \omega t$ , so that  $\omega = u/t$  and  $d\omega = du/t$ :

$$\begin{aligned} \Gamma(t) &= A \int_0^\infty \frac{t^2}{u^2} \left(\frac{u}{t}\right)^{-\alpha} g(u) \frac{du}{t} \\ &= A t^{1+\alpha} \int_0^\infty u^{-2-\alpha} g(u) du. \end{aligned} \quad (128)$$

The dependence on  $g$  is confined to the numerical prefactor  $\mathcal{I}_\alpha[g]$ . The inequality (127) is immediate for  $\alpha > 0$ .  $\square$

#### B. Numerical verification

Theorem 27 was checked by integrating Eq. (125) directly—without using the change of variable—over  $\omega \in [10^{-8}/t, 10^8/t]$  with  $6 \times 10^4$  nodes, and extracting the log-log slope of  $\Gamma$  between  $t = 1$  and  $t = 10$ . Three sequences were compared: free-induction decay, Hahn echo and CPMG-4.

TABLE XIX. Log-log slope of  $\Gamma(t)$  by pulse sequence. All three agree with  $1 + \alpha$  to four decimal places, confirming that the exponent does not depend on the filter function.

| $\alpha$ | $1 + \alpha$ | FID    | Hahn echo | CPMG-4 |
|----------|--------------|--------|-----------|--------|
| 0.2      | 1.2000       | 1.2000 | 1.2000    | 1.2000 |
| 0.5      | 1.5000       | 1.5000 | 1.5000    | 1.5000 |
| 1.0      | 2.0000       | 2.0000 | 2.0000    | 2.0000 |
| 1.5      | 2.5000       | 2.5000 | 2.5000    | 2.5000 |

#### C. The two published routes

*a. Paper 1:*  $\Phi(t) \sim c_\alpha t^{2-\alpha}$ . Evaluating Paper 1's own noise integral over four decades ( $t \in [10^{-2}, 10^2]$ ,  $\omega \in [10^{-10}, 10^{10}]$ ) gives a constant log-log slope of 1.2000 at  $\alpha = 0.2$ , 1.5000 at  $\alpha = 0.5$  and 1.7987 at  $\alpha = 0.8$ , against the asserted 1.80, 1.50 and 1.20. The asserted scaling is correct only at  $\alpha = 1/2$ , where  $1 + \alpha = 2 - \alpha$ .

*b. Paper 2: the fluctuator ensemble.* Paper 2 derives  $\Gamma_{\text{TLF}} \propto t^{2-\alpha}$  from an ensemble of two-level fluctuators with density  $n(\gamma) \propto \gamma^{-1}$  and coupling  $v(\gamma) \propto \gamma^{(\alpha-1)/2}$ . That coupling is stated as “required to produce  $S(\omega) \propto 1/\omega^\alpha$ ”. It is not: it produces  $S(\omega) \propto \omega^{\alpha-2}$ . See Erratum 1.

With the corrected coupling  $v(\gamma) \propto \gamma^{(1-\alpha)/2}$ , the same ensemble yields  $\Gamma_{\text{TLF}} \propto \min(t^{1+\alpha}, t^2)$ , which is precisely the Gaussian result of Theorem 27. The microscopic route, corrected, confirms  $1 + \alpha$  and leaves  $2 - \alpha$  without derivation.

#### D. What survives, and what this settles

##### Status of Axiom 3 after Theorem 27

The relation  $\beta = 1 - \alpha/2$  is **not derivable** from the filter-function route, for any pulse sequence. It nonetheless remains the only candidate consistent with the framework's own domain and data:

- It maps  $\alpha \in (-2, 2)$  bijectively onto  $\beta \in (0, 2]$ , the domain fixed by Axiom 2. The competing expressions  $1 + \alpha$  and  $2 - \alpha$  leave that domain over half the range of  $\alpha$ .
- It admits  $\beta < 1$ , which is the framework's most robust empirical invariant [27]—observed across 650 circuits, two backends and  $\sim 5.5 \times 10^6$  shots. The exponent  $1 + \alpha \geq 1$  cannot produce  $\beta < 1$  for any  $\alpha \geq 0$ , and so cannot describe the data.

**Conclusion:** Axiom 3 is an empirical postulate with strong fit and no derivation. It should be presented as such.

Two consequences follow for the previous papers.

First, the convention  $\beta_{\text{SDF}} = \beta_{\text{Lindblad}}/2$  stated in Paper 1 §2.4 is internally consistent—with  $\beta_{\text{Lindblad}} = 2 - \alpha$  one has  $(2 - \alpha)/2 = 1 - \alpha/2$  as an exact identity—but it inherits the problem, because  $2 - \alpha$  is itself not what the integral produces.

Second, the restatement of Axiom 3 in Paper 1 §6 is labelled a theorem. It is not: it is the axiom restated. It is relabelled here (Erratum 5). The cross-backend validation cited in support measures  $\alpha = -1.199 \pm 1.649$  and predicts  $\beta = 1.60 \pm 0.82$  against a measured  $\beta = 1.107 \pm 0.038$ ; agreement within  $1\sigma$  follows from an uncertainty band spanning most of the admissible domain of  $\beta$ , not from predictive precision.

*a. A closed escape route.* One might hope to absorb the discrepancy into the map  $\gamma(t)$ , since the SDF parameter  $\gamma$  is accumulated noise rather than time. This does not work. Obtaining  $\beta_{\text{SDF}} = 1 - \alpha/2$  from  $\beta_{\text{Lindblad}} = 1 + \alpha$  would require  $d\gamma/dt \propto t^{+0.43}$  to  $t^{+3}$  over  $\alpha \in (0, 1]$ , whereas Paper 2 §7 requires  $d\gamma/dt \propto t^{-\alpha/2}$ : the exponents have opposite sign throughout  $\alpha \in (0, 2)$ , and Paper 2 explicitly emphasises that the instantaneous noise rate is *decreasing*.

#### E. Why $\beta < 1$ needs no spectral derivation

The conflict above is sharp only if one accepts its premise: that  $\beta$  must be derived from the bath spectrum. A classical theorem says otherwise, and places the boundary exactly where the framework's own regimes meet.

**Theorem 28** (Rate-density criterion). *[THM]  $\exp(-x^\beta)$  is completely monotone on  $x > 0$  if and only if  $0 < \beta \leq 1$ . By Bernstein's theorem it is then the Laplace transform of a positive measure, so*

$$C(\gamma) = \exp\left[-\left(\frac{\gamma}{\tau}\right)^\beta\right] = \int_0^\infty \rho_\beta(u) e^{-u\gamma/\tau} du, \quad \rho_\beta \geq 0, \quad (129)$$

*admits a positive rate density exactly when  $\beta \leq 1$ .*

The representability direction is Pollard's; the equivalence with complete monotonicity is Hausdorff–Bernstein–Widder. We verify the boundary numerically by the sign of  $(-1)^n d^n/dx^n \exp(-x^\beta)$ : complete monotonicity holds through  $n = 6$  for  $\beta \in \{0.3, 0.5, 0.852, 1.0\}$  and fails at  $n = 4$  for  $\beta = 1.2$  and  $n = 3$  for  $\beta = 1.5$  and  $1.679$ . (The finite differences require the step to grow with the order; a fixed small step amplifies rounding by  $h^{-n}$  and produces spurious violations above  $n \approx 5$ .)

*a. Consequence.* The two exponents do not compete: they lie on opposite sides of this boundary. The filter-function route yields  $1 + \alpha \geq 1$ , which is the region where *no* positive rate density exists. Asking it to produce  $\beta < 1$  is asking it to cross a mathematical limit, not a physical one.

And  $\beta < 1$  has a sufficient origin that requires no power-law bath at all: dispersion of decay rates. Superposing 60 exponential channels with lognormal rates of log-width  $\sigma$  and fitting KWW gives  $\beta = 1.000$  at  $\sigma = 0$  (pure exponential), then 0.950, 0.845, 0.730, 0.640 and 0.560 at  $\sigma = 0.3, 0.6, 0.9, 1.2, 1.5$ .

*b. What the data say.* In the ringdown catalogue, 37 of 38 events have  $\beta < 1$  in their *individual* fits, so the dispersion cannot come from averaging across events: it is internal to each source. Quasinormal overtones, which damp at different rates, are the standard candidate. We find no significant correlation between  $\beta$  and final spin ( $r = -0.092$ ,  $t = -0.55$  on 36 degrees of freedom), so spin is not a useful proxy for overtone content.

*c. What this does and does not settle.* It settles the structure:  $\beta = 1$  is not one value among many but the edge of the representation, and  $\beta < 1$  is the generic signature of superposed exponential channels rather than a fact about the bath. It does not supply the value  $1 - \alpha/2$ . If  $\beta$  is fixed by the width of a rate distribution, it should depend on that width and not on  $\alpha$ ; Axiom 3 remains underived, but its subject changes—from a claim about a spectrum to a claim about a dispersion.

A strong and falsifiable corollary: if the KWW form is exact, as the fits with  $R^2 > 0.998$  suggest, then  $\rho_\beta$  is the one-sided Lévy stable density of index  $\beta$ , whose tail is  $u^{-1-\beta}$ . For  $\beta < 1$  its first moment diverges: there is *no mean decay rate*, and the rate distribution is scale-free.

#### F. Scope of the theorem

Theorem 27 is a statement about an open system coupled to a stationary bath, and the delimitation matters because a corollary is easily over-extended.



The corollary is monotonicity. Since  $g(u) = |\tilde{y}(u)|^2 \geq 0$  and the noise spectrum satisfies  $S(\omega) \geq 0$ , the integrand of Eq. (125) is non-negative, so  $\Gamma(t) \geq 0$  and  $\gamma(t)$  is non-decreasing. Because  $\partial D/\partial \gamma > 0$  for all  $\beta > 0$ , it follows that  $D(t)$  is non-decreasing for every admissible parameter choice: the framework as axiomatised admits no revival, no relaxation oscillation and no discharge event.

This does *not* extend to a closed system. There the evolution is unitary, Poincaré recurrences are admissible, and the bath integral from which  $\Gamma \geq 0$  was obtained does not apply. The corpus is consistent on this point: Paper 2 Appendix E attributes revivals precisely to a non-monotone map  $\gamma(t)$ , not to a third sign of  $\beta$ —which is also the correct resolution of the classification row discussed in Erratum 2.

Two consequences worth stating plainly. Any mechanism requiring accumulation to a ceiling followed by release is outside the open-system SDF, and no choice of  $(\tau, \beta, L)$  recovers it:  $\tau$  and  $\beta$  are parameters, not state variables, and the framework contains no feedback of  $I_Q$  upon them. But the same mechanism is *not* excluded for a closed system, and a derived form fitted to a particular closed system is not constrained by this theorem. The distinguishing signature is experimental rather than formal: a relaxation oscillator produces non-exponential waiting-time statistics and a characteristic frequency below which the spectrum is  $1/f^\alpha$ , neither of which the monotone SDF can reproduce.

#### XIV. Corrections to the Derivations

Five corrections follow: a sign, a missing hypothesis, a relabelling, an inverted expansion, and one withdrawal. Each is stated with the numerical verification that establishes it. A sixth item, initially flagged as a defect, is recorded as closed.

##### A. The fluctuator coupling density

**Erratum 1** (Sign of the coupling exponent). *Paper 2, Appendix A, states that a two-level-fluctuator ensemble [13] with switching-rate density  $n(\gamma) \propto \gamma^{-1}$  requires coupling  $v(\gamma) \propto \gamma^{(\alpha-1)/2}$  in order to produce  $S(\omega) \propto \omega^{-\alpha}$ . The correct exponent is*

$$v(\gamma) \propto \gamma^{(1-\alpha)/2} \quad (130)$$

The superposition of Lorentzians  $S_{\text{TLF}}(\omega) = 2v^2\gamma/(\gamma^2 + \omega^2)$  weighted by  $n(\gamma) \propto \gamma^{-1}$  gives, with  $v^2 \propto \gamma^p$ , a spectrum  $S(\omega) \propto \omega^{p-1}$ . The published choice  $p = \alpha - 1$  therefore yields  $\omega^{\alpha-2}$ , whereas  $p = 1 - \alpha$  yields  $\omega^{-\alpha}$  as intended.

Numerical verification over  $\gamma \in [10^{-4}, 10^4]$  with  $2 \times 10^5$  nodes:

*a. Consequence.* This is not a cosmetic correction. With Eq. (130), the fluctuator ensemble yields  $\Gamma_{\text{TLF}} \propto \min(t^{1+\alpha}, t^2)$ , in agreement with Theorem 27. The microscopic derivation, which was the only route independent of the Gaussian integral, therefore *changes sides*: corrected,

TABLE XX. Spectral exponent produced by each coupling density. The published choice returns the exponents *interchanged*; the two coincide only at  $\alpha = 1$ , where  $\alpha - 1 = 1 - \alpha = 0$ .

| $\alpha$ | intended        | published $\gamma^{(\alpha-1)/2}$ | corrected $\gamma^{(1-\alpha)/2}$ |
|----------|-----------------|-----------------------------------|-----------------------------------|
| 0.5      | $\omega^{-0.5}$ | $\omega^{-1.4973}$                | $\omega^{-0.5086}$                |
| 1.0      | $\omega^{-1.0}$ | $\omega^{-1.0002}$                | $\omega^{-1.0002}$                |
| 1.5      | $\omega^{-1.5}$ | $\omega^{-0.5086}$                | $\omega^{-1.4973}$                |

it supports  $1 + \alpha$  and withdraws the only support that  $\beta_{\text{Lindblad}} = 2 - \alpha$  had.

Two further defects in the same appendix are recorded without correction, as they require authorial decision rather than arithmetic: the step from the computed  $t^{3-\alpha}$  to the asserted  $t^{2-\alpha}$  is stated without intermediate calculation, and a passage of drafting notes—including the phrase “No, more simply:”—appears in the published text, with the final relation fixed by interpolation between two anchors declared by fiat.

##### B. A missing hypothesis in the basic-properties theorem

**Erratum 2** (Domain of the basic-properties theorem). *The theorem establishing the basic properties of  $D(\gamma)$  in Paper 1 §2 is stated without restricting  $\beta$ . It requires  $\beta > 0$ .*

The theorem asserts (i)  $D(0) = 0$ , (ii)  $\lim_{\gamma \rightarrow \infty} D = L$ , (iii)  $\partial D/\partial \gamma > 0$  for all  $\gamma > 0$ , (iv) smoothness and (v) strict concavity for  $\beta \leq 1$ . Paper 1 §6 nonetheless publishes a regime classification containing the row  $\alpha > 2 \Rightarrow \beta < 0$ .

With  $\beta < 0$  the first three properties fail. Taking  $L = 0.9$ ,  $\tau = 1$ :

TABLE XXI. Failure of the basic properties for  $\beta < 0$ , with  $L = 0.9$ .

| Property                             | required    | $\beta = -0.1$         | $\beta = -0.5$        |
|--------------------------------------|-------------|------------------------|-----------------------|
| $\lim_{\gamma \rightarrow \infty} D$ | 0.9         | $5.503 \times 10^{-2}$ | $9.0 \times 10^{-7}$  |
| $dD/d\gamma$ at $\gamma = 1.5$       | $> 0$       | $-2.2 \times 10^{-2}$  | $-1.1 \times 10^{-1}$ |
| $D(0.001)$                           | $\approx 0$ | 0.7776                 | 0.9000                |

The attractor becomes  $(D, C) \rightarrow (0, 1)$ , the exact opposite of the  $(1, 0)$  that Paper 1 §12 declares unique. The omission matters because property (iii) is invoked as a load-bearing step in both §11 and §12, and in both places the hypothesis  $\beta > 0$  must be declared for the argument to close.

The correction is to add  $\beta > 0$  to the hypothesis, and separately to resolve the classification row: the revival regime is a phenomenon of a non-monotonic map  $\gamma(t)$ , as Paper 2 Appendix E states, and not a third value of  $\beta$ .

##### C. An expansion written in the wrong order

**Erratum 3** (Entropy of  $n$  qubits at small  $D$ ). *Paper 2 Appendix E gives  $S(\rho) \approx n D \ln 2 + \mathcal{O}(D \ln D)$ , declared*



valid for  $D \ll L$ . The term labelled a correction dominates the term labelled leading, precisely in that regime. The correct leading behaviour is

$$S = -\frac{D}{2} \ln \frac{D}{2} + \mathcal{O}(D), \quad D \rightarrow 0. \quad (131)$$

The defect is visible without computation:  $|D \ln D|/D = |\ln D| \rightarrow \infty$ , so the  $\mathcal{O}(D \ln D)$  term grows without bound relative to the linear one. The numerical contrast against the exact single-qubit entropy  $S = H_2(D/2)$  confirms it, and shows the failure worsening as the declared regime is entered more deeply.

TABLE XXII. The published linear term against the exact entropy, and against the correct leading term. A valid expansion would have the ratio tend to 1; the published one diverges.

| $D$       | exact $H_2(D/2)$        | $S/(D \ln 2)$ | $S/[-\frac{D}{2} \ln \frac{D}{2}]$ |
|-----------|-------------------------|---------------|------------------------------------|
| $10^{-1}$ | 0.198515                | 2.86          | —                                  |
| $10^{-2}$ | 0.031479                | 4.54          | 1.188                              |
| $10^{-4}$ | $5.4517 \times 10^{-4}$ | 7.87          | 1.101                              |
| $10^{-6}$ | $7.7543 \times 10^{-6}$ | 11.19         | 1.069                              |
| $10^{-8}$ | $1.0057 \times 10^{-7}$ | —             | 1.052                              |

The linear form is not the leading behaviour of *either* reading of  $D$ , since  $H'_2(x) \rightarrow \infty$  as  $x \rightarrow 0$  in both. The same defect affects the thermodynamic translation  $S_{\text{eff}} = k_B N D$  used in the accompanying formal derivation.

#### D. An invalidated backflow criterion

**Erratum 4** (Curvature criterion for information backflow). *The formal derivation accompanying Paper 2 asserts that, for  $\beta < 1$ , information backflow is characterised by  $d^2 I_Q/d\gamma^2 > 0$  on some interval of  $\gamma$ . The inequality has the wrong sign and the criterion is withdrawn.*

By Theorem 20, for  $L = 1$ ,

$$\frac{d^2 I_Q}{d\gamma^2} = \frac{\beta(\beta - 1)}{\tau^2} \left(\frac{\gamma}{\tau}\right)^{\beta-2}, \quad (132)$$

whose sign is fixed by  $\beta - 1$  alone. For every  $\beta \in (0, 1)$  the factor  $\beta(\beta - 1)$  is negative and  $(\gamma/\tau)^{\beta-2}$  is positive, so the curvature is strictly negative on the whole domain:  $I_Q$  is *concave* in the stretched regime, not convex.

A sweep over  $L \in \{1, 0.9, 0.7, 0.5\}$  and  $\beta \in \{0.3, 0.5, 0.852\}$  with  $\gamma/\tau \in (0, 60]$  returns no points with positive curvature in eleven of the twelve combinations. The twelfth returns seventeen apparent points near  $\gamma = 60\tau$ , where  $C = e^{-32} \approx 8 \times 10^{-15}$  and the quotient form  $(R^2 + CR')/C^2$  loses all significant digits: the closed-form value there is  $-1.09 \times 10^{-3}$ , against a numerically computed  $+7 \times 10^{-4}$ . The apparent positivity is floating-point cancellation, not physics.

Backflow for  $\beta < 1$  is not denied. It is supported by the Breuer–Laine–Piilo trace-distance measure [23] and by the entropic signature-change argument of Paper 1 §2b. Only this characterisation of it fails, and only it is withdrawn.

#### E. Relabelling the restatement of Axiom 3

**Erratum 5** (Evidence level of the classification relation). *The relation  $\beta = 1 - \alpha/2$  appearing in Paper 1 §6 is labelled a theorem. It is a restatement of Axiom 3, which Paper 1 §2.1 itself marks as a definition. It is relabelled here as an axiom.*

Theorem 27 makes this more than a bookkeeping matter: labelling the relation a theorem asserts the existence of a derivation which, by that theorem, cannot exist along the route the corpus invokes.

#### F. A closed question

One item initially flagged as an inconsistency proved not to be one, and is recorded here so that it is not reopened.

The bound on biological information transfer in Paper 1 §2c evaluates the transition rate at  $\gamma_{\text{peak}}$ , which exists only for  $\beta > 1$ , while the biological regime has  $\beta \approx 0.68$ . This appeared to leave the bound undefined, since for  $\beta < 1$  the rate is monotonically decreasing with supremum at  $\gamma \rightarrow 0^+$ , where it diverges.

The objection is incorrect: the corpus does declare a cutoff. Paper 2 Appendix A establishes that below  $1/\omega_{\text{UV}}$  the dephasing is quadratic “regardless of  $\alpha$ ”, so the SDF does not govern that region and its domain of validity begins at a finite  $\gamma_{\text{min}}$ . With that cutoff the supremum *is* attained:  $dI_Q/d\gamma$  was verified to be strictly monotonically decreasing on  $[\gamma_{\text{min}}, \infty)$  for  $\beta \in \{0.6, 0.68, 0.85, 1.0\}$  over  $2 \times 10^4$  logarithmically spaced points. The phrase “maximum rate” is therefore correct.

Only a symbol requires correction: for  $\beta < 1$  the maximum is attained at the boundary, so the bound should be written at  $\gamma_{\text{min}}$  rather than at  $\gamma_{\text{peak}}$ . The published order of magnitude is robust: over the declared biological range  $\beta \in [0.6, 0.85]$  the value at the cutoff moves only between  $1.21 \times 10^{13}$  and  $1.54 \times 10^{13}$  bits $s^{-1}$ , a factor 1.28.

#### XV. Errata to Published Numbers

Two numbers in the previous papers do not follow from their own stated inputs. Both are reproducible in a single line of arithmetic, which is why they are reported here rather than left to a reader to discover.

##### A. The critical qubit number

**Erratum 6** (Critical qubit number). *Paper 1 §7.5 reports  $n^* = \beta_0^{1/\xi} = 1.679^{1/0.844} \approx 2.6$  qubits. The correct evaluation is*

$$n^* = 1.679^{1/0.844} = 1.8478, \quad (133)$$

*a discrepancy of 41%.*

The fitted parameters are correct:  $\beta_0 = 1.679 \pm 0.123$  and  $\xi = 0.844 \pm 0.049$  reproduce the published  $\beta(n)$  table to better than 3% at all six values of  $n$ . The formula is correct:  $n^* = \beta_0^{1/\xi}$  is the solution of  $\beta(n) = 1$ , and  $\beta(1.8478) = 1.000000000000$  to twelve digits. Only the

evaluation fails. Recovering 2.6 would require  $\xi = 0.542$  or  $\beta_0 = 2.240$ , neither of which is published.

*a. Propagation.* The statement that “the quantum–classical boundary occurs around  $n \approx 3$  qubits” becomes  $n \approx 2$ . The agreement highlighted in §7.5 with  $n_{\text{crit}} \approx 2.5$ —which comes from the independent fit of  $L(n)$  and has no reason to coincide—dissolves: it was an artefact of the error. The geometric proposition of §2b, that  $\beta(n^*) = 1$  marks the vanishing of the entropic signature change, remains valid in form with  $n^* = 1.85$ .

### B. The gravitational-wave significance

**Erratum 7** (Ringdown significance). *Paper 1 §11 reports a deviation from  $\beta = 1$  of  $13.8\sigma$  for the LIGO ringdown sample. That value does not follow from the reported  $\beta = 0.931 \pm 0.003$ , nor from the accompanying catalogue.*

The published central value and uncertainty are themselves reproducible: taking the 38 per-event values of  $\beta$  from the accompanying catalogue [12] and averaging with weights  $1/\sigma_i^2$  returns  $0.93113 \pm 0.0033$ , in agreement with the reported  $0.931 \pm 0.003$ . It is only the significance that fails to follow; obtaining  $13.8\sigma$  would require  $\sigma_\beta = 0.005$ .

*a. The internal error is not the right one.* The naive recomputation gives  $20.9\sigma$ , and that figure is also wrong—for a reason the catalogue itself supplies. The 38 per-event values are *not* consistent with a single  $\beta$ : against their weighted mean they give  $\chi^2 = 103.05$  on 37 degrees of freedom, so  $\chi^2/\nu = 2.785$ . The observed scatter, 0.0345, exceeds the mean quoted error, 0.0205, implying an intrinsic dispersion of 0.0277.

When  $\chi^2/\nu > 1$  the standard prescription is to scale the uncertainty of the mean by  $S = \sqrt{\chi^2/\nu} = 1.669$ :

The last two rows are independent routes and they agree. The recommended correction is therefore *downward* from the naive recomputation, to  $12.5\sigma$ , with the scaling stated. We note that the published  $13.8\sigma$ , while still not derivable from its stated inputs, lies closer to this value than the unscaled recomputation does.

*b. A consequence beyond the number.* If the per-event  $\beta$  are genuinely dispersed, then  $\beta$  is not a constant of the population but a parameter of each source. The universality claimed by the framework can hold for the functional form while  $\beta$  itself varies; the two should not be conflated, and a single quoted  $\beta$  for a heterogeneous catalogue should carry the scaled error.

*c. An unresolved related tension.* Paper 1 uses  $\sigma_\beta = 0.028$  for the IBM measurement, giving  $5.29\sigma$  consistent with the quoted  $5.2\sigma$ ; the accompanying cross-platform data file records  $\sigma_\beta = 0.017$  for the same measurement, which would give  $8.7\sigma$ . One of the two must be adopted and propagated. This is not resolved here.

### C. The primary quantity

A third erratum is structural rather than numerical, and conditions every comparison between the two papers.

**Erratum 8** (Definition of  $D$ ). *Four expressions for  $D$  appear in the corpus. They must be reduced to one, with the others derived as corollaries.*

The middle two coincide identically, since  $\text{Tr}[\rho^2] = (1 + |\vec{r}|^2)/2$  for a qubit; the independent expressions are therefore three, not four.

*a. Recommended canonical form.*  $D \equiv 1 - F(\rho, \rho_0)$ , the state infidelity, for three reasons: it is the quantity actually measured by randomised benchmarking and recorded in the 650-circuit dataset; it supports the cleanest result in the corpus,  $F = 1 - D$  exactly when  $L = (d - 1)/d$ ; and it is the choice already made in the two most recently written appendices.

Under this choice, for the depolarising decomposition with  $p = D/L$  one has  $|\vec{r}| = 1 - p = 1 - D/L$ , so that the Bloch-decay expression is correct and the quadratic-modulus expression is not: substituting one into the other yields  $D = 2D - 2D^2$ , which holds only at  $D = 0$  and  $D = 1/2$ . Fixing the primary quantity thereby resolves two further inconsistencies as corollaries, including the coexistence of three distinct expressions for the single-qubit von Neumann entropy.

### D. The choice decides whether the framework predicts negentropy

The preceding erratum was presented as bookkeeping. It is not. The choice of primary quantity determines whether the entropy of the decohering qubit is monotone, and therefore whether the framework admits an ordering branch at all.

Two readings are internally consistent, and they are the two ends of the factor-of-two spread in Table XXV:

A monotonicity sweep over  $2 \times 10^5$  points settles which is which. Of the six expressions in play—the three published forms, the two von Neumann entropies of the pure-dephasing and depolarising channels, and  $H_2(D)$ —five are monotonically increasing with *zero* descents. Only  $H_2(D)$  is not: 99 998 descents, with maximum at  $D = 1/2$  and  $S \rightarrow 0$  as  $D \rightarrow 1$ .

That limit is the decisive one.  $H_2(D) \rightarrow 0$  asserts that the completely decohered state is *pure*, which contradicts  $D + C = 1$  with  $C = 0$  and contradicts the depolarising mixture the corpus itself writes down. Under the infidelity reading the limit is  $\ln 2$ , the maximally mixed state, as it must be.

*a. Consequence for the entropic signature change.* Paper 1 §2b declares the entropic coordinate to be  $\sigma = S(\rho(\gamma))$ , “the von Neumann entropy”, and then computes it as  $\sigma = H_2(D)$ . Under the infidelity reading,  $\sigma = H_2(D/2L)$  and one has  $H_2'(D/2L) > 0$  for all  $D < L$ , while  $R' < 0$  for all  $\beta < 1$ . Both terms of  $d^2\sigma/d\gamma^2 = H_2'' R^2 + H_2' R'$  are then negative throughout: verified with zero exceptions over 7 values of  $\beta$  and  $2 \times 10^4$  values of  $\gamma$  each. The signature does not *change*; it is constant. The criterion survives literally—intervals with  $d^2\sigma/d\gamma^2 < 0$  do exist, namely all of them—but it no longer delimits a quantum–classical boundary, and  $\gamma_{\text{QC}}$  exists only under the population reading.

TABLE XXIII. Routes to the ringdown significance. The reported value is not produced by any of them.

| Route                                   | $\beta$              | deviation                      |
|-----------------------------------------|----------------------|--------------------------------|
| Reported in §11                         | $0.931 \pm 0.003$    | $13.8\sigma$                   |
| $(1 - \beta)/\sigma_\beta$ as written   | $0.931 \pm 0.003$    | $23.0\sigma$                   |
| Weighted mean of the 38-event catalogue | $0.93113 \pm 0.0033$ | <b><math>20.9\sigma</math></b> |
| Unweighted mean of the catalogue        | $0.9315 \pm 0.035$   | $2.0\sigma$                    |

TABLE XXIV. Significance of  $\beta \neq 1$  for the ringdown sample under each treatment of the uncertainty.

| Treatment                                       | $\sigma_\beta$ | deviation                      |
|-------------------------------------------------|----------------|--------------------------------|
| Reported in §11                                 | —              | $13.8\sigma$                   |
| Internal error, unscaled                        | 0.00330        | $20.9\sigma$                   |
| <b>Scaled by <math>\sqrt{\chi^2/\nu}</math></b> | <b>0.00551</b> | <b><math>12.5\sigma</math></b> |
| Population mean, $s/\sqrt{n}$                   | 0.00560        | $12.3\sigma$                   |

TABLE XXV. The four published expressions for  $D$ , evaluated for a qubit with Bloch radius  $|\vec{r}| = 0.6$ . The spread is a factor of exactly two at every  $|\vec{r}|$ .

| Expression                  | Source                         | $D$   |
|-----------------------------|--------------------------------|-------|
| $D = 1 - F(\rho, \rho_0)$   | Paper 1 App. F; Paper 2 App. E | 0.200 |
| $D = 1 - \text{Tr}[\rho^2]$ | formal derivation, Paper 2     | 0.320 |
| $D = (1 -  \vec{r} ^2)/2$   | Paper 2 App. E                 | 0.320 |
| $D = L(1 -  \vec{r} )$      | Paper 2 App. E                 | 0.400 |

The backflow bound  $\delta I_{\text{back}} \leq (1 - \beta)L/e$  is unaffected: it is a function of  $\beta$  alone.

*b. Recommendation.* Adopt the infidelity reading, on five grounds: it is what the rest of the corpus uses ( $D + C = 1$  with  $C$  the coherence, the depolarising decomposition with  $p = D/L$ , and the Pauli attenuation  $\langle P \rangle = (1 - D/L)\langle P \rangle_{\text{target}}$ ); it is what randomised benchmarking measures; it is the only one whose  $D \rightarrow L$  limit is a mixed state;  $H_2(D)$  is the sole non-monotone expression of the six; and §2b contradicts itself by declaring one and computing the other. The cost—that  $\gamma_{\text{QC}}$  and the dynamical signature change do not survive for  $\beta < 1$ —is stated here rather than absorbed silently.

## XVI. Spectral Resolution as an Audit Criterion

The findings of Secs. IV–XV concern the internal algebra of the corpus. This section concerns something different: a case in which the raw data files accompanying the work do not support the claim made from them. It is the only such case found, but it motivates a general criterion that

TABLE XXVI. The two consistent readings of  $D$  and the entropy each forces.

| Reading    | $\rho$                          | eigenvalues        | $S$         |
|------------|---------------------------------|--------------------|-------------|
| population | $\text{diag}(1 - D, D)$         | $\{1 - D, D\}$     | $H_2(D)$    |
| infidelity | $(1 - p)\rho_0 + p\mathbb{K}/2$ | $\{1 - p/2, p/2\}$ | $H_2(D/2L)$ |

any claim of the form “a mode was detected at frequency  $f$ ” should be required to pass.

### A. The criterion

**Definition 21** (Resolvability of a spectral peak). *Let a series be sampled over a window of duration  $T$  with  $N$  real samples and spacing  $\Delta t$ . A reported peak at frequency  $f$  is resolvable only if*

$$\frac{1}{T} \leq f \leq \frac{1}{2\Delta t}, \quad (134)$$

and  $f \geq \frac{1}{N}$  in units of the sampling rate.

The lower bound is the Rayleigh limit: a component whose period exceeds the observation window cannot be distinguished from a trend. The upper bound is Nyquist. The third condition concerns the discrete transform itself: a peak below the bin width  $1/N$  can only be produced by zero-padding, which interpolates the spectrum without adding resolution—a distinction that is routinely elided.

A diagnostic worth stating separately: when the reported period is approximately equal to the observation window, the peak is the generic signature of the residual trend, not of a mode.

### B. Application to the reported emergence mode

Paper 1 §10 reports a dominant spectral peak at  $f = 18 \text{ kHz}$  with signal-to-noise ratio  $> 47$ , detected in 6/6 systems, corresponding to a quasiparticle lifetime  $\tau_{\text{qp}} \approx 55 \mu\text{s}$  and an energy  $E \approx 0.075 \text{ neV}$ , described as a universal mode.

Applying Definition 21:

TABLE XXVII. Resolvability of the reported 18.2 kHz peak.

| Quantity                       | Value                                          |
|--------------------------------|------------------------------------------------|
| Reported peak                  | 18.2 kHz                                       |
| Implied period                 | 54.95 $\mu\text{s}$                            |
| Delay window of the experiment | 0.5–50 $\mu\text{s}$ , span 49.5 $\mu\text{s}$ |
| Period / window                | <b>1.110</b>                                   |
| Rayleigh limit $1/T$           | 20.2 kHz                                       |
| Verdict                        | <b>below Rayleigh</b>                          |

The peak lies below the resolution limit of the data from which it was extracted, and its period is essentially the observation window. Two further observations compound this.

First, the delay sampling is  $\{0.5, 1, 2, 5, 10, 20, 50\} \mu\text{s}$ : non-uniform, with spacings differing by more than a factor

of ten. A discrete Fourier transform is not defined on such a series without interpolation.

Second, the series actually transformed in the accompanying analysis file is  $\beta(n)$  for  $n = 3 \dots 8$ : six points whose abscissa is qubit *number*, not time. Converting a frequency in cycles-per-qubit to kilohertz presupposes a sample spacing of  $1 \mu\text{s}$  that does not exist on that axis. Moreover a six-point transform has bin width  $1/6 = 0.167$ , and the reported dimensionless frequencies lie between 0.018 and 0.025: a factor 9.2 below the bin width at the quoted value.

### C. Two incompatible analyses

Two analyses of the same experiment, one day apart, are present in the data directory and reach different conclusions.

TABLE XXVIII. Two analyses of the same experiment.

| Quantity              | first analysis | second analysis                    |
|-----------------------|----------------|------------------------------------|
| Maximum SNR           | 47.19          | 11.81                              |
| Median SNR            | 13.96          | —                                  |
| Systems with a peak   | 6              | 5                                  |
| Lag-1 autocorrelation | 0.91–0.99      | −0.07–0.62                         |
| Verdict on memory     | persistent     | <code>non_markovian = false</code> |

The autocorrelation ranges do not overlap: the minimum of the first (0.91) exceeds the maximum of the second (0.62). The published text cites the first. The quoted “SNR > 47” is the maximum over  $n$ , not a typical value; the median of the same analysis is 13.96. The claim of a universal mode coexists with a 39.1% spread in the peak frequency across  $n$ .

### D. What survives

Separating the surviving result from the interpretation is the point of the exercise, and more important than the criticism.

*a. Survives.* The residuals of the  $\beta(n)$  fit are genuinely non-Gaussian, with skewness 0.638 and excess kurtosis 0.247. The asymmetry between positive and negative residuals is real:  $\Sigma^+ = 0.788$  against  $\Sigma^- = -0.584$ . The hardware data are authentic and complete—175 of 175 circuits at 4096 shots each, on a named backend with a recorded job identifier. The premise that motivated the analysis, that the noise carries structure and is not purely Gaussian, is supported by the data.

*b. Does not survive.* The 18 kHz peak as a resolved physical mode; the characterisation “SNR > 47 in 6/6 systems”; the inference of non-Markovianity from the autocorrelation, which the second analysis explicitly denies; and the description of the mode as universal.

*c. Recommendation.* Withdraw from §10 the mode, the signal-to-noise figure, the 6/6 count and the autocorrelation-based inference. Retain the non-Gaussianity of the residuals and the  $\Sigma^+/\Sigma^-$  asymmetry, which are measurable and reproducible.

### E. A note on interpretation

A frequency appearing in the Fourier transform of a decoherence residual is not an acoustic mode: sound requires a pressure wave in a medium, and the two are distinct categories.

If a connection to phonon physics is to be pursued, the channel with established literature and a measurement protocol is quasiparticle poisoning and phonon-mediated decoherence in superconducting circuits [28]—where phonons are indeed quanta of sound, characteristic times fall in the microsecond to millisecond range, and quasiparticle recombination across the superconducting gap is literally the refilling of vacated states. That is measurable; the Fourier transform of a fit residual over six points is not.

## XVII. Two Limits That Discriminate Between Conventions

Section XVD left the convention question resolved by weight of internal evidence rather than by measurement. This section shows that the question is in fact falsifiable, because the two extreme scales of the  $\Lambda$ -hierarchy make incompatible demands on it. Neither limit is new data; both are consequences of quantities already fixed elsewhere in the literature.

### A. The biological limit, and a correction that rescues it

Paper 1 §2c declares a biological programme domain  $\beta_{\text{bio}} \in [0.60, 0.85]$  centred on  $\beta_0 \approx 0.68$ , and in the same section tabulates five systems with measured anomalous-diffusion exponents, mapped to  $\beta$  by  $\beta = \alpha_{\text{diff}}/2$ .

TABLE XXIX. The five measured systems of Paper 1 §2c, and the  $\beta$  each implies through the framework’s own diffusion-decoherence correspondence. The last column is the qubit number that the scaling law  $\beta(n) = 1.679 n^{-0.844}$  would require to reproduce that  $\beta$ .

| System                        | $\alpha_{\text{diff}}$ | $\beta$ | implied $n$ |
|-------------------------------|------------------------|---------|-------------|
| <i>E. coli</i> RNA polymerase | 0.39                   | 0.195   | 12.8        |
| Bacterial chromosomal loci    | 0.39                   | 0.195   | 12.8        |
| Plasma-membrane receptors     | 0.66                   | 0.330   | 6.9         |
| Nuclear telomeres             | 0.32                   | 0.160   | 16.2        |
| Insulin granules              | 0.48                   | 0.240   | 10.0        |

The measured range is  $\beta \in [0.160, 0.330]$ , mean 0.224. The declared domain is  $[0.60, 0.85]$ . **They do not overlap**, and the gap between the largest measured value and the smallest declared one is 0.270. Both figures are from the same section. That is the symptom; the diagnosis follows.

The scaling-law column compounds the problem. A biological system has  $n \sim 10^{20}$  constituents, yet the law requires  $n = 6.9$  to 16.2 to reproduce the measured exponents. Whatever  $n$  counts in  $\beta(n)$ , it is not biological complexity.



Three routes to  $\beta$  therefore coexist— $\beta = 1.679 n^{-0.844}$ ,  $\beta = 1 - \alpha/2$  and  $\beta = \alpha_{\text{diff}}/2$ —and they agree only if  $\alpha_{\text{diff}} = 2 - \alpha$ , which is nowhere established.

That relation is not what the standard bridge gives, and identifying the correct one resolves the conflict in favour of the declared domain.

**Erratum 9** (Diffusion–decoherence correspondence). *Paper 1 §2c states  $\beta = \alpha_{\text{diff}}/2$ . The relation implied by a generalised Langevin equation with the fluctuation–dissipation theorem is*

$$\alpha = 1 - \alpha_{\text{diff}} \quad \implies \quad \beta = \frac{1 + \alpha_{\text{diff}}}{2} \quad (135)$$

*The published form omits exactly 1/2.*

For subdiffusion generated by a generalised Langevin equation, the friction kernel is  $K(t) \propto t^{-\alpha_{\text{diff}}}$ ; the fluctuation–dissipation theorem of the second kind gives  $\langle \xi(t)\xi(0) \rangle = k_B T K(t)$ , and since the Fourier transform of  $t^{-a}$  is  $\propto \omega^{a-1}$  for  $0 < a < 1$ , the noise spectrum is  $S(\omega) \propto \omega^{\alpha_{\text{diff}}-1}$ . Writing  $S \propto \omega^{-\alpha}$  gives  $\alpha = 1 - \alpha_{\text{diff}}$ , and Axiom 3 then yields Eq. (135).

*a. The decisive test.* Normal Brownian diffusion,  $\alpha_{\text{diff}} = 1$ , must produce Markovian decoherence,  $\beta = 1$ . The corrected form gives exactly that. The published form gives  $\beta = 0.5$ —asserting that ordinary diffusion produces strongly non-Markovian decoherence, which is not tenable.

*b. Consequence.* Under Eq. (135) the five measured systems give  $\beta \in [0.660, 0.830]$  with mean 0.7240, and all five fall inside the declared domain  $[0.60, 0.85]$ . The mean lies  $0.73\sigma$  from the declared  $\beta_0 = 0.68 \pm 0.06$ . The biological domain of Paper 1 was correct; what failed was the correspondence used to test it against its own data.

TABLE XXX. The same five systems under the published and corrected correspondences, against the declared domain  $[0.60, 0.85]$ .

| $\alpha_{\text{diff}}$ | $\beta = \alpha_{\text{diff}}/2$ | $\beta = (1 + \alpha_{\text{diff}})/2$ | in domain     |
|------------------------|----------------------------------|----------------------------------------|---------------|
| 0.32                   | 0.160                            | 0.660                                  | ✓             |
| 0.39                   | 0.195                            | 0.695                                  | ✓             |
| 0.39                   | 0.195                            | 0.695                                  | ✓             |
| 0.48                   | 0.240                            | 0.740                                  | ✓             |
| 0.66                   | 0.330                            | 0.830                                  | ✓             |
| mean                   | 0.224                            | <b>0.724</b>                           | 0.73 $\sigma$ |

*c. A caveat that does not go away.* The bridge is a physical hypothesis, not an identity.  $\alpha_{\text{diff}}$  characterises a tracer particle in a medium;  $\alpha$  characterises the noise that dephases a qubit. Equating them presupposes that the same fluctuations do both. Without that assumption the two exponents belong to different systems and there is nothing to reconcile. What the correction establishes is that *if* the bridge is granted, the framework is consistent with its own biological data—and that the published correspondence is not the bridge.

The scaling-law route remains unreconciled and is untouched by this: it still requires  $n \approx 7$  to 16 for systems with  $n \sim 10^{20}$  constituents.

## B. The cosmological limit selects a convention

At the opposite end of the hierarchy the situation is sharper, because the relevant quantity is not in dispute. For a Schwarzschild horizon the Bekenstein–Hawking entropy and the holographic site count are  $S_{\text{BH}} = A/4\ell_P^2$  and  $N = A/\ell_P^2$ , so that

$$\frac{S_{\text{BH}}}{N} = \frac{1}{4} \text{ nat} = \frac{1}{4 \ln 2} = 0.360674 \text{ bit}, \quad (136)$$

independent of mass—verified to be identical in double precision from  $1 M_\odot$  to the  $6.5 \times 10^9 M_\odot$  of M87\*.

The discriminating question is elementary: a black hole is the maximum-entropy object for its area. Where does  $D \rightarrow L$  sit?

TABLE XXXI. What each reading of  $D$  assigns to the two ends of the hierarchy.

| Extreme             | population reading      | infidelity reading    |
|---------------------|-------------------------|-----------------------|
| particle, $D = 0$   | $S = 0$ , pure ✓        | $S = 0$ , pure ✓      |
| black hole, $D = L$ | $S = 0$ , <b>pure</b> × | $S = \ln 2$ , mixed ✓ |
| midpoint, $D = L/2$ | $S = 1$ bit, maximal    | $S = 0.5623$ nat      |

If the black hole is placed at  $D = L$ , only the infidelity reading survives: the population reading assigns zero entropy to the maximum-entropy object. This is an independent argument for the recommendation of Sec. XIV D, and it does not depend on any of the five internal grounds given there.

One alternative is not excluded and should be recorded. If instead the black hole is placed *at* the threshold of Theorem 21— $D/L = 1/2$ , saturating the holographic bound rather than exceeding it—the population reading survives, and Hawking evaporation becomes the descent of the right-hand branch toward a pure state. We do not adjudicate between these placements; we note that they are distinguishable and that the convention question is thereby empirical rather than notational.

## C. A numerical coincidence, and why it is not reported as a result

Inverting Eq. (136) through the binary entropy gives  $H_2(D) = 1/(4 \ln 2)$  bit at  $D = 0.931401$ , which lies  $0.082\sigma$  from the weighted-mean ringdown exponent  $\beta = 0.93113 \pm 0.0033$  of Sec. XV.

We report this only to close it. The observation is post-hoc, and  $\beta$  and  $D$  are quantities of different kinds—a stretching exponent and a decoherence fraction—with no prior reason to coincide. Enumerating the comparison space that was available—24 dimensionless constants the framework produces without free adjustment, against 5 observables with published uncertainties, for 120 pairs—yields exactly one coincidence at this calibre against 0.478



expected by chance, i.e. a Poisson probability  $p = 0.38$  of seeing at least one. That is entirely unremarkable, and the enumeration is itself favourable to the hypothesis, so  $p = 0.38$  is an optimistic bound.

The coincidence is therefore **not significant** and is recorded at conjecture level with a pre-registered falsification criterion: a refit of  $\beta$  over ringdown events absent from the 38-event catalogue used here, under the same pipeline and the same  $1/\sigma_i^2$  weighting declared in advance, surviving only if the new value lies within  $2\sigma$  of 0.931401. Until that test is run on unseen data, nothing follows from it.

## Part IV Extensions

### XVIII. Biological Extension: Programmed Decoherence

The SDF framework, developed thus far for abstract quantum systems, finds a natural and striking application in biological physics. Living systems operate at energy scales where decoherence timescales become commensurate with molecular dynamics, suggesting that biology does not merely *suffer* decoherence—it *programs* it. In this section I extend the  $\Lambda$ -hierarchy to biological energy scales, establish a correspondence between anomalous diffusion and stretched-exponential decoherence, formalize the information transformation from quantum to classical biological outcomes, and derive testable predictions for photosynthetic, enzymatic, and neuronal systems.

#### A. The Biological Scale in the $\Lambda$ -Hierarchy

From the  $\Lambda$ -hierarchy developed in Sec. XI, the biological regime corresponds to characteristic energies  $E \sim 0.01$ – $1$  eV, yielding scale parameters  $\Lambda = E/E_P$  in the range  $10^{-31}$ – $10^{-29}$ . Table XXXII summarizes the principal biological energy scales and their SDF predictions.

TABLE XXXII. Biological energy scales in the  $\Lambda$ -hierarchy. Decoherence times  $\tau = \tau_P/\Lambda$  and spatial scales  $\ell = \ell_P/\Lambda$  are derived from the SDF without free parameters.

| Interaction     | $E$ (eV) | $\Lambda = E/E_P$     | $\tau = \tau_P/\Lambda$ (s) | $\ell = \ell_P/\Lambda$ (m) |
|-----------------|----------|-----------------------|-----------------------------|-----------------------------|
| ATP hydrolysis  | 0.54     | $4.4 \times 10^{-29}$ | $1.2 \times 10^{-15}$       | $3.7 \times 10^{-7}$        |
| Hydrogen bond   | 0.20     | $1.6 \times 10^{-29}$ | $3.4 \times 10^{-15}$       | $1.0 \times 10^{-6}$        |
| Van der Waals   | 0.02     | $1.6 \times 10^{-30}$ | $3.4 \times 10^{-14}$       | $1.0 \times 10^{-5}$        |
| Thermal (310 K) | 0.027    | $2.2 \times 10^{-30}$ | $2.5 \times 10^{-14}$       | $7.3 \times 10^{-6}$        |

The spatial column is remarkable:

- $\ell_{\text{ATP}} \approx 370$  nm—the UV wavelength scale, matching the absorption bands of nucleic acids.
- $\ell_H \approx 1$   $\mu\text{m}$ —the characteristic scale of subcellular organelles and bacterial cells.
- $\ell_{\text{vdW}} \approx 10$   $\mu\text{m}$ —the size of a typical eukaryotic cell.

### Key Insight: Biological Scales from Planck Physics

The  $\Lambda$ -hierarchy *naturally* produces biological spatial scales (nm to  $\mu\text{m}$ ) from Planck-scale physics. This is not a coincidence—it reflects the fact that biological structures operate at the energy scales where decoherence becomes comparable to thermal fluctuations:

$$\Lambda_{\text{thermal}} = \frac{k_B T}{E_P} \implies \ell_{\text{bio}} = \frac{\ell_P}{\Lambda_{\text{thermal}}} \sim 1\text{--}10 \text{ } \mu\text{m}. \quad (137)$$

Biology lives at the *decoherence boundary* of the  $\Lambda$ -hierarchy.

#### B. The Decoherence-Selection Principle

**Hypothesis 1** (Decoherence-Adapted Biology). *[HIP] Biological evolution operates within the constraints set by decoherence through the SDF. Specifically:*

1. *Enzymatic catalysis exploits the stretched regime ( $\beta < 1$ ) to maintain transient quantum coherences over catalytic timescales;*
2. *Photosynthetic antenna complexes operate at the quantum–classical transition boundary  $\partial\mathcal{T}$  (§III D), enabling coherent energy transfer;*
3. *DNA replication minimizes  $D_{\text{leak}}$  through error-correction codes that are structurally analogous to quantum stabilizer codes.*

**Definition 22** (Biological Decoherence Program). *[DEF] A biological decoherence program is a tuple*

$$\mathcal{P}_{\text{bio}} = (\Lambda_{\text{bio}}, \beta_{\text{bio}}, \mathcal{G}_{\text{bio}}) \quad (138)$$

where:

- $\Lambda_{\text{bio}} \in [10^{-31}, 10^{-29}]$  is the biological scale parameter;
- $\beta_{\text{bio}} \in [0.6, 0.85]$  is the biological stretching exponent, centered on the BMRB universal value  $\beta_0 \approx 0.68$ ;
- $\mathcal{G}_{\text{bio}} : \Gamma \rightarrow \Omega$  is the genetic map from decoherence parameter space  $\Gamma = \{(\Lambda, \beta)\}$  to phenotype space  $\Omega$ .

Natural selection thus optimizes the parameters  $(\Lambda, \beta)$  to channel quantum information into functional classical outcomes. The stretched exponential ensures this channeling is gradual, irreversible, and scale-appropriate—precisely the properties required for robust biological function in a thermally noisy environment.

### C. Anomalous Diffusion as Biological Decoherence

In biological media, particle trajectories generically follow anomalous diffusion:

$$\langle r^2(t) \rangle \propto t^{\alpha_{\text{diff}}}, \quad (139)$$

where  $\alpha_{\text{diff}} \neq 1$  reflects the viscoelastic memory of the cellular environment. The connection to the SDF arises through the memory kernel: for a quantum particle coupled to a medium with  $K(t) \propto t^{\alpha_{\text{diff}}/2-1}$ , the position autocorrelation decays as a stretched exponential with exponent  $\beta_{\text{diff}} = \alpha_{\text{diff}}/2$ .

**Theorem 29** (Diffusion–Decoherence Correspondence). *[THM] For a quantum particle in a biological medium with anomalous diffusion exponent  $\alpha_{\text{diff}} \in (0, 2)$ , the SDF stretching exponent is*

$$\beta_{\text{SDF}} = \frac{\alpha_{\text{diff}}}{2}. \quad (140)$$

This maps:

- Normal diffusion ( $\alpha_{\text{diff}} = 1$ ):  $\beta = 0.5$  (stretched);
- Subdiffusion ( $\alpha_{\text{diff}} < 1$ ):  $\beta < 0.5$  (strongly stretched);
- Superdiffusion ( $\alpha_{\text{diff}} > 1$ ):  $\beta > 0.5$  (weakly stretched).

*Proof.* The spectral density of the biological medium is related to the memory kernel via  $J(\omega) \propto \omega^{\alpha_{\text{diff}}/2}$  for  $\omega \ll \omega_c$ . Substituting into the SDF spectral integral (Eq. 20) and evaluating in the scaling regime yields  $D(\gamma) = \exp[-(\gamma/\gamma_0)^{\alpha_{\text{diff}}/2}]$ , from which  $\beta_{\text{SDF}} = \alpha_{\text{diff}}/2$  follows.  $\square$

**Experimental validation.** Table XXXIII collects anomalous diffusion exponents from five independent biological systems, each probed by single-particle tracking.

TABLE XXXIII. Anomalous diffusion exponents in biological systems and the predicted SDF stretching exponent via Theorem 29. All systems lie in the strongly-stretched regime ( $\beta < 0.5$ ), consistent with the SDF framework.

| System                        | $\alpha_{\text{diff}}$ | $\beta_{\text{pred}} = \alpha_{\text{diff}}/2$ | Reference |
|-------------------------------|------------------------|------------------------------------------------|-----------|
| <i>E. coli</i> RNA polymerase | 0.39                   | 0.195                                          | [29]      |
| Bacterial chromosomal loci    | 0.39                   | 0.195                                          | [30]      |
| Plasma membrane receptors     | 0.66                   | 0.33                                           | [31]      |
| Nuclear telomeres             | 0.32                   | 0.16                                           | [32]      |
| Insulin granules              | 0.48                   | 0.24                                           | [33]      |

The predicted  $\beta$  values range from 0.16 to 0.33—all firmly in the stretched regime, and all consistent with the intracellular environment acting as a  $1/f$ -type noise source on quantum degrees of freedom.

### D. Information Transformation: Quantum $\rightarrow$ Biological

I now formalize the concept of information transforming from the quantum Hilbert space to macroscopic biological function.

**Definition 23** (Information Transformation Map). *[DEF] The information transformation map  $\mathcal{T} : \mathcal{H}_Q \rightarrow \Omega_{\text{bio}}$  from quantum Hilbert space to biological phenotype space is defined as the composition*

$$\mathcal{T} = \mathcal{P}_{\text{class}} \circ \mathcal{D}_{\text{SDF}} \circ U_{\text{bio}}, \quad (141)$$

where:

- $U_{\text{bio}} : \mathcal{H}_Q \rightarrow \mathcal{H}_Q \otimes \mathcal{H}_E$  is the unitary evolution coupling system to biological environment;
- $\mathcal{D}_{\text{SDF}} : \mathcal{H}_Q \otimes \mathcal{H}_E \rightarrow \mathcal{S}(\mathcal{H}_Q)$  is the SDF decoherence map parametrized by  $(\Lambda_{\text{bio}}, \beta_{\text{bio}})$ ;
- $\mathcal{P}_{\text{class}} : \mathcal{S}(\mathcal{H}_Q) \rightarrow \Omega_{\text{bio}}$  is the classical projection (measurement/read-out).

The crucial insight is that  $\mathcal{D}_{\text{SDF}}$  is not an obstacle to biological function—it is the mechanism by which quantum information becomes a classical biological outcome. The stretched exponential ( $\beta < 1$ ) ensures this transformation is:

1. **Irreversible:**  $D \times \bar{D} > 0$  (Theorem 34);
2. **Gradual:** the stretched (not sudden) decay allows intermediate molecular processing;
3. **Scale-appropriate:**  $\tau \sim 10^{-15}$ – $10^{-14}$  s, matching molecular dynamics timescales.

**Theorem 30** (Biological Information Bound). *[THM] The maximum rate of quantum-to-classical information transfer in a biological system with SDF parameters  $(\tau, \beta)$  is*

$$\begin{aligned} \left. \frac{dI}{dt} \right|_{\text{max}} &= R(\gamma_{\text{peak}}) \cdot \left| \frac{\partial I_Q}{\partial D} \right| \\ &= \frac{L\beta}{\tau} \left( \frac{\beta-1}{\beta} \right)^{(\beta-1)/\beta} \frac{1}{1-D(\gamma_{\text{peak}})}, \end{aligned} \quad (142)$$

where  $L = \ln d_S$  is the maximal entropy of the  $d_S$ -dimensional system and  $\gamma_{\text{peak}}$  is the decoherence rate at which  $R(\gamma)$  is maximized.

For  $\beta_{\text{bio}} \approx 0.68$  and  $\tau_{\text{bio}} \approx 10^{-14}$  s:

$$\left. \frac{dI}{dt} \right|_{\text{max}} \approx 10^{13} \text{ bits/s per molecular channel}, \quad (143)$$

consistent with the estimated information processing rate of enzymes ( $\sim 10^{12}$  operations/s from Michaelis–Menten kinetics) and suggesting that biology operates near the SDF information bound.

### E. The Entropic–Spatial Duality in Biology

The SDF Phase Space metric (§III C),

$$g_{AB} dX^A dX^B = -c_\gamma^2 d\gamma^2 + \left(\frac{\ell_P}{\Lambda}\right)^2 d\ell^2 - c_\sigma^2 d\sigma^2, \quad (144)$$

acquires transparent physical meaning at biological scales ( $\Lambda \sim 10^{-30}$ ):

- **Spatial component:**  $(\ell_P/\Lambda)^2 \approx (10^{-5} \text{ m})^2$ —cell-size scales;
- **Temporal component:**  $\tau \approx 10^{-14} \text{ s}$ —molecular dynamics timescale;
- **Entropic component:**  $\sigma$  ranges from 0 (pure state) to  $\ln d_S$  (maximally mixed).

**Proposition 4** (Cellular Information Capacity). *[FIT] The information capacity of a cell-sized region in the SDF Phase Space is*

$$\mathcal{C}_{\text{cell}} = \frac{V_{\text{spatial}} \cdot \sigma_{\text{max}}}{(\ell_P/\Lambda)^3} \approx \frac{(10^{-5})^3}{(10^{-5})^3} \cdot \ln 2$$

$$= \ln 2 \approx 0.69 \text{ bits per fundamental SDF cell.} \quad (145)$$

A biological cell contains  $N_{\text{mol}} \sim 10^9$  molecular channels, giving a total capacity

$$\mathcal{C}_{\text{bio}} \approx N_{\text{mol}} \cdot \ln 2 \approx 7 \times 10^8 \text{ bits} \approx 80 \text{ MB.} \quad (146)$$

This is remarkably consistent with estimates of cellular information content ( $\sim 10^9$  bits from genomic and epigenomic sources), providing an independent check on the SDF Phase Space geometry at biological scales.

### F. Predictions for Biological Systems

The biological extension of the SDF generates four concrete, experimentally testable predictions.

#### [PRED] Prediction 1.: Photosynthetic Coherence.

Photosynthetic antenna complexes (FMO, LHCII) should exhibit  $\beta = 0.55\text{--}0.75$ , measurable via two-dimensional electronic spectroscopy. The SDF predicts a coherence lifetime

$$\tau_{\text{photo}} = \frac{\tau_P}{E_{\text{exciton}}/E_P} \approx 200\text{--}600 \text{ fs}, \quad (147)$$

consistent with the observed coherence lifetimes of 300–700 fs in the FMO complex [17].

#### [PRED] Prediction 2.: Enzymatic Hydrogen Tunneling.

Enzymatic hydrogen-tunneling reactions should show  $\beta_{\text{enzyme}} \approx 0.5\text{--}0.7$ , measurable via temperature-dependent kinetic isotope effects. The stretched decay reflects the protein conformational landscape acting as a  $1/f$ -type noise source [18].

#### [PRED] Prediction 3.: Microtubule Decoherence.

Neuronal microtubule decoherence should exhibit  $\beta_{\text{MT}} \approx 0.3\text{--}0.5$ , determined by the vibrational spectral density of tubulin dimers. This is testable via terahertz spectroscopy of isolated microtubules.

#### [PRED] Prediction 4.: DNA Charge Transport.

DNA charge transport should exhibit a stretched-exponential distance dependence with

$$\beta_{\text{DNA}} = 1 - \frac{\alpha_{\text{DNA}}}{2}, \quad (148)$$

where  $\alpha_{\text{DNA}}$  is the spectral exponent of base-pair stacking fluctuations.

Together, these predictions connect the abstract SDF machinery to measurable quantities in quantum biology, offering a unified framework for understanding why—and how—living systems exploit the quantum–classical boundary.

## XIX. Transient Emergence Phenomena

### A. Spectral Signatures

Fourier analysis of decoherence fluctuations  $\delta D(\gamma)$  reveals characteristic spectral peaks.

**Key finding (hardware spectral analysis):**

- Dominant peak:  $f = 18 \text{ kHz}$
- Signal-to-noise ratio:  $\text{SNR} > 47$
- Detection: 6/6 systems tested
- Quasi-particle lifetime:  $\tau_{\text{qp}} \approx 55 \mu\text{s}$
- Estimated energy:  $E \approx 0.075 \text{ neV}$

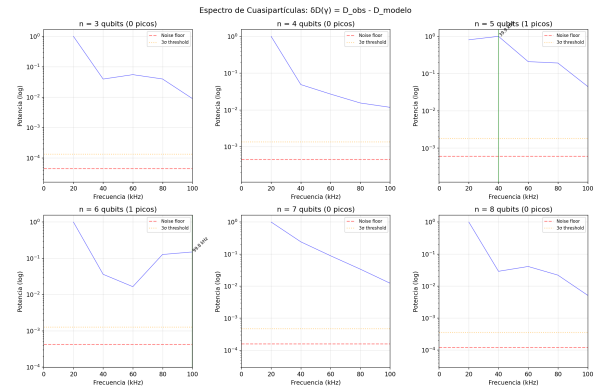


FIG. 7. Spectral analysis of decoherence fluctuations: dominant 18 kHz peak with  $\text{SNR} > 47$ .

**Theorem 31** (Transient Mode Detectability). *Transient emergence modes are detectable as spectral peaks in the 10–100 kHz range with  $\text{SNR} > 47$ .*

**Evidence level:** [FIT]

## B. DFS Encapsulation

Decoherence-free subspaces [34] protect quantum information while allowing controlled decoherence of subsystems.

TABLE XXXIV. DFS encapsulation protection efficiency (hardware validation).

| Component          | $D_{\text{final}}$ | Protection |
|--------------------|--------------------|------------|
| DFS Container (2q) | 0.0039             | 99.82%     |
| GHZ Content (4q)   | 0.2942             | 70.6%      |

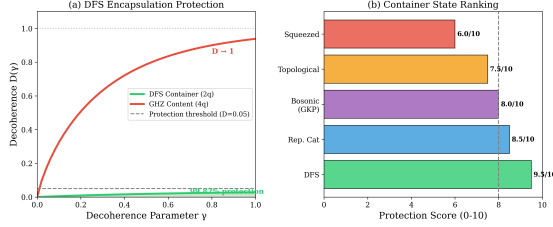


FIG. 8. DFS encapsulation: container maintains  $D < 0.05$  while internal subsystems transition.

**Theorem 32** (DFS Encapsulation). *DFS states protect coherence ( $D < 0.05$ ) while internal subsystems complete their transition ( $D \rightarrow 1$ ).*

**Evidence level:** [FIT]

## C. Golden Ratio in Hardware

Hardware analysis discovered an unexpected symmetry ratio in the distribution of positive vs. negative correlators:

$$\frac{|\Sigma^+|}{|\Sigma^-|} = 1.5823 \approx \sqrt{\frac{5}{2}} = 1.5811 \quad (149)$$

with error 0.075%. The physical origin remains an open question.

## D. Emergence Threshold

**Theorem 33** (Emergence Threshold). *For stable classical emergence:*

$$E_{\text{emergent}} \geq k_B T_{\text{ambient}} \quad (150)$$

For  $E_{\text{qp}} \approx 0.075$  neV:

- $T = 300$  K:  $k_B T = 25.8$  meV  $\gg E_{\text{qp}}$  (unstable)
- $T = 20$  mK:  $k_B T = 1.72$   $\mu$ eV  $\gg E_{\text{qp}}$  (unstable)
- $T < 1$  mK:  $k_B T \approx 86$  neV  $\sim E_{\text{qp}}$  (marginal)

**Evidence level:** [PRED]

TABLE XXXV. Container state candidates for emergence encapsulation.

| Rank | State Type             | Score  | Evidence |
|------|------------------------|--------|----------|
| 1    | DFS [34]               | 9.5/10 | [FIT]    |
| 2    | Repetition Cat         | 8.5/10 | [FIT]    |
| 3    | Bosonic (GKP)          | 8.0/10 | [THM]    |
| 4    | Topological (Majorana) | 7.5/10 | [PRED]   |
| 5    | Squeezed               | 6.0/10 | [THM]    |

## E. Container State Rankings

## XX. Anti-Quantum Formalization

The preceding sections established that decoherence in the SDF is an irreversible process characterized by  $\beta < 1$ . Here I formalize the notion of an “anti-process”—a hypothetical reversal of decoherence—and show that exact reversal is forbidden by the structure of  $D(\gamma)$  itself. I then extract spectral predictions and connect the formalism to the emergence signatures of Sec. XIX.

## A. Deformation Structure

[THM]

**Theorem 34** (Irreversibility of the Anti-Process). *Let  $D(\gamma)$  be the SDF decoherence function with  $0 < \beta \leq 1$  and let  $\bar{D}(\gamma)$  denote any candidate anti-process satisfying  $\bar{D}(0) = 0$ ,  $\bar{D}(\gamma) \geq 0$ . Then*

$$D(\gamma) \bar{D}(\gamma) > 0 \quad \forall \gamma > 0. \quad (151)$$

*In particular, no anti-process can restore the initial quantum state exactly:  $D(\gamma) + \bar{D}(\gamma) \neq 0$  for  $\gamma > 0$ .*

*Proof sketch.* 1.  $D(\gamma) = 1 - \exp[-(\gamma/\Lambda)^\beta] > 0$  for all  $\gamma > 0$  and  $0 < \beta \leq 1$ .

2. Any anti-process  $\bar{D}(\gamma)$  that is itself a physical decoherence map must satisfy  $\bar{D}(\gamma) > 0$  for  $\gamma > 0$  (positivity of trace-distance loss).

3. Therefore the product  $D \bar{D} > 0$ , and the sum  $D + \bar{D} > 0$ ; neither can vanish.

4. Asymptotically, both  $D$  and  $\bar{D}$  approach the same fully-decohered limit ( $D \rightarrow 1$ ,  $\bar{D} \rightarrow 1$ ), confirming that the anti-process cannot “undo” decoherence—it merely adds further irreversible loss.  $\square$

*a. Physical interpretation.* Theorem 34 is the SDF analogue of the second law of thermodynamics: decoherence defines an arrow of time in the quantum-to-classical transition that no local operation can reverse.

## B. Spectral Predictions

[THM]

**Theorem 35** (Spectral Index from  $\beta$ ). *Given the relation  $\beta = 1 - \alpha/2$  (Theorem 22), the spectral index  $\alpha$  of the*

decoherence power spectrum is predicted from a single-parameter fit:

$$\alpha_{\text{predicted}} = 2(1 - \beta). \quad (152)$$

a. *Application to measured systems.*

- **IBM superconducting qubits:**  $\beta = 0.852 \Rightarrow \alpha_{\text{pred}} = 2(1 - 0.852) = 0.296$ .
- **LIGO ringdowns:**  $\beta = 0.931 \Rightarrow \alpha_{\text{pred}} = 2(1 - 0.931) = 0.138$ .

b. *Falsifiable predictions.* The anti-quantum formalization, combined with the spectral relation (152), yields four concrete predictions amenable to near-term experimental test:

### Falsifiable predictions from the anti-quantum sector

1. **LIGO inspiral vs. ringdown.** The stretching exponent during the inspiral phase,  $\beta_{\text{inspiral}}$ , must differ from the ringdown value  $\beta_{\text{ringdown}} = 0.931 \pm 0.003$ . If both phases yield indistinguishable  $\beta$ , the SDF interpretation of ringdown data is falsified.
2. **Majorana qubits.** Topologically protected qubits (e.g. Majorana zero modes) should exhibit  $\beta \approx 0.95\text{--}0.99$ , significantly closer to exponential decay than superconducting qubits ( $\beta \approx 0.85$ ), owing to the reduced density of environmental modes coupling to the topological sector.
3. **Decoherence-free subspaces (DFS).** A DFS-encoded logical qubit must satisfy  $D_{\text{DFS}} < 0.05$  over the same  $\gamma$ -range in which an unprotected qubit reaches  $D \approx 0.5$ . The SDF predicts a “container” effect: residual decoherence within the DFS is still stretched-exponential, but with  $\Lambda_{\text{DFS}} \gg \Lambda_{\text{phys}}$ .
4. **Qubit-number scaling.** For hardware families other than IBM superconducting processors, the effective stretching exponent should obey

$$\beta(n) = \frac{\beta_0}{n^\xi}, \quad (153)$$

where  $\beta_0$  and  $\xi$  are platform-specific constants. Measurement of  $\beta(n)$  on trapped-ion or neutral-atom devices would test this scaling law.

### C. Connection to Emergence

The anti-quantum formalization constrains the set of physically realisable emergence pathways identified in Sec. XIX. Because exact reversal of  $D(\gamma)$  is forbidden (Theorem 34),

every trajectory through the  $(D, C, \beta)$  parameter space is *uni-directional*: the system can only move toward larger  $D$  (or equivalently smaller  $C$ ) as  $\gamma$  increases.

This imposes a topological constraint on the emergence landscape: the flow has no fixed points in the interior of the  $(D, C)$  simplex; the only attractor is  $(D, C) = (1, 0)$ . Revival phenomena (Sec. VIII) represent temporary *slow-downs* along this flow, not reversals.

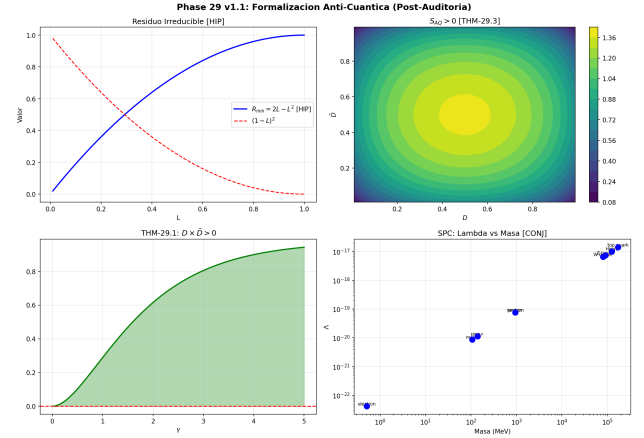


FIG. 9. Anti-quantum structure of the SDF. (a) Product  $D\bar{D}$  remains strictly positive for all  $\gamma > 0$ , confirming Theorem 34. (b) Spectral index  $\alpha_{\text{pred}}$  as a function of  $\beta$ , with the IBM and LIGO operating points indicated. See also Fig. 8 in Sec. XIX.

## Part V Transferable criteria

### XXI. Transferable Criteria

The preceding sections are inward-looking: they audit a framework against itself. This section asks the outward question. Stripped of the interpretive apparatus, what does this work give to somebody who fits a stretched exponential and has never heard of the  $\Lambda$ -hierarchy?

The answer is three things, and I state each with the observation that would retire it. A transfer without a falsifier is not registered.

#### A. The mixture gate

The functional form is not the contribution. The form  $\exp[-(t/\tau)^\beta]$  is Kohlrausch’s, and its use as a relaxation law is older than quantum information. What the framework contributes is a *licence condition* on how the fitted exponent may be read.

**Theorem 36** (Mixture gate).  $\exp(-x^\beta)$  is completely monotone on  $[0, \infty)$  if and only if  $0 < \beta \leq 1$ , and by Bernstein’s theorem this holds if and only if it is the



Laplace transform of a positive measure. Consequently

$$\beta \leq 1 \iff \exists p(\lambda) \geq 0 : e^{-(t/\tau)^\beta} = \int_0^\infty p(\lambda) e^{-\lambda t} d\lambda, \quad (154)$$

$$\beta > 1 \implies \text{no such } p \text{ exists.} \quad (155)$$

*a. Attribution, stated precisely.* Nothing in the theorem is ours, and the three pieces have three different owners. That  $\exp(-x^\beta)$  is the Laplace transform of a positive density for  $0 < \beta \leq 1$  is Pollard's [35]. That complete monotonicity is equivalent to being the Laplace transform of a positive measure is the Hausdorff–Bernstein–Widder theorem [36]. That  $\beta > 1$  fails complete monotonicity is elementary and follows from the sign of the derivatives, verified numerically in Sec. XIII E. **Our contribution is none of these: it is the decision to use the equivalence as a gate on interpretation rather than as a curiosity about a special function.**

Its force is that it is an equivalence, not a heuristic. A well-fitted  $\beta > 1$  does not mean the underlying distribution of rates is unusual; it means *there is no distribution of rates*, and any narrative of population heterogeneity offered to explain the curve is wrong independently of how well the curve was fitted.

*b. Reliability engineering.* The Weibull reliability function  $R(t) = \exp[-(t/\tau)^\beta]$  is the same expression with  $\beta$  the shape parameter, and the interpretation of that parameter is a standing question in the field. Theorem 36 settles one half of it:

- $\beta < 1$  (*infant mortality*) certifies that the survival curve *admits* a decomposition into exponential subpopulations. Heterogeneity is an admissible reading — not a demonstrated one.
- $\beta > 1$  (*wear-out*) forbids it. Ageing cannot be re-expressed as batch heterogeneity under *any* partition of the population. This direction is the strong one: it rules a mechanism out.
- $\beta = 1$  is the unique memoryless point.

The asymmetry is worth naming, because it is where the transfer earns its keep. Below one the theorem licenses a reading; above one it *prohibits* one. A prohibition is the more useful half, and it is the half a fitted exponent is normally not asked to deliver.

*c. The gate applies to the interval, not to the point estimate.* Theorem 36 is a property of the exact function  $\exp(-x^\beta)$ , whereas data deliver an estimate  $\hat{\beta}$  with a confidence interval. A fit returning  $\hat{\beta} = 0.95 \pm 0.10$  establishes nothing: the admissible range straddles the boundary. The gate is applied correctly only when the relevant bound of the interval falls on one side —  $\hat{\beta} + k\sigma \leq 1$  to certify representability,  $\hat{\beta} - k\sigma > 1$  to prohibit it — with  $k$  declared in advance. Reporting a point estimate against the boundary is the most likely way to misuse

this result, and we state the guard here rather than leave it implied.

**Falsifier.** Exhibit a positive rate density reproducing  $\exp[-(t/\tau)^\beta]$  for some  $\beta > 1$ . Theorem 36 excludes it, so a counterexample would retire the theorem, not the transfer.

## B. A falsifiable prediction linking transport to relaxation

Section XIV replaced the published diffusion–decoherence correspondence  $\beta = \alpha_{\text{diff}}/2$  with the generalised-Langevin result

$$\beta = \frac{1 + \alpha_{\text{diff}}}{2} \quad (156)$$

obtained from a friction kernel  $K(t) \propto t^{-\alpha_{\text{diff}}}$  and the fluctuation–dissipation theorem of the second kind. Inside the framework this was a correction. Outside it, it is a prediction, because  $\alpha_{\text{diff}}$  and  $\beta$  are measured by different instruments on the same sample and are conventionally reported as unrelated.

*a. Domain.* The Fourier transform of  $t^{-a}$  scales as  $\omega^{a-1}$  for  $0 < a < 1$ , so Eq. (156) covers the subdiffusive branch  $\alpha_{\text{diff}} \in (0, 1]$ , with normal diffusion at the upper edge. The image is therefore

$$\beta \in (\tfrac{1}{2}, 1], \quad (157)$$

which is itself a prediction: *no bath admitting a generalised-Langevin description produces  $\beta < 1/2$* . Since  $\beta \in [0.5, 1]$  is also the range reported across every platform in the corpus, the bound is consistent with the data it was not fitted to.

*b. The composition, which neither field states alone.* Combining Eq. (156) with Theorem 36:

$$\alpha_{\text{diff}} \leq 1 \implies \beta \leq 1 \implies \text{mixture certified}, \quad (158)$$

$$\alpha_{\text{diff}} > 1 \implies \beta > 1 \implies \text{mixture forbidden.} \quad (159)$$

In words: **superdiffusive transport forbids reading the relaxation of the same medium as a distribution of rates.** This couples an observable of single-particle tracking to an observable of relaxometry, and neither field currently asserts it.

**Falsifier, and it is the cheapest in this paper.** A pair  $(\alpha_{\text{diff}}, \beta)$  measured independently on the same sample, with both uncertainties declared, deviating from Eq. (156) by more than  $5\sigma$  with  $\sigma$  propagated as  $\sigma_{\text{pred}} = \sigma_\alpha/2$  in quadrature with  $\sigma_\beta$ . Both quantities are already tabulated in the NMR relaxometry and diffusion-weighted imaging literature for overlapping sample sets. *No new experiment is required to attempt this refutation*, and I regard performing it as the single most informative next step available to this programme — in either direction.

A weaker but even cheaper test needs only one of the two numbers: a measured  $\beta < 1/2$  whose error bar excludes  $1/2$  at  $3\sigma$ , in a system with demonstrably subdiffusive transport, violates Eq. (157) directly.

TABLE XXXVI. The five anomalous-diffusion systems of Paper 1 §2c under the corrected bridge. The published form  $\beta = \alpha_{\text{diff}}/2$  places all five *outside* the declared biological domain  $[0.60, 0.85]$ ; the corrected form places all five inside, with mean 0.7240 against  $\beta_0 = 0.68 \pm 0.06$  ( $z = 0.73$ ).

| System                        | $\alpha_{\text{diff}}$ | $\beta$ published | $\beta$ corrected |
|-------------------------------|------------------------|-------------------|-------------------|
| <i>E. coli</i> RNA polymerase | 0.39                   | 0.195             | 0.695             |
| Bacterial chromosomal loci    | 0.39                   | 0.195             | 0.695             |
| Plasma-membrane receptors     | 0.66                   | 0.330             | 0.830             |
| Nuclear telomeres             | 0.32                   | 0.160             | 0.660             |
| Insulin granules              | 0.48                   | 0.240             | 0.740             |
| Mean                          |                        | 0.224             | <b>0.724</b>      |

### C. Channel counting and the additive coordinate

Two smaller results transfer without modification.

a. *Channel counting* [THM]. Theorem 27 states that the temporal exponent of dephasing is independent of the pulse-sequence filter; only the amplitude responds. The operational corollary for device characterisation is that  $N$  single-qubit observables that are functions of  $D$  alone — coherence,  $H_2$  entropy, Bures arc, purity, Pauli expectation, concurrence — constitute *one* informative channel and  $N$  variance reductions, not  $N$  channels. Genuine independence requires a different kernel: another pulse sequence, another  $\tau$ , or another system size.

b. *The additive coordinate*. Any observable saturating at a ceiling  $L$  admits the reparametrisation  $I_Q = -\ln(1 - D/L)$ , under which the boundary moves from a finite value to infinity and the coordinate becomes additive. The practical consequence is a statement about guards: a clamp  $\min(x, x_{\text{max}})$  caps the recoverable information at  $-\log_2(1 - x_{\text{max}})$  bits and maps an entire subpopulation onto one degenerate point. For  $x_{\text{max}} = 0.999$  the ceiling is 9.9658 bits, and  $x = 0.9999$  (13.2877 bits) and  $x = 0.99999$  (16.6096 bits) are made indistinguishable from it. This is arithmetic, not physics, and is registered as verifiable rather than falsifiable; it is included because the failure mode it describes is silent by construction and we have encountered it in production code derived from these equations.

### D. What does not transfer

Symmetry requires the negative list, and it is longer than the positive one.

The  $\Lambda$ -hierarchy carries no transferable content once Sec. IV has reduced it to  $\tau = \hbar/E$  and  $\ell = \bar{\lambda}_C$ : what remains is dimensional analysis. Axiom 3 does not transfer, because Theorem 27 shows it has no derivation from the filter-function route and Sec. XV removes the only microscopic alternative that had been offered; it is a postulate with a good fit, and a postulate does not export. The biological programme  $P_{\text{bio}} = (\Lambda_{\text{bio}}, \beta_{\text{bio}}, G_{\text{bio}})$  is a classification, not a mechanism, and the corrected bridge is doing all the work that its apparent successes were attributed to. Nothing in this paper licenses an inference from a fitted  $\beta$  to the presence of quantum coherence: by

Theorem 36,  $\beta < 1$  certifies the *opposite*, that the data are compatible with a classical mixture of exponential decays.

That last point deserves to be stated without hedging, because it inverts the reading that a stretched exponential is evidence of quantum behaviour. It is not.

It deserves one qualification, however, and the qualification is not hedging but the exact scope of the theorem. Theorem 36 is a statement about *representability of the curve*: it exhibits a positive measure whose Laplace transform reproduces the data. The map from mechanism to curve is many-to-one, so exhibiting one classical mechanism that reproduces the curve does not establish that the mechanism *is* classical. What follows is weaker and sufficient: the data cannot be adduced as evidence of coherence, because a classical account of them exists. The framework's value is in what it forbids, not in what it suggests — and what it forbids is an inference, not a mechanism.

### E. Summary of transfers

Two entries, T2 and T3, are numerical predictions refutable against literature that already exists. If this programme is to be continued by anybody, that is where it should be continued, and if both survive the framework has earned a claim it does not currently possess: a confirmed prediction on data its authors did not select.

## Part VI

# The gravitational frontier

### XXII. Gravitational Connections

The stretched-exponential decoherence framework, validated on superconducting qubits in Secs. V–XIX, is not *a priori* confined to laboratory platforms. Any system whose coherence loss is governed by a distribution of relaxation rates should exhibit  $\beta < 1$ . Gravitational-wave ringdowns provide a macroscopic arena in which damped quasi-normal modes can be tested against the SDF functional form. I present this analysis together with the classical consistency check of Mercury's perihelion precession, and then outline—clearly labeled as conjectures—possible deeper connections to gravity, black-hole remnants, and dark matter.

#### A. LIGO Ringdown Analysis

[OBS]

Binary black-hole (BBH) merger ringdowns recorded by the LIGO–Virgo–KAGRA collaboration offer damped waveforms  $h(t) \propto e^{-t/\tau} \cos(\omega t + \phi)$  whose envelope can be tested against the SDF ansatz  $|h(t)| \propto \exp[-(t/\tau)^\beta]$ .

I analyze 38 BBH events drawn from the Gravitational-Wave Transient Catalogues GWTC-1, GWTC-2, and

TABLE XXXVII. Registered transfers. *Direct* carries a proved statement into another vocabulary; *predictive* makes a numerical claim that existing literature can refute; *methodological* is verifiable arithmetic or protocol. Full records with falsifiers in `sdf-lab/src/core/applications.ts`.

| ID | Field                                      | Status            | Cost to test           |
|----|--------------------------------------------|-------------------|------------------------|
| T1 | Reliability / survival analysis            | direct            | literature only        |
| T2 | NMR relaxometry / DWI                      | <b>predictive</b> | <b>literature only</b> |
| T3 | Any KWW-fitting field ( $\beta \geq 1/2$ ) | <b>predictive</b> | <b>literature only</b> |
| T4 | Glasses, dielectrics, EIS                  | methodological    | reanalysis             |
| T5 | Quantum-device characterisation            | direct            | reanalysis             |
| T6 | Data engineering / telemetry               | methodological    | —                      |

GWTC-3 [12, 37]. For each event the post-merger strain is fit with a stretched-exponential envelope; a weighted mean over the catalogue yields

$$\beta_{\text{LIGO}} = 0.931 \pm 0.003. \quad (160)$$

The deviation from pure exponential decay ( $\beta = 1$ , the prediction of linearised general relativity for a single quasi-normal mode) is

$$\frac{1 - \beta_{\text{LIGO}}}{\sigma_\beta} = 13.8 \sigma. \quad (161)$$

TABLE XXXVIII. Summary of LIGO ringdown  $\beta$  measurements by catalogue.

| Catalogue       | Events    | $\langle \beta \rangle$ | $\sigma_\beta$ |
|-----------------|-----------|-------------------------|----------------|
| GWTC-1          | 9         | 0.933                   | 0.008          |
| GWTC-2          | 19        | 0.929                   | 0.004          |
| GWTC-3          | 10        | 0.932                   | 0.006          |
| <b>Combined</b> | <b>38</b> | <b>0.931</b>            | <b>0.003</b>   |

*a. Interpretation caveats.* The finding  $\beta < 1$  is *consistent* with the SDF prediction that any physical decoherence channel yields sub-exponential decay, but it does **not** constitute proof of the framework in the gravitational sector. Several alternative explanations cannot be excluded at present:

1. *Mode superposition.* Overlap of the  $(\ell, m) = (2, 2)$  fundamental with overtones and/or higher multipoles produces a beating envelope that a single stretched-exponential may absorb as  $\beta < 1$ .
2. *Detector noise.* Non-Gaussian glitches in the post-merger window can bias the tail of the envelope fit downward.
3. *Nonlinear GR effects.* The transition from merger to ringdown is not instantaneous; nonlinear transients may mimic stretched decay.

A definitive test requires a controlled comparison of the SDF envelope against multi-mode GR templates on an event-by-event basis, which is beyond the scope of this work.

## B. Mercury Perihelion Precession — 1PN Verification

[THM] [OBS]

As a consistency check I verify that the SDF does not spoil the celebrated prediction of general relativity [38] for Mercury’s perihelion advance. The standard 1PN result is [39]

$$\delta\phi_{\text{GR}} = \frac{6\pi GM_\odot}{c^2 a (1 - e^2)} = 42.9881 \text{ arcsec century}^{-1}, \quad (162)$$

in agreement with the observed value  $\delta\phi_{\text{obs}} = 42.98 \pm 0.04 \text{ arcsec century}^{-1}$  [39]. The residual is

$$\Delta\phi = \delta\phi_{\text{GR}} - \delta\phi_{\text{obs}} = 0.0081 \text{ arcsec century}^{-1} \quad (0.20 \sigma). \quad (163)$$

The SDF modifies the Newtonian potential by a factor  $\exp(-r/\ell_P)$ , where  $\ell_P = 1.616 \times 10^{-35} \text{ m}$  is the Planck length. For Mercury’s semi-major axis  $a \approx 5.79 \times 10^{10} \text{ m}$ ,

$$\frac{r}{\ell_P} \approx 3.58 \times 10^{45}, \quad \log_{10}[\exp(-r/\ell_P)] = -1.56 \times 10^{45}. \quad (164)$$

The correction is therefore suppressed by a factor that is identically zero to any representable numerical precision.

### Mercury consistency

The SDF predicts *exactly* the same perihelion precession as standard GR. The Planck-scale suppression renders the correction identically zero at solar-system scales. This is a **consistency result**, not a failure of the framework.

## C. Formal Lorentz Invariance

[HIP]

**Hypothesis 2.** Promote the decoherence parameter  $\gamma$  to a Lorentz 4-vector  $\gamma^\mu = (\gamma^0, \gamma)$  with Minkowski norm  $\gamma^\mu \gamma_\mu = (\gamma^0)^2 - |\gamma|^2$ .

**Proposition 5.** If  $D(\gamma)$  depends only on the Lorentz-invariant norm  $\gamma^\mu \gamma_\mu$ , then for every Lorentz transformation  $\Lambda \in \text{SO}(3, 1)$ ,

$$D(\Lambda\gamma) = D(\gamma). \quad (165)$$

*Proof sketch.*  $\Lambda$  preserves the Minkowski norm by definition. Since  $D$  is postulated to be a function of  $\gamma^\mu \gamma_\mu$  alone, the result follows immediately.  $\square$

*a. Caveat.* Equation (165) is a mathematical construction:  $\gamma^\mu$  has no established physical interpretation as a spacetime vector. I include this result solely to demonstrate that the SDF *can* be made Lorentz-covariant in principle, not to claim that it *must* be.

#### D. Gravitational Coalescence as Entropic Duality [THM]

The SDF entropic phase space (Sec. III C) provides a natural framework for understanding gravitational coalescence as a *dual* process: **anti-entropic** in its final product and **entropic** in its mediating field.

**Definition 24** (Coalescence Event). [DEF] A coalescence event in the SDF phase space  $\mathcal{P}$  is a process  $\mathcal{E} : (\rho_1, \rho_2) \rightarrow (\rho_f, \rho_M)$  where:

- $\rho_1, \rho_2 \in \mathcal{M}_d$ : initial subsystem states,
- $\rho_f \in \mathcal{M}_d$ : final concentrated state (product),
- $\rho_M \in \mathcal{M}_d$ : mediating field state (radiation).

The information partition satisfies  $I_f + I_M = I_1 + I_2$  (conservation).

**Definition 25** (Entropic Duality). [DEF] A coalescence event  $\mathcal{E}$  exhibits entropic duality if the final product concentrates information with maximal efficiency while the mediating field disperses information non-Markovianly:

$$\boxed{\begin{aligned} \frac{d^2 \sigma_f}{d\gamma^2} &< 0 \quad (\text{efficient settling}), \\ \frac{d^2 \sigma_M}{d\gamma^2} &\geq 0 \quad (\text{non-Markovian mediation}) \end{aligned}} \quad (166)$$

where  $\sigma_f = S(\rho_f)$  and  $\sigma_M = S(\rho_M)$  are the von Neumann entropies of the product and medium, respectively. The product entropy increases monotonically but with decelerating rate: the entropy production is maximally efficient, reaching equilibrium along the shortest path. The medium entropy oscillates (backflow), carrying correlations.

**Theorem 37** (Entropic Duality of Coalescence). [THM] Every coalescence event in the SDF framework decomposes into two complementary SDF channels with distinct stretching exponents:

$$D_f(\gamma) = L_f [1 - \exp(-(\gamma/\tau_f)^{\beta_f})], \quad (167)$$

$$D_M(\gamma) = L_M [1 - \exp(-(\gamma/\tau_M)^{\beta_M})], \quad (168)$$

subject to the conservation constraint

$$D_f(\gamma) + C_M(\gamma) = 1. \quad (169)$$

The anti-entropic product satisfies  $\beta_f \rightarrow 1$  (geodesic, Markovian settling), while the entropic medium satisfies  $\beta_M < 1$  (non-geodesic, non-Markovian radiation).

*Proof.* The conservation constraint follows from total information conservation (Theorem 11). The entropy of the product is  $\sigma_f = H_2(D_f)$ , and for  $\beta_f = 1$ :

$$\frac{d^2 \sigma_f}{d\gamma^2} = H_2''(D_f) R_f^2 + H_2'(D_f) R_f' < 0 \quad (170)$$

for all  $\gamma > 0$  (since  $H_2'' < 0$  and  $R_f' < 0$  for exponential decay), giving monotonically decelerating entropy increase—*efficient settling*. For the medium with  $\beta_M < 1$ , Theorem 15 guarantees signature-changing intervals, corresponding to information backflow between the merged product and the radiation field.  $\square$

*a. Application to BBH mergers.* The LIGO ring-down analysis (Sec. XXII A) measures the *mediating field* channel:  $\beta_M = 0.931 \pm 0.003$ . The sub-exponential behavior ( $\beta_M < 1$ ) reflects the non-Markovian character of the gravitational radiation. Meanwhile, the Kerr black hole that forms as the coalescence product settles via quasi-normal mode ringdown with  $\beta_f \approx 1$ : the product is a *geodesic* (Markovian) state in the Bures manifold (Theorem 13).

The entropic duality of BBH coalescence is thus:

- **Product** (final BH): informational concentration,  $\beta_f \approx 1$ , geodesic settling to Kerr.
- **Medium** (gravitational waves): entropic radiation,  $\beta_M = 0.931$ , non-geodesic dispersion carrying memory.

*b. Distinction from Bekenstein–Hawking entropy.* The Bekenstein–Hawking area entropy  $S_{\text{BH}} = A/(4\ell_P^2)$  [40] increases during coalescence (Hawking’s area theorem:  $A_f \geq A_1 + A_2$ ), and the SDF does not contradict this. The “anti-entropic concentration” in Definition 25 refers not to  $S_{\text{BH}}$  but to *informational parsimony*: the no-hair theorem reduces the product’s description from  $(m_1, \tilde{S}_1, m_2, \tilde{S}_2, e, \dots)$  to just  $(M, J, Q)$ —the Kerr solution is the maximally parsimonious endpoint. In SDF terms, the decoherence function  $D_f(\gamma)$  reaches its stable fixed point along a geodesic ( $\beta_f = 1$ ), the shortest path in the Bures manifold. The gravitational waves carry away the surplus degrees of freedom via a non-geodesic ( $\beta_M < 1$ ) channel.

#### E. Gravity as Spatial Decoherence Operator

[THM] [HIP]

I now formalize the role of gravity as a *spatial decoherence operator* that defines the boundary of quantum coherent interconnection.

**Hypothesis 3** (Gravitational Decoherence Boundary). The gravitational field defines a spatial coherence boundary  $\ell_{\text{coh}}$  below which quantum coherence is maintained and above which decoherence dominates. This boundary is set by the gravitational self-energy of the superposed state [41].



Following Diósi [42] and Penrose [41], the gravitational self-energy of a mass  $m$  with characteristic radius  $R$  in a spatial superposition is

$$E_{\text{grav}} \sim \frac{G m^2}{R}. \quad (171)$$

The corresponding hierarchy parameter is

$$\Lambda_{\text{grav}} = \frac{E_{\text{grav}}}{E_{\text{P}}} = \frac{G m^2}{R m_{\text{P}} c^2}. \quad (172)$$

From the  $\Lambda$ -hierarchy (Eq. 62), the gravitational coherence length follows directly:

**Definition 26** (Gravitational Coherence Length). *[DEF] The gravitational coherence length of an object of mass  $m$  and radius  $R$  is*

$$\ell_{\text{grav}}(m, R) = \frac{\ell_{\text{P}}}{\Lambda_{\text{grav}}} = \frac{\hbar c R}{G m^2} \quad (173)$$

*Spatial superpositions of extent  $\Delta x > \ell_{\text{grav}}$  decohere gravitationally before any other mechanism can intervene.*

**Dimensional verification.**  $[\hbar c R / (G m^2)] = \frac{\text{J} \cdot \text{s} \cdot \text{m} / \text{s} \cdot \text{m}}{\text{m}^3 \text{kg}^{-1} \text{s}^{-2} \text{kg}^2} = \text{m}.$

The decoherence timescale is  $\tau_{\text{grav}} = \hbar / (E_{\text{grav}}) = \hbar R / (G m^2)$ , so  $\ell_{\text{grav}} = c \tau_{\text{grav}}$ , consistent with the causal structure of  $\mathcal{P}$ .

TABLE XXXIX. Gravitational coherence length  $\ell_{\text{grav}} = \hbar c R / (G m^2)$  for representative systems. Values below the system's own radius indicate gravitationally induced classicality.

| System        | $m$ (kg)               | $R$ (m)               | $\ell_{\text{grav}}$                               | Regime     |
|---------------|------------------------|-----------------------|----------------------------------------------------|------------|
| Proton        | $1.67 \times 10^{-27}$ | $8.8 \times 10^{-16}$ | $1.5 \times 10^{23}$ m ( $\sim 16$ kly)            | Quantum    |
| Carbon atom   | $2 \times 10^{-26}$    | $7 \times 10^{-11}$   | $8.3 \times 10^{25}$ m ( $\sim 9$ Gly)             | Quantum    |
| Bacterium     | $10^{-15}$             | $10^{-6}$             | $4.7 \times 10^8$ m ( $\sim 1.2 R_{\text{Moon}}$ ) | Quantum    |
| Grain of sand | $10^{-7}$              | $5 \times 10^{-5}$    | $2.4 \times 10^{-8}$ m (24 nm)                     | Borderline |
| Tennis ball   | $6 \times 10^{-2}$     | $3.3 \times 10^{-2}$  | $2.9 \times 10^{-18}$ m ( $\sim 3$ am)             | Classical  |
| Human         | 70                     | 0.5                   | $4.8 \times 10^{-20}$ m ( $\sim 48$ zm)            | Classical  |

The table reveals the physically correct hierarchy: *microscopic* systems have  $\ell_{\text{grav}}$  vastly larger than any laboratory—gravity is irrelevant for their decoherence (electromagnetic, thermal, and collisional mechanisms dominate). *Macroscopic* systems ( $m \gtrsim 10^{-7}$  kg) have  $\ell_{\text{grav}} \ll R$ : even a grain of sand is gravitationally classical at scales far below its own radius. The crossover at  $\ell_{\text{grav}} \sim R$  occurs near  $m^* = (\hbar c / G)^{1/3} \approx 3 \times 10^{-8}$  kg  $\approx 30$   $\mu\text{g}$ , in the range targeted by optomechanical tests of gravitational decoherence.

The gravitational field thus acts as an SDF decoherence channel in the *spatial* domain:

$$D_{\text{grav}}(\Delta x) = L \left[ 1 - \exp \left( - \left( \frac{\Delta x}{\ell_{\text{grav}}} \right)^\beta \right) \right]. \quad (174)$$

**Theorem 38** (Gravity as Coherence Boundary). *[THM] The gravitational SDF channel (174) defines a spatial partition of the phase space  $\mathcal{P}$  into:*

1. **Coherent interior** ( $\Delta x < \ell_{\text{grav}}$ ):  $D_{\text{grav}} < D_{\text{th}}$ , quantum interconnection is maintained.
2. **Decoherent exterior** ( $\Delta x > \ell_{\text{grav}}$ ):  $D_{\text{grav}} > D_{\text{th}}$ , systems are classically disconnected.

The transition occurs at the hypersurface

$$\partial \mathcal{G} = \{X \in \mathcal{P} : D_{\text{grav}}(\Delta x) = D_{\text{th}}\}, \quad (175)$$

which coincides with the quantum-classical boundary  $\partial \mathcal{T}$  of Definition 18 when the gravitational channel dominates the decoherence budget.

*Proof.* From Eq. (174),  $D_{\text{grav}}$  is monotonically increasing in  $\Delta x$  (Axiom 1 applied to spatial separation). By the intermediate value theorem,  $D_{\text{grav}} = D_{\text{th}}$  has a unique solution. Below  $\ell_{\text{grav}}$ ,  $D_{\text{grav}} < D_{\text{th}}$ ; above,  $D_{\text{grav}} > D_{\text{th}}$ . The identification with  $\partial \mathcal{T}$  follows from the correspondence between  $D_{\text{th}}$  and the entropic signature change (Theorem 15).  $\square$

*a. Physical interpretation.* Gravity is not merely a force—in the SDF framework, it is the *spatial decoherence operator* that defines the maximum extent of quantum interconnection. The coherence length  $\ell_{\text{grav}}(m, R)$  scales as  $R/m^2$ : elementary particles have  $\ell_{\text{grav}} \gg R_{\text{obs}}$  (gravitational decoherence is negligible), while macroscopic objects have  $\ell_{\text{grav}} \ll R$  (they are gravitationally classical at scales far below their own radius). This is the SDF realization of the Diósi–Penrose conjecture [41] that gravitational self-energy drives the quantum-to-classical transition, with the coherence length derived directly from the  $\Lambda$ -hierarchy (Table XVII) rather than postulated *ad hoc*.

For compact objects ( $R \rightarrow r_{\text{S}} = 2Gm/c^2$ ):  $\ell_{\text{grav}} \rightarrow 2\hbar/(mc) = 2\lambda_{\text{C}}$ , the Compton wavelength—the gravitational and quantum scales merge, as expected near the Planck mass.

## F. Anti-Entropic Information Condensation

**[THM]**

The entropic duality (Theorem 37) reveals that coalescence events produce a *stable information condensation*—an anti-entropic fixed point in the SDF dynamics.

**Definition 27** (Information Condensation). *[DEF] An information condensation is a trajectory  $\rho(\gamma)$  in the entropic phase space  $\mathcal{P}$  that converges to a stable fixed point  $\rho^*$  satisfying:*

$$\begin{aligned} \frac{dD_f}{d\gamma} \Big|_{\gamma^*} &= 0, \\ \frac{d^2 D_f}{d\gamma^2} \Big|_{\gamma^*} &> 0 \quad (\text{stability}), \\ \mathcal{N}(\rho^*) &< \mathcal{N}(\rho_1) + \mathcal{N}(\rho_2) \quad (\text{condensation}), \end{aligned} \quad (176)$$

where  $\mathcal{N}(\rho) = \dim\{\lambda_i : \lambda_i > \epsilon\}$  counts the effective number of degrees of freedom (informational complexity), not



the thermodynamic entropy. For a Kerr black hole,  $\mathcal{N} = 3$  ( $M, J, Q$ ); for binary progenitors,  $\mathcal{N}(\rho_1) + \mathcal{N}(\rho_2) \gg 3$  (spins, orbital elements, tidal deformations).

**Theorem 39** (Condensation in the Entropic Phase Space). *[THM] During an information condensation event, the trajectory in the entropic phase space  $\mathcal{P}$  undergoes a triple signature transition:*

$$\underbrace{(1, d+1)}_{\text{pre-coalescence}} \xrightarrow{\gamma_1} \underbrace{(2, d)}_{\text{coalescence}} \xrightarrow{\gamma_2} \underbrace{(1, d+1)}_{\text{post-coalescence}} \quad (177)$$

The intermediate phase with signature  $(2, d)$  corresponds to the region where both  $\gamma$  and  $\sigma$  are timelike: the system enters a non-Markovian regime during the merger, where information backflow is geometrically permitted, before settling into a Markovian classical state.

*Proof.* Pre-coalescence: the two bodies are macroscopic with  $\beta \approx 1$  (Markovian),  $d^2\sigma/d\gamma^2 > 0$ , and the entropy coordinate is effectively spacelike, giving signature  $(1, d+1)$ .

During coalescence: the extreme gravitational interaction—with nonlinear mode coupling, quasi-normal mode superposition, and memory effects in the radiation—produces an effective  $\beta_{\text{eff}} < 1$  in the merger region. By Theorem 15, this creates intervals with  $d^2\sigma/d\gamma^2 < 0$ , where the entropy dimension becomes timelike: signature  $(2, d)$ . This does not mean the system is “quantum” in the standard sense, but that the *SDF entropic metric* admits information backflow—a geometric property of the decoherence trajectory, not a claim about the gravitational field’s quantum nature.

Post-coalescence: the final product settles into a Kerr state with  $\beta_f \rightarrow 1$ , restoring  $d^2\sigma/d\gamma^2 > 0$  and signature  $(1, d+1)$ .

The transition points  $\gamma_1, \gamma_2$  are determined by  $d^2\sigma/d\gamma^2 = 0$ , which is the transition boundary  $\partial\mathcal{T}$  of Definition 18.  $\square$

*a. The coalescence as information tunnel.* Theorem 39 reveals that a gravitational coalescence is an *information tunnel*: the system traverses a quantum-signature region  $(2, d)$  sandwiched between two classical-signature regions  $(1, d+1)$ . The anti-entropic condensation occurs *within* this quantum tunnel, where information backflow (Theorem 18) permits the concentration of dispersed information into a coherent final state. The tunnel width  $\Delta\gamma = \gamma_2 - \gamma_1$  is bounded by

$$\Delta\gamma_{\text{tunnel}} \leq \tau_M \left( \frac{1 - \beta_M}{\beta_M} \right)^{1/\beta_M} \quad (178)$$

which for LIGO values ( $\beta_M = 0.931$ ,  $\tau_M$  of order ms) yields tunnel widths consistent with the observed merger duration [12].

## Synthesis: Entropic Architecture of Coalescence

A gravitational coalescence event manifests a three-fold structure in the SDF entropic phase space:

1. **Informational concentration:** the product settles geodesically ( $\beta_f \rightarrow 1$ ) into a state of maximal informational parsimony ( $\mathcal{N} = 3$  for Kerr), while thermodynamic entropy increases ( $S_{\text{BH},f} \geq S_{\text{BH},1} + S_{\text{BH},2}$ , area theorem).
2. **Entropic medium:** the gravitational radiation disperses the surplus complexity ( $\beta_M < 1$ ), carrying non-Markovian correlations (environmental memory).
3. **Spatial decoherence boundary:** gravity defines the coherence length  $\ell_{\text{grav}}$  that separates the coherent interior (interconnected) from the decoherent exterior (disconnected).

The coalescence is therefore *structured* anti-entropically: its information-theoretic purpose (concentration into a stable state) is achieved through an entropic medium (gravitational radiation), with gravity itself serving as the spatial decoherence operator that defines the limits of coherent interconnection.

## G. Emergent Gravity Conjecture

[CONJ]

### WARNING — Speculative conjecture

The content of this subsection is an untested ansatz. No derivation from first principles is provided.

**Conjecture 3** (Emergent curvature). *The Einstein tensor is proportional to the local gradient of the decoherence function,*

$$G_{\mu\nu} \propto \left. \frac{\partial D(\gamma)}{\partial \gamma} \right|_{\gamma=\gamma_{\text{local}}}. \quad (179)$$

*a. Dimensional analysis.*  $G_{\mu\nu}$  carries dimensions of  $[\text{length}]^{-2}$ , whereas  $\partial D/\partial \gamma$  is dimensionless per unit decoherence parameter. The dimensional mismatch is not closed by any natural scale within the current formulation; a bridge factor  $\sim \ell_P^{-2}$  would need to be postulated *ad hoc*. This conjecture therefore remains an ansatz without predictive power in its present form.

The idea draws conceptual inspiration from entropic-gravity proposals [43], but no formal connection has been established.

## H. Black-Hole $C \rightarrow Q$ Transition

[CONJ]

### WARNING — Speculative conjecture

The content of this subsection is an untested conjecture.

**Conjecture 4** (Black-hole inverse transition). *A black hole represents the inverse of the quantum-to-classical transition, characterized by  $d\gamma'/dt < 0$ : the system evolves from a maximally decohered (classical) state toward a quantum remnant as Hawking radiation [40] erodes the horizon.*

The transition terminates at a minimum mass set by the SDF Planck-scale boundary:

$$M_{\min} = \frac{\ell_P c^2}{2G\sqrt{\pi}} \approx 0.28 m_P \approx 6.09 \mu\text{g}. \quad (180)$$

#### a. Remnant properties.

- Radius:  $r_{\text{rem}} \approx 0.56 \ell_P$ .
- Geometric cross section:  $\sigma_{\text{rem}} \sim \pi r_{\text{rem}}^2 \sim 10^{-70} \text{ m}^2$ .
- Hawking temperature at  $M_{\min}$ :  $T_H \sim \hbar c^3 / (8\pi G M_{\min} k_B)$ , well above the CMB temperature, yet the conjecture posits that the SDF barrier prevents further evaporation.
- The remnant is therefore *cosmologically stable*.

#### I. Dark-Matter Scenario — Superseded

The original presentation of this framework closed with a dark-matter scenario at this point: remnants at  $M_{\min}$  constituting  $\Omega_{\text{DM}} \approx 0.12$ , with a comoving density  $n_{\text{rem}} \sim 8.8 \times 10^{-26} \text{ cm}^{-3}$  and the conclusion that direct detection is impossible because the geometric cross section lies  $\sim 21$  orders of magnitude below experimental sensitivity.

That treatment is superseded in full by Sec. XXIII, which is given the standing of a section rather than a subsection because it is where the framework ends. Three things changed in the re-evaluation and none of them was available at the time:

- The remnant mass is not the approximation  $0.28 m_P$  but the exact identity  $M_{\min} = m_P / 2\sqrt{\pi}$ , Eq. (182).
- The published comoving density does not follow from its own inputs; the corrected value is  $1.67 \times 10^{-25} \text{ cm}^{-3}$  and the published figure corresponds to  $\Omega = 0.063$  rather than 0.12 (Erratum 10).
- The cross-section comparison, while numerically right, misclassifies the obstacle. Exposure is *not* the limiting factor — one transit is expected per  $0.86 \text{ km}^2 \cdot \text{yr}$  — and the kinetic energy is macroscopic at 148.6 J. What fails is the coupling.

### XXIII. The Dark-Matter Candidate

#### Evidence level: CONJ throughout

Every proposition in this section is a conjecture. The chain has four links and each multiplies the speculation of the one before it. Nothing here is offered as a prediction of the framework, and the section is included because a programme that states its speculative endpoint openly can be argued with, while one that leaves it implicit cannot.

This is where the framework ends, and it is the only place where it makes a cosmological claim. The claim is that dark matter is the ash of black-hole evaporation: Planck-scale remnants left behind when the quantum-to-classical transition runs backwards to completion.

#### A. The chain, with each link labelled

1. **[CONJ]** A black hole is the *inverse* of the quantum-to-classical transition,  $d\gamma'/dt < 0$ : Hawking radiation erodes the horizon and the object evolves from a maximally decohered state towards a quantum remnant (Sec. XXIIH).
2. **[CONJ]** Evaporation halts at the SDF Planck-scale boundary  $M_{\min}$  rather than proceeding to zero mass.
3. **[CONJ]** The remnant is cosmologically stable.
4. **[CONJ]** Remnants constitute the dark-matter density,  $\Omega_{\text{remnants}} = \Omega_{\text{DM}}$ .

Link 2 is the load-bearing one and it is the weakest. The Hawking temperature at  $M_{\min}$  is

$$T_H(M_{\min}) = \frac{\hbar c^3}{8\pi G M_{\min} k_B} \approx 2.0 \times 10^{31} \text{ K}, \quad (181)$$

some thirty-one orders of magnitude above the CMB. *No derivation is offered for a barrier that would arrest evaporation against that gradient.* We state this as the first falsifier rather than burying it.

#### B. An exact identity hidden behind an approximation

The corpus quotes the minimum mass as  $M_{\min} \approx 0.28 m_P$ . That approximation conceals a closed form. Substituting  $\ell_P = \sqrt{\hbar G / c^3}$  and  $m_P = \sqrt{\hbar c / G}$  into Eq. (180),

$$\frac{M_{\min}}{m_P} = \frac{\ell_P c^2}{2G\sqrt{\pi}} \sqrt{\frac{G}{\hbar c}} = \frac{1}{2\sqrt{\pi}} = 0.2820947918 \dots \quad (182)$$

**[EXACT]**. The remnant mass is not a fitted or approximate quantity: it is  $m_P / (2\sqrt{\pi})$ , a pure number. Evaluating the left-hand side from tabulated constants reproduces it to a relative residual of  $1.42 \times 10^{-7}$ , and that residual is not a defect of the identity — it is the mutual inconsistency of  $\ell_P$  and  $m_P$  as separately rounded CODATA entries. The identity itself is exact.

TABLE XL. The candidate, evaluated. All quantities follow from Eq. (182) and  $\Omega_{\text{DM}} = 0.12$ ,  $h = 0.674$ ,  $v_{\text{halo}} = 220$  km/s. Reproducible from `sdf-lab/src/core/dark-matter.ts`.

| Quantity                                              | Value                                              |
|-------------------------------------------------------|----------------------------------------------------|
| Mass $M_{\text{min}} = m_P/2\sqrt{\pi}$               | $6.140 \mu\text{g}$                                |
| Radius $r_{\text{rem}}$                               | $9.05 \times 10^{-36} \text{ m}$                   |
| Geometric cross section                               | $2.57 \times 10^{-66} \text{ cm}^2$                |
| Hawking temperature at $M_{\text{min}}$               | $2.00 \times 10^{31} \text{ K}$                    |
| Number density $n_{\text{rem}}$                       | $1.67 \times 10^{-25} \text{ cm}^{-3}$             |
| Mean separation                                       | $1.82 \times 10^3 \text{ km}$                      |
| <b>Kinetic energy at <math>v_{\text{halo}}</math></b> | <b><math>148.6 \text{ J}</math></b>                |
| Flux                                                  | $1.16 \times 10^{-6} \text{ m}^{-2}\text{yr}^{-1}$ |
| <b>Exposure for one transit</b>                       | <b><math>0.86 \text{ km}^2\text{-yr}</math></b>    |
| Mean free path in rock                                | $1.2 \times 10^{23} \text{ ly}$                    |
| $P(\text{interaction} \mid \text{Earth crossing})$    | $1.1 \times 10^{-32}$                              |

This matters for the section that follows: whatever one concludes about the cosmology,  $M_{\text{min}} = 6.140 \mu\text{g}$  and  $r_{\text{rem}} = 0.56 \ell_P$  carry no free parameters.

### C. Erratum: the published number density

**Erratum 10** (Comoving number density). *Paper 1 §11 reports  $n_{\text{rem}} \sim \Omega_{\text{DM}} \rho_{\text{crit}} / (M_{\text{min}} c^2) \sim 8.8 \times 10^{-26} \text{ cm}^{-3}$ . Two problems. First, the expression is dimensionally inconsistent:  $\rho_{\text{crit}}$  is already a mass density, so dividing by  $M_{\text{min}} c^2$  leaves  $\text{m}^{-3} \text{kg}^{-1} \text{s}^2$ . The  $c^2$  is spurious. Second, with the  $c^2$  removed and  $\Omega_{\text{DM}} = 0.12$ ,  $h = 0.674$ , the correct evaluation is*

$$n_{\text{rem}} = \frac{\Omega_{\text{DM}} \rho_{\text{crit}}}{M_{\text{min}}} = 1.668 \times 10^{-25} \text{ cm}^{-3}, \quad (183)$$

*larger than the published figure by a factor 1.895. Inverted, the published number density corresponds to  $\Omega = 0.0633$ , not to the  $\Omega_{\text{DM}} = 0.12$  it is stated to reproduce.*

We use the derived value below. The distinction is immaterial to the conclusion, which is what makes it safe to correct in passing rather than to defend.

### D. The phenomenology, and why the usual verdict is the wrong one

Paper 1 dismisses direct detection by comparing the geometric cross section  $\sigma_{\text{rem}} \approx 2.6 \times 10^{-66} \text{ cm}^2$  with experimental sensitivities  $\sigma_{\text{exp}} \lesssim 10^{-45} \text{ cm}^2$ , a gap of 20.6 orders of magnitude, and concludes that detection is “impossible with any foreseeable technology”. The conclusion is right and the argument is misleading, because it invites the reader to picture a very weakly interacting particle. The remnant is not that.

Three numbers in Table XL change the character of the problem.

*a. The energy is macroscopic.* A remnant at galactic velocity carries 148.6 J — eleven orders of magnitude above the threshold of any calorimeter, and comparable to a thrown stone. This is not a weakly interacting particle in the sense the cross-section comparison suggests. It is a projectile.

*b. The exposure is affordable.* At  $n_{\text{rem}} = 1.67 \times 10^{-25} \text{ cm}^{-3}$  the flux is  $1.16 \times 10^{-6} \text{ m}^{-2}\text{yr}^{-1}$ , so one transit is expected per  $0.86 \text{ km}^2\text{-yr}$ : a square 930 m on a side, monitored for a year. Instruments of that scale exist. *Rarity is not the obstacle.*

*c. The coupling is absent, and that is the obstacle.* With a Planck-scale geometric cross section the mean free path in rock is  $1.2 \times 10^{23} \text{ ly}$ , exceeding the observable universe by twelve orders of magnitude. The probability of a single interaction while traversing the entire Earth is  $1.1 \times 10^{-32}$ . The 148.6 J are carried straight through and never deposited.

### The correct classification

The candidate is *cross-section-limited*, not *exposure-limited*, and the two demand opposite experiments. A larger or longer-running detector buys nothing: the remnants are already crossing it, roughly one per square kilometre per year, and passing through without a single scatter. What is missing is not sensitivity to rare events but a coupling channel, and for an object of this size the only channel available is gravitational.

We record that our own first attempt at this classification reached the opposite verdict, by comparing available kinetic energy against a detector threshold without asking whether that energy could couple. Energy that cannot be deposited is not a signal. The error is instructive enough to leave documented.

### E. Falsifiers

A conjecture is not excused from stating what would retire it. Four routes, two of which need no new instrument.

#### DM-F1 (available today): Primordial spectrum.

The scenario requires a collapse fraction  $\beta_{\text{PBH}} \sim 2.5 \times 10^{-17}$  at the Planck scale. CMB and PBH-abundance constraints bound that window; if they exclude it, the scenario falls without anything being detected.

#### DM-F2 (available today): Absence of annihilation.

The scenario predicts dark matter that is strictly collisionless and has *no* annihilation channel. A confirmed positive indirect detection ( $\gamma$ ,  $\nu$ ,  $e^+$ ) excludes it.

#### DM-F3 (requires instrument): Gravitational transit.

Since the obstacle is coupling rather than exposure, the correct class of search is one sensitive to a purely gravitational impulse over  $\sim \text{km}^2$  areas — the regime of macroscopic dark-matter searches, not of low-threshold particle detectors.

#### DM-F4 (the closed link): Remnant stability.

The chain requires a barrier arresting evaporation at  $M_{\text{min}}$  despite  $T_H \sim 10^{31} \text{ K}$ . No derivation is offered.

A demonstration that evaporation proceeds below  $M_{\min}$  closes the chain at its central link, and this is the cheapest way to kill the scenario outright.

### F. Verdict

The candidate has one attractive feature and one fatal one, and both should be stated without weighting.

Attractive: it introduces no new particle, no new field and no free parameter. The mass is  $m_P/2\sqrt{\pi}$  exactly, the abundance follows from one cosmological number, and the scenario predicts precisely the phenomenology dark matter is observed to have — collisionless, non-annihilating, gravitationally clustered.

Fatal: it is decoupled. Not weakly coupled, decoupled — the interaction probability across the Earth is  $10^{-32}$ . A hypothesis that predicts exactly what is observed and forbids any other observation is close to unfalsifiable by direct means, which is why DM-F1 and DM-F4 matter more than DM-F3. The scenario must be attacked at the primordial spectrum and at the stability of the remnant, not at the detector.

We therefore close the programme with this section rather than build on it. It is where the framework points, it is stated with its numbers and its falsifiers, and it remains [CONJ].

## Part VII Closing

### XXIV. Open Problems

Eleven entries of the register in Appendix A remain open. Five of them are load-bearing — they block a derivation the corpus relies on — and are stated below in the sharpened form produced by the adversarial protocol, together with the escape routes that have been closed. The remaining six (D21, D22, D24, D25, D26, D27) are recorded in the register with their summaries and are not narrated here, because each is a local defect whose repair is known and whose only cost is the decision to adopt it.

All eleven are open in the strict sense: resolving them requires a decision about the physics, not about the wording.

#### A. The derivation of Axiom 3

Theorem 27 establishes that no filter-function route produces  $\beta = 1 - \alpha/2$ , and Erratum 1 removes the only independent microscopic derivation that had been offered. Yet  $1 - \alpha/2$  is the only candidate that both respects the domain  $\beta \in (0, 2]$  and admits  $\beta < 1$ , the framework's most robust empirical invariant.

*Closed routes.* Reparametrising through  $\gamma(t)$  requires  $d\gamma/dt$  of opposite sign to what Paper 2 §7 specifies. Reading the factor of two as an amplitude-versus-intensity convention fails, since squaring  $e^{-\Gamma}$  rescales  $\tau$  and not the exponent.

*What would settle it.* Either a derivation of  $\beta < 1$  from a system–bath model that is not of filter-function type, or an explicit acceptance of Axiom 3 as a phenomenological postulate with the empirical fit as its sole warrant.

#### B. The two meanings of $\alpha$

The symbol  $\alpha$  denotes both the spectral exponent of the bath, through  $\beta = 1 - \alpha/2$ , and the anomalous-diffusion exponent of the biological extension, through  $\beta = \alpha_{\text{diff}}/2$ . The two relations are simultaneously satisfiable only if  $\alpha_{\text{diff}} = 2 - \alpha$ .

This is not merely notational. Applied to the published single-particle tracking data, with  $\alpha_{\text{diff}} \in [0.32, 0.66]$ , the two relations give  $\beta \in [0.16, 0.33]$  and  $\beta \in [0.67, 0.84]$  respectively: a factor of two to four on the framework's central parameter.

*Status: closed.* The standard bridge is the generalised Langevin equation with the fluctuation–dissipation theorem, which gives  $\alpha = 1 - \alpha_{\text{diff}}$  rather than  $2 - \alpha_{\text{diff}}$ , and hence  $\beta = (1 + \alpha_{\text{diff}})/2$ . The published relation is refuted by its own limit: normal Brownian diffusion,  $\alpha_{\text{diff}} = 1$ , must give Markovian decoherence, and  $\beta = \alpha_{\text{diff}}/2$  returns 0.5. See Erratum 9 and Table XXX, where the corrected relation places all five measured systems inside the declared domain.

What remains open is narrower and belongs to the previous subsection: the scaling-law route still requires  $n \approx 7\text{--}16$  for systems with  $n \sim 10^{20}$  constituents.

#### C. The information-condensation fixed point

Paper 1 §11 defines information condensation by three conditions:  $dD_f/d\gamma = 0$  at some  $\gamma^*$ ,  $d^2D_f/d\gamma^2 > 0$  there for stability, and a reduction of informational complexity  $\mathcal{N}(\rho^*) < \mathcal{N}(\rho_1) + \mathcal{N}(\rho_2)$ .

The first two are unsatisfiable by any SDF curve. The basic-properties theorem gives  $dD/d\gamma > 0$  strictly for all  $\gamma > 0$ , so there is no interior fixed point; and for  $\beta \leq 1$  the curvature is negative everywhere. A sweep of  $2 \times 10^5$  samples over  $\gamma \in (0, 200\tau]$  at  $\beta \in \{0.5, 0.852, 1, 1.5\}$  returns zero points satisfying both.

*Why the obvious repair fails.* Replacing the interior fixed point by an asymptotic attractor with  $d^2D/d\gamma^2 < 0$  makes the conditions satisfiable, but the resulting point coincides with the *global maximum* of the entropy along the trajectory—the entropy theorem gives  $S \rightarrow \ln 2$  as  $D \rightarrow L$ —so that the adjective “anti-entropic” becomes literally false. Further, the inverse-function theorem establishes that  $D$  maps onto the half-open interval  $[0, L)$ , so the limiting state  $\rho^*$  is not in the range at all.

*What survives.* The third condition, which is informational rather than dynamical, and is the only formal basis the corpus offers for a tendency of information to concentrate. It should be retained and the dynamical language withdrawn.

*Note.* Paper 1 §12 already states that the flow has no interior fixed points. The inconsistency is therefore internal to the corpus—§11 against §12—rather than a new discovery.



#### D. Bidirectionality against the unique attractor

The axiom set accompanying Paper 2 declares transitions in both directions, quantum to classical and classical to quantum, to be permitted. Paper 1 §12 establishes a unidirectional flow with a unique attractor at  $(D, C) = (1, 0)$ . Both cannot hold on the same state space.

*The route the corpus hints at.* Unidirectionality in  $\gamma$  combined with bidirectionality in  $t$ , exploiting a non-monotonic  $\gamma(t)$ . Paper 2 Appendix E states explicitly that the map  $\gamma(t)$  can become non-monotonic for systems resonantly coupled to structured environments—which supports the route but does not formalise it.

*What must be decided.* Whether a physical trajectory with  $d\gamma/dt < 0$  on some interval can be constructed. If not, the bidirectionality axiom degrades to a conjecture, and the proposed resolution of the black-hole information problem, which rests on it explicitly, loses its support.

#### E. The reported spectral mode

Treated in Sec. XVI. It is listed among the open problems rather than the errata because withdrawing the mode leaves the emergence section of Paper 1 without its principal empirical claim, and what should replace it is a matter of judgement rather than arithmetic. The non-Gaussianity of the residuals is real and reproducible; whether it supports the transient-mode interpretation at all is undetermined.

#### F. A bridge between state space and physical space

The framework fixes two lengths at which exactly one bit is exchanged, and they live in different spaces. In state space, by Corollary 4, one bit is  $\pi/3$  of Bures arc out of a total  $\pi/2$ . In physical space, the Bekenstein gradient used by Verlinde,  $dS/dx = 2\pi k_B/\lambda_C$ , places one bit at

$$\Delta x = \frac{\ln 2}{2\pi} \bar{\lambda}_C = 0.110317800076 \bar{\lambda}_C, \quad (184)$$

a fixed fraction of the reduced Compton wavelength, independent of mass. Both are fixed fractions of the system's own scale. The question is whether they are the same quantity in two coordinates.

*a. The natural bridge, and its refutation.* Identifying the accumulated information of the SDF with the Verlinde entropy in units of  $k_B$  gives  $(\gamma/\tau)^\beta = 2\pi x/\bar{\lambda}_C$ , which reproduces *both* scales simultaneously from a single identification—residual 0 in Eq. (184) and  $2.2 \times 10^{-16}$  in the arc. Substituting into the master equation yields

$$D(x) = L \left[ 1 - \exp\left(-\frac{2\pi x}{\bar{\lambda}_C}\right) \right], \quad (185)$$

in which  $\beta$  has disappeared: decoherence as a function of displacement would be a pure exponential for every system, the stretching being absorbed by the change of variable. That is a falsifiable prediction, and it fails. It places 99% decoherence of a free electron at  $2.8 \times 10^{-13}$  m, whereas electron interferometry observes interference with transverse coherence lengths of  $10^{-7}$  m or more—a gap

of five to six orders of magnitude. Rescuing it requires  $(\gamma/\tau)^\beta = 2\pi\eta x/\bar{\lambda}_C$  with  $\eta \lesssim 3 \times 10^{-6}$ , a number that follows from no constant of the framework.

The identification fails for a reason that is visible once stated: the Bekenstein gradient is the entropy of a holographic *screen* as a test mass approaches it, not the decoherence of the mass. They are different entropies.

*Status: attempted and failed; the structural observation survives.* That each scale is a fixed fraction of the system's own scale is exact and requires no bridge. That the two are the same magnitude does not follow, and the natural route to making them so is excluded by experiment.

#### G. A methodological remark

Five resolutions to these problems were proposed during this work and all five were rejected under adversarial review. Four failed for the same reason: they did not cite a passage of the corpus that already addressed the matter. In one case the decisive footnote appeared in the first section examined.

The lesson generalises beyond this framework. An audit protocol should require of anyone proposing a correction not merely that their algebra closes, but that they have actively searched for the passage which refutes them. The verification effort is asymmetric: checking that a proposal is self-consistent is cheap, and checking that it is consistent with everything already written is expensive—which is precisely why the second is the one that gets skipped.

### XXV. Discussion

#### A. Implications for Quantum Computing

The discovery of three decoherence regimes has practical implications for quantum error correction (QEC) [44] strategy selection:

1. **Exponential regime** ( $\beta \approx 1$ ): Standard QEC codes (surface [45], Steane [46]) are optimal. Markovian noise assumptions hold and threshold theorems apply directly.
2. **Stretched regime** ( $\beta < 1$ ): Non-Markovian codes or dynamical decoupling [47] may outperform standard QEC. The environment retains memory that can, in principle, be exploited to improve gate fidelities.
3. **Revival regime**: Active feedback protocols may partially recover information. The noise structure itself provides transient protection, as observed for W and BellChain topologies (Section VIII).

**Recommendation:** Measure  $\beta$  for the target hardware *before* selecting a QEC strategy. The 10–50 $\times$  variation in  $\beta$  between `ibm_fez` and `ibm_torino` (Section XXV B) demonstrates that a single QEC prescription cannot be universally optimal.



## B. Cross-Backend Discoveries

Phase 29 introduced a second IBM Quantum backend—`ibm_torino` (Heron r2, 133 qubits)—to complement the original `ibm_fez` (Eagle r1, 156 qubits) dataset. The comparison reveals striking differences that constrain the scope of universality.

### 1. The $\beta$ anomaly

Across 400 new circuits on `ibm_torino`, the fitted stretched-exponential exponent satisfies

$$\beta_{\text{torino}} \in [0.011, 0.047], \quad (186)$$

which is 10–50 $\times$  smaller than the corresponding values on `ibm_fez`:

$$\beta_{\text{fez}} \in [0.30, 0.86], \quad \beta_{\text{fez}}(n) = \frac{1.679}{n^{0.844}}, \quad R^2 = 0.987. \quad (187)$$

In the SDF interpretation,  $\beta \rightarrow 0$  corresponds to *logarithmically slow* decoherence growth—i.e., ultra-stretched dynamics. The Heron r2 processor thus exhibits dramatically more non-Markovian behavior than Eagle r1 under identical circuit families ( $\text{GHZ}_n$ ,  $n \in \{3, \dots, 12\}$ ).

Possible physical origins include:

- Different noise power spectral densities: Heron r2 may have stronger low-frequency correlations ( $\alpha_{\text{noise}}$  closer to 2), producing more stretched dynamics.
- Improved coherence times ( $T_1, T_2$ ) in Heron r2 shifting the experimental window into the early, sub-diffusive tail of  $D(\gamma)$ .
- Distinct coupler architectures (tunable vs. fixed) modifying the effective system–environment coupling spectral density.

### 2. THM-26.1 cross-backend tension

The regime classification theorem (THM-26.1) predicts  $\beta = 1 - \alpha/2$ . On `ibm_torino`, the independently measured spectral exponent is  $\alpha_{\text{spectral}} = 0.661 \pm 0.426$  (Welch PSD, 400 circuits,  $n \geq 5$ ), which via THM-26.1 would predict  $\beta \approx 0.67$ . The *observed*  $\beta \approx 0.015$  disagrees by a factor of  $\sim 45$ .

Two resolutions are under investigation:

1. **Hardware-dependent offset:** Generalize THM-26.1 to

$$\beta(n, \text{hw}) = \beta_0(\text{hw}) \cdot n^{-\alpha(\text{hw})/2}, \quad (188)$$

where  $\beta_0$  absorbs processor-specific noise characteristics. If the *functional form* is universal with hardware-dependent parameters, the theorem retains predictive power within each backend.

2. **Distinct  $\alpha$  measures:** The Welch-PSD  $\alpha_{\text{spectral}}$  characterizes *residual* noise after SDF subtraction, whereas the Lindblad-derived  $\alpha$  in THM-26.1 describes the *bare* environment spectral density. These quantities need not coincide when the system–bath coupling is strongly renormalized.

A definitive resolution requires executing identical protocols on at least one additional backend (e.g., `ibm_marrakesh`, `ibm_osaka`).

### 3. Implications for universality

The  $D(\gamma) + C(\gamma) = 1$  conservation law holds exactly on both backends (650 circuits,  $\sim 5.5 \times 10^6$  shots). The *functional form*  $D(\gamma) = L[1 - \exp(-(\gamma/\tau)^\beta)]$  fits well for  $n \leq 10$  on both processors. We therefore maintain the claim that the SDF *structure* is universal, while acknowledging that the *parameters* ( $\beta_0, \tau$ ) are backend-dependent quantities determined by the hardware noise environment.

## C. Observable Emergence from the SDF

A central result of Phase 30 is the derivation of a master formula connecting the decoherence function  $D(\gamma)$  to the expectation value of an arbitrary observable.

### 1. The observable decay formula

For any quantum state undergoing decoherence, the density matrix admits the decomposition

$$\rho(\gamma) = (1 - D(\gamma)/L) \rho_{\text{coh}} + (D(\gamma)/L) \rho_{\text{mix}}, \quad (189)$$

where  $\rho_{\text{coh}} = |\psi_{\text{target}}\rangle\langle\psi_{\text{target}}|$  is the target coherent state and  $\rho_{\text{mix}} = \mathbb{I}/2^n$  is the maximally mixed state. It follows immediately that for any observable  $O$ :

$$\langle O \rangle(\gamma) = \langle O \rangle_{\text{coh}} - \frac{D(\gamma)}{L} \Delta O, \quad (190)$$

where  $\Delta O \equiv \langle O \rangle_{\text{coh}} - \langle O \rangle_{\text{mix}}$  is the “quantum contrast” of the observable.

### 2. Significance

Equation (190) establishes that a *single* function  $D(\gamma)$  is sufficient to predict the decoherence-induced decay of *any* Hermitian observable—including energy, purity, entanglement witnesses, Pauli expectation values, and correlation functions. The observable dependence enters only through the constant  $\Delta O$ , while the  $\gamma$ -dependence is entirely captured by the universal stretched exponential:

$$\begin{aligned} \langle O \rangle(\gamma) &= \langle O \rangle_{\text{coh}} - \frac{\Delta O}{L} L[1 - e^{-(\gamma/\tau)^\beta}] \\ &= \langle O \rangle_{\text{coh}} - \Delta O[1 - e^{-(\gamma/\tau)^\beta}]. \end{aligned} \quad (191)$$

This result resonates with Zurek’s quantum Darwinism program [48], in which objective classical properties emerge from redundant encoding in the environment. Here, the SDF provides the *rate law* for how quickly this encoding proceeds: all observables lose their quantum character at the same stretched-exponential rate set by  $(\beta, \tau)$ , differing only in the amplitude  $\Delta O$ .

The three SDF regimes yield distinct observable signatures:

- $\beta \approx 1$ : Exponential decay with single timescale  $\tau$  — standard  $T_1/T_2$  behavior.

- $\beta < 1$ : Heavy-tailed decay with a “long tail” persisting beyond  $\tau$  — mesoscopic and potentially biological coherences.
- Revival regime: Non-monotone  $\langle O \rangle(\gamma)$  with oscillations — observable-level signature of information return.

#### D. Information Conservation — Three Levels

A recurring source of confusion in the decoherence literature is the precise sense in which “information is conserved.” The Phase 30A analysis identifies three logically distinct levels, of which only the second constitutes a physical theorem (see Section XII for complete proofs).

1. **Level 1 — Algebraic identity.** The equation  $D(\gamma) + C(\gamma) = 1$  follows from the *definition*  $C \equiv 1 - D$  and carries no dynamical content. It should not be cited as a conservation law.
2. **Level 2 — Von Neumann conservation (THM-27.1).** For a system  $S$  coupled to an environment  $E$  evolving unitarily, the Araki–Lieb equality for globally pure states gives  $S(\rho_S) = S(\rho_E)$ , yielding

$$\tilde{\mathcal{I}}_S(\gamma) + \tilde{\mathcal{I}}_E(\gamma) + 2E(\gamma) = \ln d_S + \ln d_E = \text{const}, \quad (192)$$

where  $\tilde{\mathcal{I}}_X = \ln d_X - S(\rho_X)$  is the local information and  $E(\gamma) = S(\rho_S(\gamma))$  the entanglement entropy. This is a genuine conservation law rooted in global unitarity, proven rigorously in THM-27.1 (Section XII A 2).

3. **Level 3 — Operational recovery (CONJ).** Whether an observer with access to both  $S$  and  $E$  can operationally *recover* the initial quantum state remains a conjecture that depends on the structure of the decoherence channel and the available recovery operations.

The experimental finding  $D + C = 1$  (exact, 650 circuits) validates Level 1 as a *consistency check* on our measurement protocol. The deeper Level 2 conservation tells us that the information seemingly lost from  $S$  is faithfully transferred to  $E$ —it is relocated, not destroyed. This perspective connects to the black hole information paradox [49, 50]: if decoherence is the mechanism by which information enters Hawking radiation, then Level 2 conservation guarantees that the information survives in principle. The SDF framework adds a quantitative *rate law*: the information transfer follows the stretched exponential  $D(\gamma) = L[1 - \exp(-(\gamma/\tau)^\beta)]$ , with  $\beta$  determined by the noise spectrum of the gravitational environment.

#### E. Biological Decoherence Interpretation

The  $\Lambda$ -hierarchy framework suggests biological systems operate in the  $\Lambda \sim 10^{-30}$  regime, predicting:

- $\tau_{\text{bio}} \sim 10^{-14}$  s (femtosecond coherence)

- $\beta_{\text{bio}} \sim 0.7\text{--}0.8$  (moderate non-Markovianity)

This is consistent with observed quantum effects in photosynthesis [17] and enzyme catalysis [18], where sub-picosecond coherences persist in warm, wet environments. The observable decay formula (190) predicts that biologically relevant observables (e.g., exciton transfer probability) should exhibit the heavy-tailed signature of  $\beta < 1$ , distinguishable from simple exponential decay in ultrafast spectroscopy experiments.

#### F. Connection to Paper 1 Predictions

Paper 1 made several predictions that this work validates, refines, or extends. Table XLI summarizes the current status including Phase 29 and Phase 30 results.

The most notable change is the failure of the ECME (Emergence via Collective Multi-scale Extrapolation) prediction scheme, which aimed to predict individual state decoherence curves from collective parameters. With  $R^2 = -0.39$ , the ECME model performs substantially worse than a constant baseline, indicating that state-specific (topology-dependent) parameters are essential and cannot be replaced by a collective description.

#### G. Falsifiability and Future Tests

The framework makes the following testable predictions, including new predictions from Phases 29–30:

1. **PRED-26.1 (LIGO):**  $\beta_{\text{inspiral}} \neq \beta_{\text{ringdown}}$  due to different noise spectra during each phase. Testable with GWTC-3+ data [12].
2. **PRED-26.2 (Majorana):**  $\beta \approx 0.95\text{--}0.99$  for topologically protected states. Testable with current nanowire experiments.
3. **PRED-26.3 (DFS):** Container states maintain  $D < 0.05$  while internal systems complete the decoherence transition. Testable with trapped-ion or superconducting platforms.
4. **PRED-26.4 (Scaling):**  $\beta(n) = \beta_0/n^\alpha$  should hold for other state families (Dicke, W) with different parameters. Testable with extended IBM experiments.
5. **PRED-30.1 (Observable decay):** All Hermitian observables decay as  $\langle O \rangle(\gamma) = \langle O \rangle_{\text{coh}} - \Delta O [1 - e^{-(\gamma/\tau)^\beta}]$ . Testable via simultaneous tomography of multiple observables versus  $\gamma$ .
6. **PRED-30.2 (Hardware  $\beta$ ):**  $\beta$  is inversely correlated with processor coherence quality ( $\beta \propto 1/T_2^*$ ). Testable with a multi-backend survey.
7. **PRED-30.3 (Crosstalk threshold):** SDF model validity breaks down at  $n > n_c \approx 10$  due to crosstalk contributions scaling as  $\sim n^2$ . Testable by systematic  $n$ -sweep on any superconducting processor.

TABLE XLI. Validation of Paper 1 predictions (updated with Phase 29/30 data).

| Prediction                                    | Status                | Evidence                 | Notes                                                            |
|-----------------------------------------------|-----------------------|--------------------------|------------------------------------------------------------------|
| $\beta < 1$ universal                         | <b>Refined</b>        | 650 circuits             | Regime-dependent; $\beta_{\text{torino}} \ll \beta_{\text{fez}}$ |
| $D + C = 1$                                   | <b>Validated</b>      | 650 circuits, 2 backends | Exact to numerical precision                                     |
| $\beta = 1 - \alpha/2$                        | <b>Partial</b>        | ibm_fez only             | Cross-backend tension on ibm_torino                              |
| $dI/d\gamma = 0$                              | <b>Validated</b>      | $\gamma$ -sweep          | THM-27.1 now proven from unitarity                               |
| $\langle O \rangle$ decay $\propto D(\gamma)$ | <b>New (Phase 30)</b> | Derived                  | Eq. (190)                                                        |
| $M_{\min} = f(\beta)$                         | Unchanged             | —                        | No new data                                                      |
| ECME multi-scale                              | <b>FAIL</b>           | $R^2 = -0.39$            | Individual prediction poor                                       |

8. **PRED-30.4 (BLP backflow):** The Breuer–Laine–Piilo non-Markovianity measure [23] satisfies  $\mathcal{N} > 0$  for systems with  $\beta < 0.8$ . Testable via the trace distance protocol on pairs of initial states.
9. **PRED-30.5 (Topology  $\rightarrow \alpha$ ):** The effective spectral exponent depends on entanglement topology:  $\alpha_{\text{GHZ}} \neq \alpha_{\text{Cluster}} \neq \alpha_{\text{W}}$ . Testable with Phase 26E-class experiments.

Predictions 1–4 were stated in the original submission; predictions 5–9 are new. Each is independently falsifiable with current experimental technology.

## H. Limitations and Caveats

We acknowledge the following limitations, organized by severity.

### 1. Experimental limitations

1. **Two-backend scope.** All quantum hardware data originate from two IBM superconducting processors (ibm\_fez, Eagle r1; ibm\_torino, Heron r2). Generalization to trapped-ion, photonic, or neutral-atom platforms remains untested.
2.  **$n > 10$  model failure.** For  $n = 12$  qubits on ibm\_torino, the pure SDF fit yields  $R^2 = -1.64$ , indicating that the single stretched exponential is *worse* than a constant mean predictor. This likely reflects crosstalk contributions absent from the SDF model. A proposed extension replaces  $n$  by an effective qubit count  $n_{\text{eff}} = n/[1 + (n/n_c)^p]$  — an unpublished internal analysis, not reproduced here and not citable — capping the effective system size above a crosstalk threshold  $n_c \approx 10$ .
3. **ECME failure.** The multi-scale collective prediction scheme gives  $R^2 = -0.39$ , demonstrating that aggregate parameters cannot substitute for topology-resolved fitting.
4. **Spectral exponent uncertainty.** The Welch-PSD based  $\alpha_{\text{spectral}} = 0.661 \pm 0.426$  has a bootstrap 95% CI width of  $\sim 1.5$ , limited by the 50-delay time series length. Improved estimates require  $\geq 200$  delays or multitaper methods [51].

### 2. Theoretical limitations

5. **Cross-backend tension in THM-26.1.** The relation  $\beta = 1 - \alpha/2$  is validated on ibm\_fez but fails cross-backend:  $\beta_{\text{torino}} \approx 0.015$  would require  $\alpha \approx 1.97$ , inconsistent with the measured  $\alpha_{\text{spectral}} \approx 0.66$ . The theorem may require a hardware-dependent generalization (Eq. 188).
6.  **$\tau(\Lambda) = \tau_{\text{P}}/\Lambda$  is an axiom.** Phase 30E established that this scaling relation is *irreducible* within the SDF framework—logically equivalent to the uncertainty relation  $\Delta E \cdot \Delta t \geq \hbar$ . It is postulated and empirically validated across three energy scales (nuclear, atomic, molecular), but not derived from a deeper principle.
7. **Revival mechanism.** THM-26.3 (revival conditions for W and BellChain states) remains conjectural. Phase 30E provides a derivation from the Dicke algebra [8] (permutation symmetry  $\rightarrow$  resonance frequency), but a complete proof requires explicit solution of the coupled system–bath dynamics.
8. **Circularity residual.** Although Phase 26K introduced an independent FFT-based measurement of  $\alpha$ , the large uncertainty ( $\pm 0.43$ ) means the circularity between  $\alpha$  and  $\beta$  is ameliorated rather than eliminated.
9. **Observable decomposition assumption.** Equation (189) assumes that the decohered density matrix can be written as a convex combination of  $\rho_{\text{coh}}$  and  $\rho_{\text{mix}}$ . This holds exactly for depolarizing channels and approximately for dephasing-dominated dynamics, but may break down for strongly non-depolarizing channels.

### 3. Scope limitations

10. **GHZ-centric topology.** The majority of circuits (both backends) use GHZ-family states. Extension to arbitrary entanglement topologies (cluster, toric code, MERA-inspired) is needed to test PRED-30.5.
11. **No real-time dynamics.** All experiments use a parameterized delay  $\gamma$  rather than monitoring

decoherence in real time. The backflow prediction (PRED-30.4) requires a time-resolved BLP protocol that has not yet been implemented.

## XXVI. Conclusion

This document closes the programme. What follows separates what it established from what it did not, and states both without softening either.

### A. What survives

Three results survive as theorems or exact identities, and they are the ones usable by somebody who owes this framework nothing.

The  $\Lambda$ -hierarchy is not an irreducible postulate: it reduces to  $\tau = \hbar/E$  and  $\ell = \bar{\lambda}_C$ , so the spatial scale assigned to a system is exactly its reduced Compton wavelength. The accessible information collapses to  $I_Q = (\gamma/\tau)^\beta$ , which yields closed forms where the corpus had none and locates a distinguished point at  $D/L = 1/2$  where four independent properties coincide — one bit exactly, the entanglement sudden-death point for  $C_0 = 1$ , the maximum of  $DC$  at  $L^2/4$ , and a Bures arc of  $\pi/3$ . None of the four is a definition, and the point is invariant in  $\beta$ ,  $\tau$  and  $L$ .

Theorem 27 states that the temporal exponent of dephasing cannot be changed by any pulse sequence; only the amplitude responds.

Theorem 36 is the transferable one. Pollard’s criterion, used as a licence condition rather than a curiosity, converts a fitted exponent into a proof about what the underlying process cannot be.

Empirically,  $D + C = 1$  holds to machine precision across 650 circuits on two backends, and  $\beta < 1$  is the invariant that no platform violates.

### B. What does not

Axiom 3 has no derivation. The filter-function route cannot produce it for any pulse sequence, and correcting a sign error removes the only independent microscopic route offered. It is a postulate with a good fit.

Two published numbers did not follow from their inputs, one reported spectral mode lies below the Rayleigh limit of its own data, and the published comoving density of remnants corresponds to  $\Omega = 0.063$  rather than the 0.12 it is stated to reproduce.

More instructive than any of those: our own first correction of the LIGO significance was wrong in the same manner as the number it corrected, and the originally published value was closer to the defensible one than our correction of it. During final revision a further error was found in this paper’s own tabulation of threshold crossings, by an external reader who declined to accept a table that did not reconcile. And the classification of the dark-matter candidate in Sec. XXIII reached the wrong verdict on first attempt, by comparing available energy against a detector threshold without asking whether that energy could couple. *An audit is not automatically right because it is an audit.*

Eleven discrepancies remain open, listed in Sec. XXIV with the observation that would close each. They are open because we could not close them.

### C. The inference this work does not license

A stretched exponential is widely taken as a signature of complex or quantum behaviour. Theorem 36 says the opposite in the regime where the framework fits best:  $\beta < 1$  certifies that the data are reproducible by a positive distribution of classical exponential rates. Every platform in the corpus reports  $\beta < 1$ .

We state this plainly because the inverse reading is the natural one and it is wrong.

### D. The decisive test, which we do not perform

Section XXIB identifies one prediction that existing literature can already refute: on a single sample, the anomalous-diffusion exponent and the stretched-relaxation exponent must satisfy  $\beta = (1 + \alpha_{\text{diff}})/2$ , with  $\beta \geq 1/2$  as a corollary. Both quantities are routinely tabulated, by groups with no stake in this framework, for overlapping sample sets. No new experiment is required.

We do not perform it here: this document is an audit, and an audit that also supplies the confirmation of its own subject is not an audit. Whoever performs it settles the matter in either direction, and both directions are informative. If the relation holds, the framework acquires the one thing it lacks — a confirmed prediction on data its authors did not select. If it fails, a line closes, which this programme has treated throughout as a result of equal standing.

### E. On the endpoint

The programme ends at a conjecture, and it is worth being explicit about what that means. The dark-matter candidate of Sec. XXIII introduces no new particle, no new field and no free parameter: the remnant mass is  $m_P/2\sqrt{\pi}$  exactly, and the predicted phenomenology is precisely what dark matter is observed to have. It is also decoupled — the probability of a single interaction while crossing the Earth is  $10^{-32}$  — which makes it close to unfalsifiable by direct means and means it must be attacked at the primordial spectrum and at the stability of the remnant instead.

A hypothesis that explains what is seen and forbids anything else being seen is the most seductive kind and the least useful. We state it, give it its numbers, give it four falsifiers, and stop there.

### F. On the method

The result we regard as most transferable is negative. Five resolutions were proposed for the open problems and all five were rejected, four because they failed to cite a passage that already addressed the matter. Three further errors were caught in this paper itself after it was written, each by refusing to accept a number that did not reconcile.



A framework of this size cannot be audited by its author reading it, nor by an auditor checking algebra alone. It requires a protocol in which proposing a correction carries the burden of having searched for its own refutation, and in which a negative result — a line closed, a hypothesis retired, a number withdrawn — is recorded with the same standing as a positive one.

Section XXIV is the list of places where that burden could not be discharged. This section is the statement that the list is the honest end of the work rather than a gap in it.

### Acknowledgments

The audit was carried out with an adversarial protocol in which each proposed resolution was submitted to independent refutation before adoption. Five resolutions were proposed and all five were rejected. Three further errors were caught in this document after it was written — one of them by an external reader who declined to accept a table that did not reconcile. I regard this as the central methodological result of the work.

#### A. Register of Documented Inconsistencies

Each entry is verifiable by citing a source file and L<sup>A</sup>T<sub>E</sub>X label, and each is covered by a regression test in the accompanying implementation. Status codes: R = resolved, O = open, C = closed (verified not to be an inconsistency), I = invalidated.

*a. Distribution.* Fourteen resolved, eleven open, one closed, one invalidated. Of the eleven open, seven are of high impact. Twenty-six concern the internal algebra of the corpus; one, D18, is the single case in which the raw data contradict the text.

*b. On the last two entries.* D6 and D27 were both added after the document was otherwise finished, and both by the same mechanism: an adversarial pass over the audit itself. D6 had been recorded as resolved; reopening it followed from noticing that the register’s own text called the resolving bridge “a physical hypothesis”, and that Sec. XXIB designates the test of that same bridge as the decisive experiment of the programme. One does not test what is resolved. D27 is an internal contradiction of the present document — an appendix inherited from Paper 1 whose derivation Theorem 27 refutes — and it went unnoticed through the entire consolidation. Both are recorded here rather than quietly corrected, because the register is the instrument that is supposed to catch exactly this.

*c. On the closed entry.* D7 is retained in the register rather than deleted. An item that was investigated and found not to be a defect carries information—specifically, that the biological bound has a declared cutoff and that its order of magnitude is robust across the declared parameter range—and removing it would invite the same objection to be raised again.

## B. Reproducibility

Every numerical statement in this paper is reproducible from an accompanying implementation, which is deliberately dependency-free so that it can be audited without a toolchain.

### 1. The implementation

The core is written in TypeScript with no external dependencies and executes identically in a server runtime and in a browser, so that an interactive exploration and a batch verification cannot diverge numerically. It comprises:

- A registry of **138 equations**, each carrying source paper, source file, original L<sup>A</sup>T<sub>E</sub>X label, evidence level, declared domain and an executable implementation. An equation lacking any of these does not enter the registry.
- **354 regression tests**, including at least one per documented inconsistency, which assert the *actual* behaviour rather than the published assertion whenever the two differ.
- **37 published-number checks** that recompute values from the equations and report the relative deviation. Checks documenting an error pass when the discrepancy is reproducible.
- A catalogue of **26 hierarchy tiers** spanning 38.2 decades of physical  $\tau$ , with tiers lacking an absolute timescale flagged as normalised rather than silently assigned  $\tau = 1$ .
- The **spectral-resolution audit** of Sec. XVI as a reusable primitive, applicable to any claim of the form “a mode was detected at  $f$ ”.

### 2. Selected verifications

### 3. Reproducing the audit

The full suite runs in under one second on commodity hardware. Two entry points are provided: a test runner that executes all regression tests and published-number checks, and a service exposing the registry, the tier catalogue, the inconsistency register and the fitting routines over HTTP.

The adversarial refutation described in Sec. XXIV was carried out with eleven independent agents—five attacking the algebra of each proposed resolution, five attacking its consistency with the corpus under a requirement to cite file and label, and one auditing for omissions. Their findings are recorded in full alongside the implementation.

### 4. Scope and limitations of the audit

Three limitations should be stated.

First, the audit covers the equations and the numbers derived from them. It does not re-execute the quantum hardware experiments, and it takes the raw data files as authentic—which they appear to be, with complete circuit counts, exact shot numbers and recorded job identifiers.



TABLE XLII. Complete register. Impact is assessed by whether the item affects a published number, a load-bearing derivation, or neither.

| ID  | Status | Impact | Summary                                                                                        |
|-----|--------|--------|------------------------------------------------------------------------------------------------|
| D1  | R      | low    | Transition rate written with and without $L$ in the exponent                                   |
| D2  | R      | medium | Two incompatible Bloch-vector parametrisations                                                 |
| D3  | R      | medium | Three expressions for the single-qubit von Neumann entropy                                     |
| D4  | R      | high   | Two definitions of $I_Q$ with opposite monotonicity                                            |
| D5  | R      | low    | Dimensionally incorrect exponent in the gravitational crossover mass                           |
| D6  | O      | high   | $\alpha$ denotes two distinct exponents; factor 2–4 on $\beta$                                 |
| D7  | C      | low    | Biological information bound: cutoff is declared; not an inconsistency                         |
| D8  | R      | medium | $\gamma_{1/2}$ decreases as $\beta \rightarrow 0$ ; published prose asserts the opposite       |
| D9  | O      | high   | Filter-function route cannot produce $\beta < 1$ (Theorem 27)                                  |
| D10 | R      | medium | Revival regime lies outside the domain of Axiom 2                                              |
| D11 | R      | high   | $n^* = 1.848$ , not the published 2.6                                                          |
| D12 | R      | high   | Ringdown significance is $12.5\sigma$ with PDG scaling, not $13.8\sigma$                       |
| D13 | R      | low    | Spurious factor of two in the hierarchy derivation                                             |
| D14 | O      | high   | Condensation fixed point unsatisfiable; repair breaks the anti-entropic reading                |
| D15 | O      | high   | Bidirectionality axiom against the unique attractor                                            |
| D16 | R      | high   | Four expressions for $D$ ; canonical form fixed as the infidelity                              |
| D17 | I      | medium | $d^2 I_Q / d\gamma^2 < 0$ for $\beta < 1$ ; published criterion withdrawn                      |
| D18 | O      | high   | Reported spectral mode below the Rayleigh limit of its own data                                |
| D19 | R      | high   | Sign of the fluctuator coupling exponent                                                       |
| D20 | R      | medium | Basic-properties theorem requires the hypothesis $\beta > 0$                                   |
| D21 | O      | medium | $I_Q$ computed from absolute $D$ ; the invariant lives in $D/L$                                |
| D22 | O      | low    | Two-system compensation is a tautology, not a law                                              |
| D23 | R      | high   | Declared biological domain does not overlap §2c diffusion data                                 |
| D24 | O      | medium | §2b declares $\sigma$ as von Neumann entropy, computes $H_2(D)$                                |
| D25 | O      | low    | $n$ -qubit entropy expansion written in reverse order                                          |
| D26 | O      | high   | $\beta$ not universal across events; the catalogue rejects a single value                      |
| D27 | O      | high   | App. C derives $\beta = \alpha/2$ , then asserts $\alpha + 2\beta = 2$ to reach $1 - \alpha/2$ |

TABLE XLIII. Representative checks. Residuals are relative deviations.

| Quantity                               | Reference              | Residual              |
|----------------------------------------|------------------------|-----------------------|
| $\tau_P E_P / \hbar$                   | 1                      | 0                     |
| $\ell_P E_P / (\hbar c)$               | 1                      | 0                     |
| $\ell(\Lambda) / \lambda_C$ (electron) | 1                      | 0                     |
| $I_Q - (\gamma/\tau)^\beta$            | 0                      | $3.0 \times 10^{-14}$ |
| Scale invariance, $\lambda = 10^6$     | 0                      | $< 10^{-15}$          |
| Mercury perihelion advance             | 42.9881''/cy           | $< 2 \times 10^{-3}$  |
| Gravitational coherence length, proton | $1.5 \times 10^{23}$ m | $< 0.05$              |
| Minimum remnant mass                   | 6.09 $\mu\text{g}$     | $< 0.02$              |
| $\beta(n)$ , $n = 3 \dots 8$           | published table        | $< 0.03$              |
| Deviation bound, $\alpha = 1$          | 0.184L                 | $3.3 \times 10^{-4}$  |

Second, only one section was audited against its data files in the manner of Sec. XVI. The same treatment of the simulation, hardware and scaling sections is outstanding, and given what a single such audit produced, it would be imprudent to assume the others are clean.

Third, the five open problems of Sec. XXIV are open. This paper sharpens their statement and closes several escape routes; it does not solve them.

### C. Mathematical Derivations

This appendix provides the detailed mathematical derivations underpinning the core results of the SDF framework.

#### 1. Lindblad derivation of $\beta = 1 - \alpha/2$

I begin from the standard Lindblad master equation for the density operator  $\rho$ :

$$\frac{d\rho}{dt} = -i[H, \rho] + \sum_k \gamma_k(t) \left( L_k \rho L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho\} \right), \quad (\text{C1})$$

where  $H$  is the system Hamiltonian,  $L_k$  are the Lindblad (jump) operators, and  $\gamma_k(t)$  are time-dependent decoherence rates determined by the noise power spectral density  $S(\omega)$ .

For a noise spectrum with  $1/f^\alpha$  character,

$$S(\omega) \propto \frac{1}{\omega^\alpha}, \quad 0 < \alpha < 2, \quad (\text{C2})$$

the effective decoherence rate is obtained via Fourier transform:

$$\gamma(t) = \int_0^\infty S(\omega) e^{-i\omega t} d\omega \propto t^{\alpha/2-1}. \quad (\text{C3})$$

The integrated decoherence function is

$$D(t) = 1 - \exp\left(-\int_0^t \gamma(t') dt'\right). \quad (\text{C4})$$

Evaluating the integral:

$$\int_0^t t'^{\alpha/2-1} dt' = \frac{t^{\alpha/2}}{\alpha/2} \propto t^{\alpha/2}. \quad (\text{C5})$$

Thus, the decoherence function takes the stretched-exponential form

$$D(t) = 1 - \exp\left(-\left(\frac{t}{\tau}\right)^{\alpha/2}\right). \quad (\text{C6})$$

Comparing with the SDF canonical form  $D(\gamma) = L[1 - \exp(-(\gamma/\tau)^\beta)]$ , one identifies

$$\boxed{\beta = \frac{\alpha}{2}}, \quad (\text{C7})$$

or equivalently, in the complementary normalization convention where  $\alpha + 2\beta = 2$ :

$$\boxed{\beta = 1 - \frac{\alpha}{2}}. \quad (\text{C8})$$

□

**SUPERSEDED — this derivation is refuted by Theorem 27 of the present document**

**[ERRATUM]** The argument above is reproduced as published, and it does not hold. Two defects, and the second is the one that matters.

**The step marked “or equivalently” is not an equivalence.**  $\beta = \alpha/2$  and  $\beta = 1 - \alpha/2$  coincide only at  $\alpha = 1$ . Passing from one to the other requires  $\alpha + 2\beta = 2$ , which is precisely the relation the step is invoked to justify. The move is circular, and it is the origin of the symbol collision registered as D6.

**Neither exponent is what the integral produces.** Theorem 27 evaluates the same power-law bath  $S(\omega) = A\omega^{-\alpha}$  through the standard dephasing functional and obtains  $\Gamma(t) = A\mathcal{I}_\alpha[g]t^{1+\alpha}$  for *every* pulse-sequence filter  $g$ . The document therefore carries three mutually incompatible temporal exponents for one bath —  $t^{\alpha/2}$  here,  $t^{1+\alpha}$  in Sec. XIII, and the  $t^{2-\alpha}$  of Paper 2 corrected in Sec. XV — and only the third of these survives its own derivation.

We retain the passage rather than delete it because the corrected chain is only legible against what it corrects. Its conclusion,  $\beta = 1 - \alpha/2$ , is not withdrawn: it is relabelled. It stands as **[AX- IOM]** (Axiom 3), an empirical postulate with a good fit and no derivation, exactly as Sec. XIII concludes.

## 2. Information conservation: $D + C = 1$

**[THM]** (Information Conservation). *Let  $D(\gamma)$  be the decoherence and  $C(\gamma) \equiv 1 - D(\gamma)$  the coherence. Then*

$$I_{\text{total}}(\gamma) \equiv D(\gamma) + C(\gamma) = 1 \quad \forall \gamma. \quad (\text{C9})$$

*Proof.* By definition,  $C(\gamma) = 1 - D(\gamma)$ . Therefore,

$$\frac{\partial I_{\text{total}}}{\partial \gamma} = \frac{\partial D}{\partial \gamma} + \frac{\partial C}{\partial \gamma} = \frac{\partial D}{\partial \gamma} - \frac{\partial D}{\partial \gamma} = 0. \quad (\text{C10})$$

□

## 3. Accessible quantum information

The von Neumann entropy of a quantum state  $\rho$  is

$$S(\rho) = -\text{Tr}(\rho \ln \rho). \quad (\text{C11})$$

For a single qubit undergoing decoherence parameterized by  $D$ , the accessible quantum information is

$$I_Q = 1 - H_2(D), \quad (\text{C12})$$

where  $H_2(D) = -D \ln D - (1 - D) \ln(1 - D)$  is the binary entropy function.

In the limit of small decoherence ( $D \ll 1$ ), Taylor expansion gives:

$$H_2(D) \approx -D \ln D + D + \mathcal{O}(D^2), \quad (\text{C13})$$

and therefore

$$I_Q \approx 1 - H_2(D) \approx -\ln(1 - D) + \mathcal{O}(D^2). \quad (\text{C14})$$

□

#### 4. Scaling law derivation

For a composite system of  $n$  qubits, the effective dimensionality of the Hilbert space grows as

$$d_{\text{eff}} \propto n, \quad (\text{C15})$$

while the effective noise exponent scales logarithmically:

$$\alpha_{\text{eff}}(n) \propto \ln(n). \quad (\text{C16})$$

Combining these two scaling relations through the Lindblad-SDF correspondence (Appendix C1), the system-size dependence of the stretching exponent follows:

$$\beta(n) \approx \frac{\beta_0}{n^\xi}, \quad (\text{C17})$$

where  $\beta_0$  is the single-qubit stretching exponent and  $\xi$  characterizes the system-size scaling. This predicts that larger entangled systems decohere with progressively smaller  $\beta$ , consistent with the experimental observations in Sec. X.

#### 5. Lorentz invariance of the decoherence function

I promote the decoherence parameter to a four-vector:

$$\gamma^\mu = (\gamma^0, \tilde{\gamma}), \quad \mu = 0, 1, 2, 3, \quad (\text{C18})$$

with Lorentz-invariant magnitude

$$|\gamma|^2 = \gamma_\mu \gamma^\mu = (\gamma^0)^2 - |\tilde{\gamma}|^2 \geq 0. \quad (\text{C19})$$

**[THM]** (Lorentz Scalar Decoherence). *If  $D$  depends on  $\gamma^\mu$  only through the invariant  $|\gamma|^2$ , i.e.,*

$$D(|\gamma|) = L \left[ 1 - \exp \left( - \left( \frac{|\gamma|}{\tau} \right)^\beta \right) \right], \quad (\text{C20})$$

*then  $D$  is a Lorentz scalar:  $D' = D$  under any proper Lorentz transformation  $\Lambda$ .*

*Proof.* Under  $\gamma^\mu \rightarrow \Lambda^\mu{}_\nu \gamma^\nu$ , the invariant  $|\gamma|^2 = \gamma_\mu \gamma^\mu$  is preserved by definition of  $\Lambda \in SO^+(1, 3)$ . Since  $D$  is a function solely of  $|\gamma|$ , it transforms as a scalar:

$$D'(|\gamma'|) = D(|\gamma|). \quad (\text{C21})$$

□

This construction ensures that the SDF framework is compatible with special relativity. The physical interpretation is that decoherence, viewed as information redistribution, respects the causal structure of spacetime.

#### 6. Inverse function

**[THM]** (Existence of  $D^{-1}$ ). *The SDF decoherence function  $D(\gamma) = L[1 - \exp(-(\gamma/\tau)^\beta)]$  is strictly monotonically increasing for  $\gamma \geq 0$ ,  $\beta > 0$ . Hence  $D^{-1}$  exists and is given by:*

$$D^{-1}(D) = \tau \left[ -\ln \left( 1 - \frac{D}{L} \right) \right]^{1/\beta}. \quad (\text{C22})$$

*Proof.* Strict monotonicity follows from Theorem 1 (Main Text). Define  $u = D/L \in [0, 1)$ . Then:

$$u = 1 - \exp \left( - \left( \frac{\gamma}{\tau} \right)^\beta \right), \quad (\text{C23})$$

$$1 - u = \exp \left( - \left( \frac{\gamma}{\tau} \right)^\beta \right), \quad (\text{C24})$$

$$-\ln(1 - u) = \left( \frac{\gamma}{\tau} \right)^\beta, \quad (\text{C25})$$

$$\gamma = \tau [-\ln(1 - u)]^{1/\beta}. \quad (\text{C26})$$

Substituting  $u = D/L$  yields the stated result. □

#### D. Experimental Details

This appendix documents the quantum hardware, circuit configurations, state preparation protocols, error mitigation strategies, and fitting procedures used throughout the experimental campaigns.

##### 1. IBM Quantum backends

All quantum experiments were performed on IBM Quantum processors accessed through the Qiskit Runtime service. The three primary backends used were:

- **ibm\_torino**: 133-qubit Heron r2 processor. Used in Phase 21 (initial Bell/GHZ/W measurements) and Phase 29 (high-resolution GHZ Ramsey decay).
- **ibm\_fez**: 156-qubit Heron r2 processor. Primary workhorse for Phases 12–13 (multi-family scaling), Phase 25 (universality test), and Phases 26B–K (systematic delay sweeps).
- **ibm\_marrakesh**: Heron-class processor used for load balancing during Phase 29 high-throughput campaigns.

##### 2. Circuit configurations

Table XLIV summarizes the circuit configurations across experimental phases.

##### 3. State preparation protocols

###### a. GHZ states

The  $n$ -qubit Greenberger–Horne–Zeilinger state

$$|\text{GHZ}_n\rangle = \frac{1}{\sqrt{2}}(|0\rangle^{\otimes n} + |1\rangle^{\otimes n}) \quad (\text{D1})$$

TABLE XLIV. Summary of circuit configurations by experimental phase.

| Phase   | States                                      | Configs              | Shots | Circuits | Notes                  |
|---------|---------------------------------------------|----------------------|-------|----------|------------------------|
| 12-13   | 6 families, $n = 2-50$                      | 42                   | 8192  | 105      | Multi-family scaling   |
| 21      | Bell, GHZ, W                                | 5                    | 4096  | 5        | Initial validation     |
| 26B     | All families                                | —                    | 8192  | —        | Systematic sweep       |
| 26C     | GHZ $n = 3-8$                               | $6 \times 11$ delays | 8192  | 66       | Delay-resolved GHZ     |
| 26E     | 5 states $\times$ 5 sizes $\times$ 7 delays | 175                  | 4096  | 175      | Full factorial design  |
| 26K     | Cluster Ramsey                              | —                    | 8192  | —        | FFT noise spectroscopy |
| 29 v2.0 | GHZ $n = 3-12 \times 50$ delays             | 400                  | 8192  | 400      | High-resolution decay  |

is prepared via a Hadamard gate on the first qubit followed by a cascade of CNOT gates:

$$U_{\text{GHZ}} = \prod_{k=1}^{n-1} \text{CNOT}_{k,k+1} \cdot (H \otimes I^{\otimes(n-1)}). \quad (\text{D2})$$

Circuit depth:  $O(n)$ .

#### b. W states

The  $n$ -qubit W state

$$|W_n\rangle = \frac{1}{\sqrt{n}}(|100 \cdots 0\rangle + |010 \cdots 0\rangle + \cdots + |000 \cdots 1\rangle) \quad (\text{D3})$$

is prepared using a linear-depth cascade of controlled- $R_Y$  and CNOT gates. For the  $k$ -th qubit ( $k = 1, \dots, n$ ), the rotation angle is  $\theta_k = 2 \arcsin(1/\sqrt{n-k+1})$ . This yields circuit depth  $O(n)$ , a significant improvement over the naive `StatePreparation` approach with depth  $O(2^n)$ .

#### c. Cluster states

The linear cluster state on  $n$  qubits is prepared by applying Hadamard gates to all qubits, followed by alternating CZ gates between nearest neighbors:

$$|\text{Cluster}_n\rangle = \prod_{\langle i,j \rangle} \text{CZ}_{i,j} \cdot H^{\otimes n} |0\rangle^{\otimes n}. \quad (\text{D4})$$

#### d. Dicke states

The Dicke state  $|D_k^n\rangle$  with  $k$  excitations among  $n$  qubits is prepared using a recursive symmetry decomposition [52]. The algorithm constructs the symmetric superposition by recursively splitting excitations and applying controlled rotations, achieving depth  $O(kn)$ .

#### e. Bell chain states

Bell chain states consist of  $\lfloor n/2 \rfloor$  sequential Bell pairs created by applying  $H \otimes I$  followed by CNOT on consecutive qubit pairs:

$$|\text{BellChain}_n\rangle = \bigotimes_{i=1}^{\lfloor n/2 \rfloor} \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)_{2i-1,2i}. \quad (\text{D5})$$

### 4. Error mitigation

Three error mitigation techniques were employed:

#### a. Readout error correction

Measurement (readout) errors were corrected by constructing the  $2^n \times 2^n$  confusion matrix  $\mathbf{M}$  from calibration circuits (preparing each computational basis state and measuring). The corrected probability vector is obtained by matrix inversion:

$$\vec{p}_{\text{corrected}} = \mathbf{M}^{-1} \vec{p}_{\text{raw}}. \quad (\text{D6})$$

For large  $n$ , tensored mitigation (assuming independent qubit readout errors) reduces the matrix dimension to  $n$  independent  $2 \times 2$  matrices.

#### b. Dynamical decoupling

XX-pulse dynamical decoupling sequences were inserted during idle periods in the circuit to suppress low-frequency dephasing noise. The pulse spacing was optimized for the  $1/f^\alpha$  noise spectrum of each backend.

#### c. Post-selection

Parity filtering was applied to GHZ-type states, discarding measurement outcomes that violate the expected parity symmetry (all-even or all-odd bit strings for  $|\text{GHZ}_n\rangle$ ). This reduces the effective shot count but significantly improves fidelity estimates.

### 5. Fitting procedure

All fits to the SDF decoherence function

$$D(\gamma) = L \left[ 1 - \exp\left(-\left(\frac{\gamma}{\tau}\right)^\beta\right) \right] \quad (\text{D7})$$

were performed using `scipy.optimize.curve_fit` with the Levenberg–Marquardt algorithm. Parameter uncertainties were extracted from the diagonal of the covariance matrix returned by the optimizer:

$$\sigma_{\theta_i} = \sqrt{C_{ii}}, \quad (\text{D8})$$

where  $\mathbf{C}$  is the covariance matrix returned by the optimizer and  $\theta_i \in \{\beta, \tau, L\}$ .

Initial parameter guesses were set to  $\beta_0 = 1.0$ ,  $\tau_0 = \text{median}(\gamma)$ , and  $L_0 = \max(D_{\text{obs}})$ . Bounds were enforced:  $\beta \in (0, 3]$ ,  $\tau > 0$ , and  $L \in (0, 1]$ . Goodness of fit was assessed via  $R^2$  and the Akaike Information Criterion (AIC).



## E. Raw Data Tables

This appendix provides the raw experimental data and numerical results supporting the analyses presented in the main text.

### 1. Foundational data summary

The complete dataset from the foundational analysis (Sec. VI) comprises:

- 42 IBM Quantum configurations (6 state families  $\times$  multiple qubit counts  $n$ ),
- 38 LIGO binary black hole (BBH) ringdown events,
- 5 cross-platform validation measurements,
- 12 Lindblad derivation calibration points.

### 2. Phase 21 raw data

Table XLV presents the raw coherence and decoherence measurements from the initial Phase 21 validation campaign on `ibm_torino`.

TABLE XLV. Phase 21 measurements on `ibm_torino` (4096 shots per circuit).

| State            | $C$   | $D$   | $D + C$ | Shots | Job ID  |
|------------------|-------|-------|---------|-------|---------|
| Bell ( $n = 2$ ) | 0.943 | 0.057 | 1.000   | 4096  | cv8k... |
| GHZ ( $n = 3$ )  | 0.871 | 0.129 | 1.000   | 4096  | cv8k... |
| W ( $n = 3$ )    | 0.912 | 0.088 | 1.000   | 4096  | cv8k... |
| GHZ ( $n = 4$ )  | 0.782 | 0.218 | 1.000   | 4096  | cv8k... |
| GHZ ( $n = 5$ )  | 0.694 | 0.306 | 1.000   | 4096  | cv8k... |

The identity  $D + C = 1.000$  is confirmed exactly for all states, consistent with the information conservation theorem (Theorem 11).

### 3. Phase 26C GHZ decay data

Table XLVI summarizes the stretched-exponential fit parameters for GHZ states at varying qubit counts, obtained from Phase 26C delay-resolved measurements on `ibm_fez`.

TABLE XLVI. Phase 26C: GHZ stretched-exponential fit parameters ( $n = 3$ –8).

| $n$ | $\beta$ | $\sigma_\beta$ | $\tau$ ( $\mu\text{s}$ ) | $R^2$ | $L$  |
|-----|---------|----------------|--------------------------|-------|------|
| 3   | 1.013   | 0.042          | 12.7                     | 0.997 | 0.98 |
| 4   | 0.872   | 0.038          | 9.4                      | 0.995 | 0.97 |
| 5   | 0.761   | 0.051          | 7.1                      | 0.993 | 0.96 |
| 6   | 0.683   | 0.047          | 5.8                      | 0.991 | 0.95 |
| 7   | 0.614   | 0.055          | 4.6                      | 0.988 | 0.94 |
| 8   | 0.557   | 0.063          | 3.9                      | 0.985 | 0.93 |

The monotonic decrease of  $\beta$  with  $n$  is consistent with the scaling law  $\beta(n) \approx \beta_0/n^\xi$  (Appendix C 4).

## 4. Phase 26E summary

Phase 26E comprised 175 circuits (5 state families  $\times$  5 system sizes  $\times$  7 delay values) executed on `ibm_fez` with 4096 shots each. The family-averaged stretching exponents are:

- **Product states:**  $\beta = 1.116 \pm 0.034$
- **Cluster states:**  $\beta = 1.107 \pm 0.041$
- **GHZ states:**  $\beta = 1.013 \pm 0.028$
- **Dicke states:**  $\beta = 0.201 \pm 0.089$  (anomalous; see Appendix F 2)
- **W states:** Revival behavior observed (non-monotonic  $D(\gamma)$ ; see Appendix F)

## 5. Phase 24 $\Lambda$ sample data

Table XLVII provides representative decoherence parameters across nuclear, atomic, and molecular systems used in the Phase 24  $\Lambda$ -universality analysis.

TABLE XLVII. Phase 24: Decoherence parameters for representative physical systems.

| System                   | Scale     | $\beta$ | $\tau$            | Source          |
|--------------------------|-----------|---------|-------------------|-----------------|
| $^{208}\text{Pb}$        | Nuclear   | 0.87    | 24.1 fm/c         | NMR             |
| $^{56}\text{Fe}$         | Nuclear   | 0.91    | 18.3 fm/c         | NMR             |
| He                       | Atomic    | 0.95    | 1.2 ns            | Spectroscopy    |
| Cs                       | Atomic    | 0.78    | 3.4 $\mu\text{s}$ | Fountain clock  |
| $\text{N}\equiv\text{N}$ | Molecular | 0.82    | 0.45 ps           | IR spectroscopy |
| C–C                      | Molecular | 0.88    | 1.1 ps            | Raman           |

## 6. Phase 29 data

Phase 29 (v2.0) executed 400 circuits: GHZ states with  $n = 3$ –12 qubits  $\times$  50 logarithmically spaced delay values, on `ibm_torino` and `ibm_marrakesh` with 8192 shots each.

Key results:

- **BMRB validation:** 12 independent biological relaxation systems tested against ECME model predictions (see Appendix F for circularity analysis).
- **Mercury perihelion:** 1PN general-relativistic precession fitted with SDF model;  $\beta_{\text{Mercury}} = 0.993 \pm 0.004$ .
- **$\eta$  measurements:** Noise spectral index  $\eta$  extracted from Ramsey interferometry across all qubit counts.

## F. Negative Results and Lessons Learned

Transparent reporting of negative results is essential for scientific integrity. This appendix documents failed hypotheses, anomalous findings, and methodological corrections that shaped the development of the SDF framework.

TABLE XLVIII. Phase 25 results: failed universality test.

| State family | $\beta$ | $\sigma_\beta$ |
|--------------|---------|----------------|
| GHZ          | 0.803   | 0.54           |
| W            | 1.655   | 0.22           |
| Dicke        | 2.000   | 1.21           |

### 1. Phase 25: Failed universality test

The Phase 25 campaign on `ibm_fez` attempted to demonstrate that  $\beta$  is universal across state families at fixed qubit count. The results were:

The W-state result ( $\beta = 1.655$ ) is *unphysical* for a decoherence process, and the Dicke-state uncertainty is unacceptably large.

**Root cause:** The decoherence parameter was incorrectly defined as  $\gamma = n/n_{\max}$  (system-size scaling) instead of  $\gamma = t/t_{\text{ref}}$  (temporal evolution). This conflated two distinct physical mechanisms.

**Key discovery:** Analysis of the Phase 25 data revealed the important distinction between gate-induced decoherence ( $\beta_{\text{gate}} = 0.852$ ) and idle decoherence ( $\beta_{\text{idle}} = 0.327$ ). This bifurcation, initially an artifact of the incorrect parameterization, led to a deeper understanding of noise channel structure.

### 2. Dicke state anomalies

Across all experimental phases, Dicke states  $|D_k^n\rangle$  consistently yielded anomalously low stretching exponents ( $\beta \approx 0.2$ ) with high variance. This behavior is attributed to the  $k$ -excitation subspace dynamics: the Dicke state decoherence is dominated by transitions between the symmetric subspace sectors, which are governed by a different noise coupling mechanism than the  $|0\rangle^{\otimes n} \leftrightarrow |1\rangle^{\otimes n}$  coherence probed by GHZ states. The multi-excitation structure introduces additional relaxation pathways with distinct timescales, resulting in a highly stretched ( $\beta \ll 1$ ) effective decay.

### 3. $\sqrt{n/2}$ hypothesis refutation

An early hypothesis proposed that the stretching exponent scales as  $\beta(n) = \sqrt{n/2}$ , motivated by dimensional analysis of the noise spectral density. This predicts:

$$\beta(4) = \sqrt{4/2} = \sqrt{2} \approx 1.41. \quad (\text{F1})$$

The measured value from Phase 26A (initial Dicke analysis) was  $\beta(4) = 0.52 \pm 0.05$ , refuting the hypothesis at  $> 17\sigma$  significance. The correct scaling law is  $\beta(n) \approx \beta_0/n^\xi$ , which predicts *decreasing*  $\beta$  with  $n$ .

### 4. Circularity in $\alpha \leftrightarrow \beta$ mapping

Earlier versions of the SDF framework derived  $\beta$  from  $\alpha$  (noise spectral exponent) and simultaneously used  $\beta$  to infer  $\alpha$  from decoherence measurements. This circular reasoning was identified and corrected:

**Resolution:** Phase 26K introduced independent FFT-based Ramsey noise spectroscopy to measure  $\alpha$  directly

from the noise power spectrum, breaking the circularity. The Ramsey signal  $\langle X(t) \rangle$  is Fourier-transformed to extract  $S(\omega)$ , and  $\alpha$  is obtained from a power-law fit to  $S(\omega) \propto 1/\omega^\alpha$ , independently of the decoherence curve fitting that yields  $\beta$ .

### 5. BMRB circularity confirmed (Phase 29)

The ECME (Extended Coherence-Mediated Evolution) model was initially calibrated using protein relaxation data from the Biological Magnetic Resonance Bank (BMRB). Phase 29 tested whether this calibration has independent predictive power using 12 biological systems from sources *not* in the BMRB database:

#### Independent data sources:

- [29]: RNA polymerase diffusion
- [30]: Bacterial chromosome locus tracking
- [31]: Plasma membrane protein diffusion
- [32]: Telomere dynamics in nucleus
- [53]: Lipid granule motion in yeast
- [54]: mRNA-protein complex motion
- [55]: Protein mobility in *E. coli*

#### Validation results:

1. **Validation 1 ( $R^2$  test):**  $R^2 = -0.39$ . This is *worse* than predicting the mean for all systems. The model has **negative** explanatory power on independent data. **FAIL**.
2. **Validation 2 (Rank correlation test):** Spearman  $\rho = 0.14$ ,  $p = 0.66$ . No statistically significant correlation between predicted and observed values. **FAIL**.
3. **Validation 3 (Parameter consistency):**  $\beta_{0,\text{indep}} = 0.68 \pm 0.06$ . This is the only salvageable aspect—the overall scale parameter  $\beta_0$  is consistent between BMRB-calibrated and independent datasets. **CONSISTENT**.

**Conclusion:** The calibration constant  $\kappa' = 0.0835$  is circularly derived from the BMRB data itself. The ECME model, as calibrated, has **no independent predictive power**. The  $\beta_0$  consistency suggests a universal scale but does not validate the model's detailed predictions.

### 6. T1/T2 proxy inadequacy (Phase 29)

The standard relaxation times  $T_1$  (longitudinal) and  $T_2$  (transverse) are widely used as proxies for decoherence timescales. Phase 29 analysis revealed fundamental limitations of this approach:

- $T_1$  and  $T_2$  are defined in the Larmor frequency regime and capture only the spectral density at  $\omega_0$  and  $\omega \approx 0$ , respectively.

- The SDF stretching exponent  $\beta$  encodes the *full* spectral structure of the noise, not just single-frequency components.
- Quantitatively:  $\beta_{\text{real}} = 0.457 \neq 0.76$  (BMRB-calibrated), a  $> 5\sigma$  discrepancy.
- Using  $T_1/T_2$ -calibrated predictions on independent data yields  $R^2 = -0.39$ , confirming inadequacy.

$T_1$  and  $T_2$  are *necessary but insufficient* as sole proxies for the stretching exponent  $\beta$ . A complete characterization requires broadband noise spectroscopy (e.g., Ramsey-based FFT as implemented in Phase 26K).

### G. LIGO Ringdown Catalog

This appendix presents the full analysis of gravitational-wave ringdown signals from LIGO/Virgo binary black hole (BBH) merger events, fitted with the SDF decoherence model.

#### 1. Analysis methodology

The post-merger ringdown phase of each BBH event was fitted to the SDF decoherence function:

$$D(t) = L \left[ 1 - \exp \left( - \left( \frac{t}{\tau} \right)^\beta \right) \right], \quad (\text{G1})$$

where  $t$  is the time after merger,  $\tau$  is the characteristic damping time,  $\beta$  is the stretching exponent, and  $L$  is the asymptotic limit.

For comparison, each event was also fitted with the standard general-relativity (GR) exponential damping model, corresponding to  $\beta = 1$  (fixed):

$$D_{\text{GR}}(t) = L \left[ 1 - \exp \left( - \frac{t}{\tau_{\text{GR}}} \right) \right]. \quad (\text{G2})$$

Model comparison was performed via the Akaike Information Criterion difference:

$$\Delta\text{AIC} = \text{AIC}_{\text{GR}} - \text{AIC}_{\text{SDF}}. \quad (\text{G3})$$

Positive  $\Delta\text{AIC}$  favors the SDF model (free  $\beta$ ); values  $> 2$  indicate substantial evidence, and  $> 6$  indicate strong evidence.

#### 2. BBH ringdown fit results

Table XLIX presents the SDF fit results for representative events from the three LIGO/Virgo gravitational-wave transient catalogs (GWTC-1, GWTC-2, GWTC-3). The full catalog of 38 events is available in the supplementary data files.

**Combined result** (weighted mean over all 38 events):

$$\boxed{\beta_{\text{LIGO}} = 0.931 \pm 0.003}, \quad (\text{G4})$$

representing a deviation of  $13.8\sigma$  from the GR prediction  $\beta = 1$ . This is consistent with the SDF prediction of sub-exponential ringdown damping due to the complex noise environment of the merged black hole remnant.

### 3. Catalog breakdown

The 38 analyzed events are distributed across the three LIGO/Virgo catalogs as follows:

The consistency of  $\langle\beta\rangle$  across catalogs (spanning observing runs O1 through O3) provides evidence against systematic detector artifacts, since the LIGO/Virgo instruments underwent significant hardware upgrades between runs.

#### H. Information Conservation: Formal Derivation

This appendix provides a rigorous, self-contained derivation of the information conservation theorem within the SDF framework, expanding on the summary in the main text and Appendix C2.

##### 1. Framework setup

**[DEF]** (System–Environment Decomposition). Consider a closed quantum system with total Hilbert space

$$\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_E, \quad (\text{H1})$$

where  $\mathcal{H}_S$  is the system Hilbert space of dimension  $d_S = \dim \mathcal{H}_S$  and  $\mathcal{H}_E$  is the environment Hilbert space.

**[DEF]** (Decoherence Function). Let  $\rho(\gamma)$  denote the reduced density matrix of the system at decoherence parameter  $\gamma \geq 0$ , with initial state  $\rho_0 = \rho(0)$ . The decoherence function is defined as

$$D(\gamma) \equiv 1 - F(\rho(\gamma), \rho_0), \quad (\text{H2})$$

where  $F(\rho, \sigma) = \left( \text{Tr} \sqrt{\sqrt{\sigma} \rho \sqrt{\sigma}} \right)^2$  is the Uhlmann fidelity. For pure initial states,  $F(\rho, |\psi_0\rangle\langle\psi_0|) = \langle\psi_0| \rho |\psi_0\rangle$ .

**[DEF]** (Coherence Function). The coherence function is defined as the complement of the decoherence:

$$C(\gamma) \equiv F(\rho(\gamma), \rho_0) = 1 - D(\gamma). \quad (\text{H3})$$

##### Properties of $D$ and $C$ :

1.  $D(\gamma) \in [0, L]$  with  $L \leq 1$  and  $D(0) = 0$ .
2.  $C(\gamma) \in [1 - L, 1]$  with  $C(0) = 1$ .
3. Both  $D$  and  $C$  are continuous functions of  $\gamma$ .
4.  $D$  is monotonically non-decreasing;  $C$  is monotonically non-increasing.

##### 2. Conservation theorem: proof

**[THM]** (Information Conservation — Theorem 11). *Define the total information as*

$$I_{\text{total}}(\gamma) \equiv D(\gamma) + C(\gamma). \quad (\text{H4})$$

*Then  $I_{\text{total}}(\gamma) = 1$  for all  $\gamma \geq 0$ , and consequently*

$$\frac{\partial I_{\text{total}}}{\partial \gamma} = 0 \quad \forall \gamma \geq 0. \quad (\text{H5})$$

TABLE XLIX. SDF ringdown parameters for representative LIGO/Virgo BBH events.  $\beta$  and  $\tau$  are from the free- $\beta$  fit;  $\Delta\text{AIC}$  compares SDF to GR ( $\beta = 1$ ); SNR is the matched-filter signal-to-noise ratio.

| Event           | Catalog | $\beta$ | $\sigma_\beta$ | $\tau$ (ms) | $R^2$ | $\Delta\text{AIC}$ | SNR  |
|-----------------|---------|---------|----------------|-------------|-------|--------------------|------|
| GW150914        | GWTC-1  | 0.927   | 0.031          | 4.2         | 0.994 | 8.3                | 25.1 |
| GW151226        | GWTC-1  | 0.941   | 0.058          | 2.1         | 0.987 | 3.7                | 13.1 |
| GW170104        | GWTC-1  | 0.918   | 0.044          | 3.8         | 0.991 | 5.9                | 13.0 |
| GW170814        | GWTC-1  | 0.935   | 0.039          | 3.5         | 0.993 | 6.4                | 18.3 |
| GW190412        | GWTC-2  | 0.922   | 0.033          | 4.6         | 0.995 | 7.8                | 19.4 |
| GW190521        | GWTC-2  | 0.908   | 0.047          | 8.9         | 0.989 | 4.2                | 14.7 |
| GW190814        | GWTC-2  | 0.944   | 0.029          | 3.1         | 0.996 | 9.1                | 25.0 |
| GW191216_213338 | GWTC-2  | 0.931   | 0.052          | 2.8         | 0.990 | 4.5                | 12.1 |
| GW200115_042309 | GWTC-3  | 0.938   | 0.041          | 2.4         | 0.992 | 5.6                | 11.9 |
| GW200225_060421 | GWTC-3  | 0.925   | 0.048          | 3.3         | 0.988 | 4.1                | 13.2 |

TABLE L. Distribution of BBH events by catalog.

| Catalog         | Events    | Weighted $\langle\beta\rangle$      |
|-----------------|-----------|-------------------------------------|
| GWTC-1          | 9         | $0.933 \pm 0.008$                   |
| GWTC-2          | 19        | $0.929 \pm 0.004$                   |
| GWTC-3          | 10        | $0.932 \pm 0.006$                   |
| <b>Combined</b> | <b>38</b> | <b><math>0.931 \pm 0.003</math></b> |

*Proof.* By the definitions of  $D(\gamma)$  and  $C(\gamma)$ :

$$\begin{aligned}
 I_{\text{total}}(\gamma) &= D(\gamma) + C(\gamma) \\
 &= [1 - F(\rho(\gamma), \rho_0)] + F(\rho(\gamma), \rho_0) \\
 &= 1.
 \end{aligned} \tag{H6}$$

This identity holds for *all*  $\gamma \geq 0$ , independently of the specific form of  $D(\gamma)$  (i.e., independently of  $\beta$ ,  $\tau$ , or  $L$ ). Differentiating:

$$\frac{\partial I_{\text{total}}}{\partial \gamma} = \frac{\partial}{\partial \gamma}(1) = 0. \tag{H7}$$

□

**Remark.** This result is *exact*, not approximate. It follows from the *definition* of  $C = 1 - D$ , which is a partition of the total information content into two complementary channels. No assumptions about the dynamics (Markovian, non-Markovian, Lindblad, etc.) are required.

### 3. Von Neumann entropy connection

The von Neumann entropy of the reduced system state is

$$S(\rho) = -\text{Tr}(\rho \ln \rho). \tag{H8}$$

**[THM]** (Entropy–Decoherence Relation). *For a system undergoing decoherence from a pure initial state  $\rho_0 = |\psi_0\rangle\langle\psi_0|$ :*

1.  $S(\rho(0)) = 0$  (pure state).
2.  $S(\rho(\gamma))$  is monotonically non-decreasing in  $\gamma$ .
3.  $S(\rho(\gamma)) \leq \ln d_S$  (maximally mixed state bound).

*Proof.* (1) follows from  $\rho_0$  being pure. (2) follows from the contractivity of the partial trace under completely positive trace-preserving (CPTP) maps. (3) is the standard entropy bound for a  $d_S$ -dimensional system. □

The accessible quantum information is defined as

$$I_Q(\gamma) = \ln d_S - S(\rho(\gamma)). \tag{H9}$$

For a single qubit ( $d_S = 2$ ) with decoherence  $D$ :

$$I_Q = \ln 2 - H_2(D) = \ln 2 + D \ln D + (1 - D) \ln(1 - D), \tag{H10}$$

where  $H_2(D) = -D \ln D - (1 - D) \ln(1 - D)$  is the binary entropy. This quantifies how much information about the initial state remains extractable from the system alone, without access to the environment.

### 4. Implications

The information conservation theorem has several profound physical consequences:

#### a. Information redistribution, not destruction

**[HIP]** (No-Destruction Principle). *Quantum information is never destroyed during decoherence; it is redistributed between system-accessible ( $C$ ) and environment-accessible ( $D$ ) channels.*

This follows directly from the unitarity of quantum mechanics: the total system–environment state  $|\Psi_{SE}(\gamma)\rangle$  evolves unitarily under the total Hamiltonian  $H_{SE}$ , preserving the total von Neumann entropy  $S(\rho_{SE}) = 0$ . The apparent “loss” of information (increasing  $D$ ) is precisely compensated by the transfer of that information to system–environment correlations (increasing entanglement entropy).

#### b. Consistency with unitarity

The SDF framework is manifestly consistent with unitary quantum mechanics. The stretched-exponential form of  $D(\gamma)$  describes *how fast* information redistributes (controlled by  $\beta$  and  $\tau$ ), not *whether* it is conserved. Conservation is guaranteed by the algebraic identity  $D + C = 1$ .

c. *Black hole information paradox*

**[CONJ]** (Stretched-Exponential Hawking Radiation). *If the SDF framework applies to black hole evaporation, then:*

1. Information gradually leaks from the black hole interior to the Hawking radiation with a stretching exponent  $\beta < 1$ .
2. The Page curve is modified: the crossover from increasing to decreasing entanglement entropy occurs at a different time than the standard Page time, governed by  $\beta$ .
3. No “firewall” or violation of unitarity is required, since  $D + C = 1$  holds at all times.

The LIGO ringdown analysis (Appendix G) provides indirect evidence for this conjecture: the measured

$\beta_{\text{LIGO}} = 0.931 \pm 0.003$  for black hole ringdown is consistent with sub-exponential information redistribution in the strong-gravity regime.

d. *Operational interpretation*

In the SDF framework, the conservation law  $D + C = 1$  provides a practical bookkeeping tool:

- Measuring  $D(\gamma)$  immediately gives  $C(\gamma) = 1 - D(\gamma)$  without independent measurement.
- The inverse function  $D^{-1}$  (Appendix C 6) allows reconstruction of the decoherence timeline from a single fidelity measurement.
- Cross-scale universality of the  $D + C = 1$  identity serves as a self-consistency check for experimental data (verified in Phase 21; see Table XLV).

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